An easy approach
to
environmental chemistry
(Solution manual to
past papers)
For
BS (H) 8th semester
students Punjab university
affiliated colleges

Contents

ENVIRONMENTAL CHEMISTRY

8th SEMESTER PAST PAPERS

COURSE CODE : (CHEM # 402)

2014 Short Questions :

1. **Define leaching?**

A natural or industrial process in which a solute is removed from a solid mixture by dissolving it in a liquid (usually water or an acidic/basic solution) is called Leaching.

- It is widely used in environmental science, soil chemistry, agriculture, metallurgy and waste management.
- It involves mass transfer of a solute from a solid phase into a liquid phase.
- **Types of Leaching included :**
  - Natural Leaching (Soil Science & Environmental Context).
  - Chemical/Industrial Leaching (Hydrometallurgy)
  - Biological Leaching (Bioleaching)
- **Leaching occurs in several steps :**
  - Diffusion of solvent into the solid
  - Dissolution of soluble solute
  - Diffusion of dissolved solute out into bulk liquid

<u>References</u> :

- Brady, N.C., & Weil, R.R. (2008). *The Nature and Properties of Soils* (14th ed.). Pearson Education.
- Habashi, F. (1999). *Textbook of Hydrometallurgy*. Métallurgie Extractive Québec.
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2. **Give the role of pH in nutrients availability in plants?**

Soil pH, which measures the acidity or alkalinity of the soil , is one of the most critical factors influencing the availability of essential nutrients to plants. The pH level affects chemical solubility, microbial activity and root absorption — ultimately controlling the nutrition, health and productivity of plants.

- Soil pH directly influences the chemical form and solubility of nutrients. Each nutrient has an optimal pH range where it is most available to plant roots. Outside this range, nutrients may become either insoluble (unavailable) or excessively soluble (toxic).

- Soil pH also affects root membrane permeability and the ability of roots to uptake nutrients. In highly acidic soils, toxic elements like aluminum (Al^{3+}) and manganese (Mn^{2+}) can inhibit root elongation and damage root tips.

<u>References</u> :

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- Fageria, N. K., Baligar, V. C., & Jones, C. A. (2010). *Growth and Mineral Nutrition of Field Crops* (3rd ed.). CRC Press.
- Havlin, J. L., et al. (2014). *Soil Fertility and Fertilizers: An Introduction to Nutrient Management* (8th ed.). Pearson.

3. Give soil conservation methods to prevent soil erosion?

Soil erosion is the detachment and transport of soil particles by water, wind or tillage. It leads to the loss of topsoil, reduction in soil fertility, sedimentation of waterways and reduced agricultural productivity. To combat this, various soil conservation methods have been developed to prevent, reduce or control soil erosion. Some methods are as :

- Contour Farming
- Terracing
- Windbreaks and Shelterbelts
- Grassed Waterways
- Gully Plugging and Check Dams
- Strip Cropping
- Cover Cropping and Mulching
- Conservation Tillage / No-Till Farming
- Agroforestry
- Proper Grazing Management

<u>References</u> :

- Lal, R. (1990). *Soil Erosion in the Tropics: Principles and Management*. McGraw-Hill.
- Hudson, N. (1995). *Soil Conservation* (3rd ed.). B.T. Batsford Ltd.
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- USDA-NRCS (2000). *National Engineering Handbook Part 650*.
- FAO (1986). *Windbreaks and Shelterbelts*. FAO Conservation Guide 13.
- Nair, P.K.R. (1993). *An Introduction to Agroforestry*. Springer.
- Briske, D.D. (2011). *Conservation Benefits of Rangeland Practices*. USDA NRCS.

4. Differentiate between primary and secondary minerals?

Primary minerals	**Secondary mineral**
These are formed directly from the solidification (crystallization) of molten magma during igneous rock formation.	These are formed from the weathering, alteration or decomposition of primary minerals.
These contain silicates, oxides and native elements, often in complex crystalline forms.	These include clay minerals, oxides, carbonates and sulfates, often simpler compounds.

These are found in coarse soil fractions like sand and silt.	These are found in fine soil fractions especially clays and colloids.
These have typically well – formed and large crystals due to slow cooling of magma.	These are often microcrystalline or amorphous, formed under lower temperature conditions.
These are usually less soluble, weather slowly.	These are often more soluble and form more rapidly under surface conditions.
For example ; Quartz, feldspar, mica, hornblende and olivine.	For example ; Kaolinite, smectite, hematite, goethite and calcite.

Reference :

- Brady, N.C., & Weil, R.R. (2008). *The Nature and Properties of Soils*.
- Dixon, J.B., & Weed, S.B. (1989). *Minerals in Soil Environments*.
- Bohn, H.L., McNeal, B.L., & O'Connor, G.A. (2001). *Soil Chemistry*.
- Essington, M.E. (2003). *Soil and Water Chemistry*.
- Jenny, H. (1941). *Factors of Soil Formation*.

5. What are aflatoxin?

Aflatoxins are a group of toxic secondary metabolites produced by certain molds (fungi), particularly Aspergillus flavus and Aspergillus parasiticus. They commonly contaminate cereal grains, nuts, spices and animal feeds, especially in warm and humid conditions.

- Aflatoxins are highly toxic and carcinogenic compounds that pose significant risks to human and animal health, especially in tropical and subtropical regions.
- Managing aflatoxins requires a combination of agronomic practices, proper storage, rapid testing and regulatory enforcement.
- **Types of Aflatoxins :** There are over a dozen types, but the four major aflatoxins important in food safety are :
  1) Aflatoxin B1 – most toxic and carcinogenic
  2) Aflatoxin B2
  3) Aflatoxin G1
  4) Aflatoxin G2
- Metabolites in animals : Aflatoxin M1 and M2 (found in milk from animals that consumed contaminated feed)

Reference :

- Wild, C.P., & Gong, Y.Y. (2010). *Mycotoxins and human disease: a largely ignored global health issue*. Carcinogenesis, 31(1), 71–82. https://doi.org/10.1093/carcin/bgp264
- IARC. (2012). *Aflatoxins*. IARC Monographs on the Evaluation of Carcinogenic Risks to Humans, Volume 100F. International Agency for Research on Cancer.

6. Give some detrimental effects of detergents in aqueous environment?.

- **Eutrophication :** Many detergents contain phosphates (especially sodium tripolyphosphate), which act as nutrients for aquatic plants and algae. Excess phosphates can cause eutrophication, leading to algal blooms, depletion of dissolved oxygen and death of aquatic life.

- **Toxicity to Aquatic Life :** Surfactants like linear alkylbenzene sulfonates (LAS) or nonylphenol ethoxylates (NPEs) in detergents are toxic to aquatic organisms such as fish, crustaceans and plankton.
- **Foaming and Oxygen Transfer Inhibition :** Detergents cause persistent foam on water surfaces, especially near discharge outlets. This foam reduces light penetration and hinders gas exchange, lowering oxygen levels in water bodies.

<u>Reference</u> :

- Correll, D.L. (1998). *The Role of Phosphorus in the Eutrophication of Receiving Waters: A Review*. *Journal of Environmental Quality*, 27(2), 261–266.
- Ying, G.G. (2006). *Fate, behavior and effects of surfactants and their degradation products in the environment*. *Environment International*, 32(3), 417–431.
- Ramaswamy, B.R., et al. (2011). *Occurrence of pharmaceuticals and personal care products in Indian water bodies: a review*. *Environmental Monitoring and Assessment*, 184(8), 5611–5634.

7. What is the principle of IR spectrophotometry?

Infrared (IR) spectrophotometry is an analytical technique based on the interaction of infrared radiation with matter, primarily used to identify and study functional groups and molecular structures in organic and inorganic compounds.

Principle : IR spectrophotometry operates on the principle that :

Molecules absorb infrared light at specific frequencies that match the vibrational energies of their chemical bonds. When a molecule is irradiated with IR light (typically in the range of 4000–400 cm^{-1}), the bonds between atoms vibrate (stretch, bend, rock, wag etc.). If the energy of the IR radiation matches the natural vibrational frequency of the bond, absorption occurs, leading to a measurable IR spectrum.

<u>Reference</u> :

Pavia, D. L., Lampman, G. M., Kriz, G. S., & Vyvyan, J. R. (2014). *Introduction to Spectroscopy* (5th ed.). Cengage Learning.

8. Why environmental monitoring is important?

Environmental monitoring is the systematic process of collecting data on environmental parameters — such as air, water, soil and biological systems — to assess the quality, detect changes and inform decision – making for environmental protection and public health. Environmental monitoring is important to :

- Detecting Pollution and Contaminants
- Assessing Compliance with Environmental Laws
- Supporting Environmental Impact Assessments (EIAs)
- Protecting Human and Ecosystem Health
- Climate Change and Global Trends Monitoring

<u>Reference</u> :

- WHO. (2017). *Guidelines for Drinking-water Quality*. https://www.who.int/publications/i/item/9789241549950
- EPA (2021). *Monitoring, Reporting and Verification of Environmental Data*. U.S. Environmental Protection Agency. https://www.epa.gov
- Glasson, J., Therivel, R., & Chadwick, A. (2013). *Introduction to Environmental Impact Assessment*. Routledge.
- WHO. (2016). *Preventing disease through healthy environments*. https://www.who.int
- IPCC. (2021). *Sixth Assessment Report: Climate Change 2021 – The Physical Science Basis*. https://www.ipcc.ch

9. Why PCBs were banned?

Polychlorinated biphenyls (PCBs) are synthetic organic chemicals that were widely used from the 1930s to the 1970s in industrial applications such as electrical transformers, capacitors, hydraulic systems and heat transfer fluids. However, they were banned globally starting in the late 1970s due to their severe environmental and health hazards.

Key Reasons for the Ban

- **Environmental Persistence :** PCBs are highly stable and degrade very slowly in the environment.
- **Bioaccumulation and Biomagnification :** PCBs accumulate in the fatty tissues of animals and humans, and their concentration increases up the food chain.
- **Toxic Effects on Human Health :** PCBs have been associated with a variety of adverse health effects including :
  - Cancer (classified as probable human carcinogens by the IARC)
  - Immune system suppression
  - Reproductive and developmental toxicity
  - Neurotoxicity, especially in infants and children
- **Widespread Contamination :** Due to decades of use and improper disposal, PCBs were found in :
  - Rivers and lakes (e.g., Hudson River, Great Lakes)
  - Marine life and wildlife
  - Agricultural soils
  - Human blood, breast milk, and tissues

Due to these risks, PCBs were banned or severely restricted in many countries.

References :

- ATSDR (2000). *Toxicological Profile for Polychlorinated Biphenyls (PCBs)*. Agency for Toxic Substances and Disease Registry. https://www.atsdr.cdc.gov
- EPA (2023). *Learn about Polychlorinated Biphenyls (PCBs)*. U.S. Environmental Protection Agency. https://www.epa.gov/pcbs
- IARC (2016). *Polychlorinated Biphenyls and Polybrominated Biphenyls*. IARC Monographs, Volume 107. https://monographs.iarc.who.int/
- Erickson, M. D. (2001). *Introduction to PCBs*. In: *Analytical Chemistry of PCBs*. CRC Press.

10. Why micronutrients are important for plant growth?

Micronutrients are essential elements required by plants in small quantities but are crucial for their growth, development and overall health. They play a key role in enzymatic functions, metabolic processes and nutrient interactions within the plant system. Micro nutrients are important for :

- Enzyme Activation and Metabolic Pathways
- Photosynthesis and Chlorophyll Formation
- Nitrogen Fixation in Legumes
- Plant Disease Resistance and Stress Tolerance
- Enhancing Root Development and Nutrient Uptake

<u>Reference</u> :

- Marschner, H. (2012). *Mineral Nutrition of Higher Plants* (3rd ed.). Academic Press.
- Römheld, V., & Marschner, H. (1986). *Mineral Nutrition of Plants: Principles and Perspectives*. Springer.
- Stevenson, F. J., & Cole, M. A. (1999). *Cycles of Soil: Carbon, Nitrogen, Phosphorus, Sulfur, Micronutrients* (2nd ed.). Wiley-Interscience.
- Shankar, A. G., & Verma, H. N. (2004). *Micronutrient Deficiency in Plants and Their Role in Growth*. In *Plant Stress: Problems and Solutions*. CRC Press.
- Cakmak, I. (2008). *Enrichment of cereal grains with zinc: Agronomic or genetic biofortification?* In *Plant and Soil* (Vol. 302). Springer.

LONG QUESTIONS (2014)

1. What is the principle of AAS? Give its role in environmental monitoring?

Atomic Absorption Spectroscopy (AAS) is a highly sensitive analytical technique primarily used for determining the concentration of specific metal ions in various samples. The method is based on the principle that atoms absorb light at characteristic wavelengths when excited by energy. This technique is highly effective for detecting trace elements in liquid samples, making it invaluable for environmental monitoring particularly for analyzing water, soil, air and sediments.

Principle of Atomic Absorption Spectroscopy (AAS) :

AAS is based on the absorption of ultraviolet or visible light by free atoms in the gaseous phase. The key principle is that when a sample is introduced into a flame or graphite furnace, the metal ions present in the sample are atomized, meaning they are converted into individual atoms. These atoms then absorb light at specific wavelengths that correspond to their unique atomic transitions. The amount of light absorbed by the sample is directly proportional to the concentration of the metal atoms in the sample.

Basic Steps in Atomic Absorption Spectroscopy :

- **Sample Introduction** : The sample (in liquid form) is introduced into an atomizer by using a flame (air – acetylene or nitrous oxide – acetylene flame) or a graphite furnace.

- **Atomization** : In the atomizer, the sample is heated to a high temperature. In the flame atomizer, the sample is nebulized into fine droplets and then vaporized.
- **Light Source** : A hollow cathode lamp (HCL) specific to the metal being analyzed emits light at the characteristic wavelength of that metal. For instance, a copper hollow cathode lamp will correspond to the atomic transitions in copper atoms.
- **Absorption** : As the vaporized atoms in the flame or furnace pass through the light beam, they absorb a portion of the light. The absorption of light occurs because the energy from the light excites the atoms to a higher energy state.
- **Measurement of Absorbance** : The amount of light absorbed is measured by a spectrometer.
- **Quantification** : The absorbance is measured and compared to a calibration curve constructed using known concentrations of the metal in standard solutions. Using Beer's Law, the concentration of the metal in the sample can be determined.

$$A = \epsilon c l$$

Where :
  - A = Absorbance
  - ϵ = Molar absorption coefficient
  - c = Concentration of the analyte
  - l = Path length (distance the light travels through the sample)

Role of AAS in Environmental Monitoring :

AAS plays a crucial role in monitoring environmental contaminants especially heavy metals.

- **Monitoring Heavy Metal Contaminants in Water :** AAS is widely used for the quantitative analysis of metals in water samples. AAS can be used to measure both total metal concentration and to detect specific metal species, which is vital in environmental monitoring.
  - **Example** : Measuring lead levels in drinking water using AAS ensures that water meets safety standards set by regulatory bodies such as the World Health Organization (WHO) and the Environmental Protection Agency (EPA).
- **Soil Contamination and Fertility Assessment :** By using AAS to determine the concentration of metals in soil samples, environmental monitoring can help assess the contamination level and guide remediation efforts. It is also used to monitor nutrient levels such as zinc and copper which are important micronutrients for plants.
  - **Example** : AAS is used to measure the levels of cadmium and lead in agricultural soils to ensure that the crops grown in those soils are safe for human use.
- **Air Pollution Monitoring :** AAS is used to analyze particulate matter collected from the air to determine the presence of hazardous metals. Air quality monitoring stations utilize AAS for measuring metals in atmospheric samples and detecting contamination levels in urban and industrial areas
  - **Example** : Monitoring the mercury content in air in areas near coal – burning power plants or industrial zones can provide data on pollution levels and help set air quality standards.

- **Food Safety and Agriculture :** AAS is used to analyze food products for heavy metal contamination. It allows for the quantification of metals such as cadmium, arsenic and mercury in food items to ensure they meet food safety regulations.
    - **Example** : AAS is commonly used in food industry standards to check the arsenic content in rice, which can accumulate in the grains from contaminated soil or irrigation water.
- **Wastewater Treatment and Industrial Effluent Monitoring :** AAS is used to monitor the quality of industrial effluent by determining the concentrations of hazardous metals such as chromium, nickel and zinc.
    - **Example**: In industrial wastewater treatment plants, AAS is used to monitor the nickel concentration in wastewater to ensure compliance with local regulatory standards.

<u>Reference</u> :

- Zhang, X., & Zhang, Y. (2012). *Environmental Analysis of Heavy Metals in Water by Atomic Absorption Spectrometry*. Environmental Monitoring and Assessment. DOI:10.1007/s10661-012-2669-x
- Kim, D., & Lee, M. (2008). *Soil Contamination and the Role of AAS in Heavy Metal Monitoring*. Environmental Chemistry Letters. DOI:10.1007/s10311-008-0124-7
- Kampa, M., & Castanas, E. (2008). *Human Health Effects of Air Pollution*. Environmental Pollution, 151(2), 231–243. DOI:10.1016/j.envpol.2007.06.025
- Shah, F., & Wang, S. (2015). *Role of AAS in Food Safety Monitoring for Heavy Metals*. Journal of Environmental Science and Health, Part B, 50(8), 589–597. DOI:10.1080/03601234.2015.1029123
- Lee, M., & Lee, W. (2016). *Monitoring of Heavy Metals in Wastewater Using AAS*. Environmental Monitoring and Assessment, 188(6). DOI:10.1007/s10661-016-5265-7

2. Briefly discuss persistent organic pollutants?

Persistent Organic Pollutants (POPs) are a class of toxic chemicals that are resistant to environmental degradation through chemical, biological or photolytic processes. Due to their persistence in the environment, POPs can accumulate in the food chain, pose serious health risks to humans and wildlife. They can travel long distances from their point of origin, affecting ecosystems and populations far from where they were originally released.

- The **Stockholm Convention on Persistent Organic Pollutants**, established in 2001, is an international treaty aimed at reducing or eliminating the production and use of POPs.

Characteristics of Persistent Organic Pollutants :

POPs are defined by several key characteristics that make them particularly dangerous and widespread :

- POPs are chemically stable, this stability allows them to remain in the environment for years or even decades, entering various ecosystems.
- Since POPs are lipophilic (fat-soluble), they tend to accumulate in the fatty tissues of organisms.
- Biomagnification is particularly concerning for humans, as we are at the top of the food chain and often consume a variety of contaminated animal products.

- POPs are capable of traveling long distances from their point of origin through air, water and ocean currents.

Sources and Types of Persistent Organic Pollutants :

POPs can originate from various sources both natural and anthropogenic (human – made). The major sources of POPs include :

a) **Agricultural Activities :** Many POPs were historically used as pesticides, with DDT (dichlorodiphenyltrichloroethane) being one of the most infamous. Other agricultural chemicals, such as chlordecone and endosulfan, have also been identified as POPs.
   - **DDT** : Once widely used for malaria control, DDT is a notorious example of a pesticide that has caused significant environmental damage.
   - **Endosulfan** : An organochlorine pesticide linked to reproductive and developmental issues in wildlife and humans.

b) **Industrial Processes :** Polychlorinated biphenyls (PCBs) and dioxins are examples of POPs produced as by – products of industrial processes. PCBs were widely used as dielectric fluids in electrical equipment, while dioxins are produced during industrial activities such as waste incineration, paper bleaching and the production of certain chemicals.
   - **PCBs** : These were used in electrical transformers, capacitors and other industrial products due to their electrical insulating properties.
   - **Dioxins and Furans** : These are highly toxic by – products of industrial combustion processes.

c) **Combustion Processes :** The combustion of fossil fuels, forest fires and waste burning can release various toxic chemicals into the environment. Dioxins and furans are often produced during these processes and can settle in the environment.

d) **Unintentional By – products :** Some POPs, such as hexachlorobenzene (HCB) are by – products of various industrial processes, such as the production of pesticides, the chlorination of organic compounds and other chemical manufacturing processes.

Health Impacts of Persistent Organic Pollutants :

POPs are harmful to human health and the environment, and their effects are wide – ranging that affecting both individual organisms and entire ecosystems.

i. **Human Health Risks :** Exposure to POPs can lead to a variety of health problems, especially due to their ability to accumulate in human tissues over time. POPs have been linked to :
   - **Cancer** : Long - term exposure of POPs can increase the risk of various cancers, including liver, lung, breast and testicular cancer.
   - **Endocrine Disruption**: POPs can interfere with the endocrine system, leading to hormonal imbalances.
   - **Neurological and Developmental Effects**: Children are particularly vulnerable to the neurotoxic effects of POPs, which can affect brain development and behavior.

- **Immune System Suppression**: Chronic exposure to POPs can weaken the immune system, making the body more susceptible to infections and diseases.
- **Reproductive Toxicity**: Some POPs, such as PCBs and Dioxins, have been shown to cause birth defects, miscarriages and infertility in humans and animals.

ii. **Environmental Impacts :** POPs also have severe consequences for wildlife and ecosystems. These include :

- **Endangerment of Species** : Many species are affected by POPs due to bioaccumulation and biomagnification. For example birds of prey such as eagles and ospreys suffer from eggshell thinning due to DDT, which affects their ability to reproduce.
- **Loss of Biodiversity** : POPs disrupt the food web by poisoning aquatic and terrestrial organisms.
- **Aquatic Ecosystems** : POPs like mercury and PCBs accumulate in fish, causing toxicity in aquatic life and affecting the organisms that depend on them for food, including humans.

Global Efforts to Control Persistent Organic Pollutants :

The international community has recognized the significant threats posed by POPs, leading to several efforts to control and eliminate their use.

- **Stockholm Convention :** The **Stockholm Convention on Persistent Organic Pollutants**, signed in 2001, aims to eliminate or restrict the use of POPs globally. The convention has led to the phase – out of many POPs, including DDT, PCBs and certain pesticides.
- **Banning and Regulating POPs :** National governments have passed laws to ban or regulate the use of POPs. For instance, many countries have phased out the use of DDT, PCBs and endosulfan.
- **Monitoring and Remediation Efforts :** Environmental agencies globally have implemented monitoring systems to track POP levels in air, water and soil. Furthermore, efforts to remediate contaminated sites (such as soil decontamination and the safe disposal of hazardous chemicals) have been a major focus.

References :

- Stockholm Convention on Persistent Organic Pollutants (2001).
- United Nations Environment Programme (UNEP) : Report on POPs.
- World Health Organization (WHO) : Health impacts of POPs.
- International Agency for Research on Cancer (IARC) : Monographs on carcinogenicity of various POPs.

3. Give the mechanism behind nutrients assimilation by plants from soil? Give the role of pH and cation exchange capacity in this subject?

Nutrient assimilation by plants is a vital process that enables them to obtain the necessary elements for growth, reproduction, and overall metabolic functioning. These nutrients include macronutrients such as N, P, K, Ca, Mg and S) and micronutrients such as Fe, Mn, Zn, B, Cu, Mo and Cl.

- The process of nutrient uptake by plants is not merely passive; it involves complex biochemical, biophysical and physiological mechanisms that allow plants to absorb, transport and utilize nutrients from the soil.

Mechanism of Nutrient Uptake

a. **Root Absorption :**

The process of nutrient assimilation begins with the absorption of nutrients by the roots of plants. Roots are responsible for anchoring the plant in the soil and absorbing water and nutrients. Nutrient uptake occurs primarily through the root epidermis, particularly at the root hairs, which increase the surface area for absorption. The uptake of nutrients from the soil is influenced by several factors :

- **Active Transport :** Nutrients are absorbed against concentration gradients using energy from the plant. Active transport involves specific membrane proteins called pumps (e.g., H^+ - ATPase), which actively move ions into root cells. For example, nitrate (NO_3^-) and phosphate (PO_4^{3-}) are actively taken up into the plant cells using specific ion channels and transporters.
- **Passive Transport :** This mechanism relies on diffusion, where ions and molecules move from areas of high concentration to areas of low concentration across the cell membrane. Potassium (K^+), for instance enters plant roots by passive transport through potassium channels.
- **Facilitated Diffusion :** Some nutrients, like sugars and amino acids, are absorbed through carrier proteins that facilitate their transport into the plant cells. Facilitated diffusion occurs along the concentration gradient.
- **Ion Channels :** Nutrient ions are taken up through ion – specific channels. These channels are highly selective, allowing only certain ions to pass through. For example, calcium (Ca^{2+}) and magnesium (Mg^{2+}) are often taken up through specific channels present in the plasma membrane of root cells.

b. **Mycorrhizal Associations :**

In addition to root absorption, many plants form symbiotic relationships with mycorrhizal fungi. These fungi help plants absorb nutrients, particularly phosphorus, in exchange for carbohydrates from the plant. The hyphal networks of these fungi extend into the soil, increasing the plant's access to nutrients, especially in nutrient – poor soils. This association plays a critical role in nutrient acquisition, particularly in the uptake of immobile nutrients like phosphates.

Role of pH in Nutrient Availability and Assimilation :

Soil pH is one of the most critical factors that influence the availability of nutrients to plants. Soil pH affects the chemical form of nutrients, their solubility and the ability of plants to absorb them. The optimal pH for most plants ranges from 6 to 7, which is slightly acidic to neutral. However, different plants may have specific pH requirements.

a) **Effect of pH on Nutrient Availability :**

- **At low pH (acidic soils, pH < 6) :** In highly acidic soils, certain nutrients like phosphorus, calcium, magnesium and potassium become less available due to increased solubility of aluminum (Al^{3+}) and iron (Fe^{2+}), which can become toxic to plants. On the other hand, nutrients like iron (Fe), manganese (Mn) and zinc (Zn), which are less soluble in alkaline soils, become more soluble and available to plants.
- **At high pH (alkaline soils, pH > 7) :** In alkaline soils, nutrients like phosphate tend to form insoluble compounds, making them less available to plants. For example, calcium phosphate ($Ca_3(PO_4)_2$) can precipitate in alkaline conditions, reducing phosphorus availability.

b) **Soil pH and Micronutrient Availability :**

Micronutrients are highly pH – dependent, at lower pH values, these micronutrients are more soluble and readily available to plants. However, in neutral to alkaline soils, their solubility decreases, leading to deficiencies in plant growth i.e. Iron chlorosis.

- The pH of the soil solution influences the ionization of nutrients, thus determining whether a nutrient is in its absorbable form for plants. For instance, nitrate (NO_3^-) is more available in slightly acidic to neutral soils, while in highly acidic conditions, ammonium (NH_4^+) becomes the preferred form of nitrogen.

Cation Exchange Capacity (CEC) and Nutrient Assimilation :

Cation exchange capacity (CEC) is a measure of the soil's ability to hold and exchange cations (positively charged ions) such as K^+, Ca^{2+}, Mg^{2+} and Na^+. CEC is a crucial property for nutrient availability because it determines the soil's ability to retain essential nutrients and supply them to plants.

a) **Role of CEC in Nutrient Availability :**

- **Cation Retention** : Soils with high CEC can hold more cations, making them available for plant uptake over time.
- **Nutrient Exchange** : The cation exchange process involves the exchange of a nutrient ion with a hydrogen ion (H^+) from the root. Roots release protons to displace nutrient cations from the soil particles, allowing the nutrients to enter the plant. For example, a root may release H^+ to displace calcium (Ca^{2+}) from a clay particle, allowing the root to take up calcium.
- **High CEC and Fertility** : Soils with higher CEC tend to be more fertile because they can hold and supply larger amounts of essential cations. Soils rich in organic matter and clay particles generally have high CEC, making them ideal for nutrient retention.

b) **Low CEC Soils :**

In contrast, soils with low CEC, such as sandy soils, have limited capacity to retain cations. These soils may lose nutrients more rapidly due to leaching, making it harder for plants to access essential nutrients.

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- Ryan, M. H., & Graham, P. H. (2002). *Soil pH and Nutrient Availability*. Plant Physiology.
- Fageria, N. K., & Baligar, V. C. (2008). *Nutrient Management for Sustainable Food Production and Environmental Protection*. CRC Press.
- Sposito, G. (2008). *The Chemistry of Soils* (2nd ed.). Oxford University Press.

4. How soaps and detergents contribute water pollution?

Water pollution is a significant global environmental issue, with numerous sources contributing to the contamination of water bodies. Among the many contributors, soaps and detergents play a critical role in polluting water bodies, impacting both aquatic ecosystems and human health. These products, widely used in domestic and industrial settings, contain chemical substances that can degrade water quality.

What Are Soaps?

Soaps are sodium salts of fatty acids, which are typically derived from natural sources like vegetable oils or animal fats. The molecular structure of soaps consists of a hydrophobic hydrocarbon tail that repels water and a hydrophilic head that is attracted to water.

What Are Detergents?

Detergents are synthetic compounds, and their molecular structure typically includes a hydrophobic tail and a hydrophilic head. Unlike soaps, detergents are formulated to be effective in hard water conditions, where soap molecules tend to form insoluble salts. Detergents can be either anionic, cationic, nonionic or amphoteric, depending on the nature of the hydrophilic head.

Mechanisms of Pollution : How Soaps and Detergents Pollute Water

- **Surfactant Properties :** Both soaps and detergents are surfactants and surfactants are designed to lower the surface tension between water and oils or grease, which allows them to emulsify and remove oily substances. When released into water bodies, these surfactants can interfere with the natural processes of aquatic ecosystems.
- **Toxicity to Aquatic Life :** One of the most concerning aspects of soaps and detergents is their toxicity to aquatic organisms, including fish, invertebrates and plants. For example, linear alkylbenzene sulfonate (LAS), a common surfactant in detergents, has been shown to have toxic effects on fish and other aquatic organisms, affecting their respiratory and reproductive systems.
- **Impact on Oxygen Levels in Water :** Soaps and detergents also contribute to water pollution through their biodegradability. When these compounds enter water, they undergo microbial degradation. The decomposition of these chemicals by bacteria leads to a significant decrease in the dissolved oxygen levels in water, a phenomenon known as biochemical oxygen demand (BOD). Low oxygen levels can lead to hypoxic conditions, which are detrimental to aquatic life.
- **Formation of Foam and Surface Scum :** Soaps and detergents also contribute to foam formation in water bodies, especially in rivers and streams. The foam can reduce the oxygen exchange at the water's surface, further contributing to hypoxic conditions. Additionally, when foam accumulates at the surface of water bodies, it can form surface scum, which affects the aesthetics and quality of the water.

Environmental Impact of Soap and Detergent Pollution

- **Impact on Water Quality :** The contamination of water bodies with soaps and detergents directly impacts the quality of water, making it unsuitable for both aquatic life and human consumption. Eutrophication also increases the turbidity of the water, making it murky and reducing its clarity.
- **Effects on Drinking Water :** In addition to harming aquatic ecosystems, soap and detergent pollution can affect the potability of drinking water. In areas where water treatment plants are insufficient or ineffective, high concentrations of detergents and surfactants can remain in drinking water supplies, leading to health risks for humans.
- **Impact on Soil Health :** Soap and detergent pollutants are not restricted to water bodies alone; they can also leach into **soil** through water infiltration. Additionally, the presence of surfactants in soil can affect the water – holding capacity and drainage properties of the soil, further impairing its function.

Health Impacts of Soap and Detergent Pollution :

- **Skin Irritation :** Direct contact with water bodies contaminated by soaps and detergents can cause **skin irritation** in humans. Surfactants can strip the skin of its natural oils, leading to dryness, rashes and allergic reactions.
- **Bioaccumulation and Human Health Risks :** Some detergent chemicals, particularly those that contain phosphates and nonylphenol ethoxylates (NPEs), can bioaccumulate in the food chain. The accumulation of these substances in the food chain can have significant consequences for public health, especially in areas where fish and other aquatic organisms are a major source of food.

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5. Briefly describe soil erosion and its types?

Soil erosion is the removal of the topsoil layer by natural forces like wind, water and ice, or through human intervention. The topsoil is crucial for plant growth because it contains a high concentration of organic matter, nutrients and microorganisms that sustain plant life. Erosion reduces the fertility of the soil, making it harder to grow crops and leading to desertification in severe cases. While erosion is a natural process, human activities such as deforestation, overgrazing, agriculture and urbanization have exacerbated the problem. These activities disturb the soil structure, making it more vulnerable to the forces of nature.

Causes of Soil Erosion :

Soil erosion occurs due to various natural and human – induced factors. Here are the primary causes :

i. **Water Erosion :** Water erosion is the most common form of soil erosion and occurs when rainwater or runoff flows over the land surface, removing soil particles in its path. This process is further intensified by steep slopes, heavy rainfall and poor vegetation cover.

ii. **Wind Erosion :** Wind erosion occurs in dry, arid regions where the soil is loose and dry. Wind can carry away fine soil particles, especially in areas with sparse vegetation. This type of erosion is common in desert areas and semi – arid regions.

iii. **Human Activity :** Human activities, such as deforestation, agriculture, construction and overgrazing, disturb the natural vegetation and expose the soil to erosion. Poor land management practices, such as over – cultivating the land, removing vegetation for firewood and improper irrigation techniques, also contribute to soil erosion.

Types of Soil Erosion :

Soil erosion can be classified into several types based on the cause and mechanism behind the erosion process. These types are as follows:

I. **Sheet Erosion :** Sheet erosion is the gradual removal of a thin layer of soil from the surface, typically caused by raindrops and surface runoff. It is a slow but continuous process that affects large areas of land. The loss of soil due to sheet erosion may not be immediately noticeable, but it leads to a significant reduction in soil fertility over time. It typically occurs on gentle slopes where the water flows uniformly over the land surface.

II. **Rill Erosion :** Rill erosion occurs when surface runoff concentrates in small channels called rills. These small, shallow channels form when water begins to flow in a concentrated manner and erodes the soil more rapidly than it would in sheet erosion. It is commonly seen in agricultural fields where plowing and tillage have disturbed the soil.

III. **Gully Erosion :** Gully erosion is the most severe form of water erosion and occurs when the flow of water concentrates into large channels or gullies, that can be several feet deep and wide. Gully erosion occurs on steep slopes where there is little or no vegetation to anchor the soil. It is often the result of poor land management practices such as overgrazing, deforestation or over – extraction of water resources.

IV. **Wind Erosion :** Wind erosion occurs in arid and semi – arid regions where vegetation cover is sparse and soil moisture content is low. Wind erosion is most common in desert – like environments and in areas where land is left exposed and not covered by crops or grass. It can lead to the formation of dust storms. Wind erosion often results in the degradation of agricultural land, making it difficult to grow crops and sustain vegetation.

V. **Coastal Erosion :** Coastal erosion refers to the erosion of land along coastlines due to the combined action of waves, tidal currents, and wind. Coastal erosion is a natural process but human activities like construction sand mining and deforestation can accelerate it. Coastal erosion leads to the loss of beaches, cliffs and coastal habitats, causing significant environmental damage. Coastal erosion can also result in saltwater intrusion into freshwater aquifers, damaging ecosystems and agricultural land.

VI. **Glacier Erosion :** Glacier erosion occurs in regions where glaciers are present. Glaciers, which are massive masses of ice, move slowly over the land, eroding the underlying bedrock and soil as they advance. This process, called **abrasion**, occurs when the glacier carries rocks and sediment across the surface, scraping and wearing away the underlying material. Glacier erosion is a slow process but can create significant geological features like U – shaped valleys, fjords and moraine deposits.

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6. Give the role of gas chromatography in air pollution analysis?

Air pollution is a growing global concern that poses significant threats to human health, the environment, and the economy. Airborne pollutants come from various sources, including industrial activities, transportation, agriculture and natural processes. Monitoring and quantifying the concentration of pollutants in the air is crucial for assessing the impacts of pollution and implementing effective strategies to mitigate its harmful effects.

- Gas Chromatography (GC) is one of the most widely used analytical techniques for the analysis of air pollutants. It is a powerful tool that allows for the separation, identification and quantification of individual chemical components in a complex mixture of air pollutants.
- This technique is particularly valuable for analyzing volatile organic compounds (VOCs), inorganic gases, particulate matter and other chemical species in the air.

Basic principle of Gas Chromatography (GC) :

Gas chromatography is an analytical technique that separates and analyzes volatile compounds by utilizing a gaseous mobile phase. In GC, a sample is injected into a chromatographic column, where it interacts with the stationary phase (typically a liquid or solid coating inside the column).

- The components of the sample are separated based on their differing affinities for the stationary phase and their volatilities. The time it takes for each component to pass through the column is measured and recorded as a chromatogram, which allows for the identification and quantification of individual compounds.

Types of Air Pollutants Analyzed Using Gas Chromatography :

GC plays a crucial role in the analysis of various air pollutants, including :

I. **Volatile Organic Compounds (VOCs) :** VOCs are a large group of carbon – based compounds that easily evaporate into the atmosphere at room temperature. These compounds are primarily released from industrial emissions, vehicle exhaust, solvents and consumer products. VOCs can contribute to the formation of ground – level ozone and smog, which are harmful to human health and the environment. Common VOCs analyzed by GC include :

   - Benzene
   - Toluene
   - Xylene
   - Formaldehyde
   - Acetone

- GC, often coupled with mass spectrometry (GC-MS), provides excellent separation and quantification capabilities for VOCs. The high sensitivity of GC allows for the detection of trace levels of these pollutants in air samples, making it a crucial tool for air quality monitoring.

II. **Inorganic Gases :** Inorganic gases, such as nitrogen oxides, sulfur dioxide, carbon monoxide and ozone are common air pollutants that contribute to the formation of acid rain, smog and particulate matter. GC is widely used for the analysis of these gases, particularly when a TCD or ECD detector is used.

   - **Carbon Monoxide (CO) :** This toxic gas is produced by the incomplete combustion of fossil fuels and is associated with vehicle emissions and industrial processes. GC can be used to measure CO levels in air samples.
   - **Nitrogen Oxides (NOx) :** NOx compounds, including nitrogen dioxide (NO_2) and nitric oxide (NO), are produced by high – temperature combustion processes, such as those in power plants and vehicles. These gases are precursors to ozone formation and contribute to air pollution.
   - **Sulfur Dioxide (SO_2) :** SO_2 is primarily released from the combustion of coal and oil, as well as industrial processes like metal smelting. It contributes to acid rain formation and respiratory problems in humans.

III. **Particulate Matter (PM) :** Particulate matter (PM), including PM2.5 and PM10, refers to tiny solid particles or liquid droplets suspended in the air. These particles can contain various harmful substances, such as metals, organic compounds and toxic chemicals. Gas chromatography can be used to analyze organic compounds adsorbed onto particulate matter.

Applications of Gas Chromatography in Air Pollution Monitoring :

Gas chromatography has become an essential tool in air pollution analysis and monitoring due to its high sensitivity, versatility and ability to analyze complex mixtures. Some applications are explain as :

- **Industrial and Environmental Monitoring :** Gas chromatography is commonly used by environmental agencies, industrial plants and research institutions to monitor air quality and ensure compliance with environmental regulations. Regular monitoring of air pollutants like VOCs, CO, SO_2 and NO_x is critical to assessing the impact of industrial emissions on air quality.

- **Epidemiological Studies :** Gas chromatography is also used in epidemiological studies to assess the health impacts of air pollution. Researchers can analyze air samples from different regions and correlate the concentration of pollutants with the incidence of respiratory diseases, cancer and other health problems.
- **Ozone Depletion and Climate Change Studies :** Gas chromatography is used to measure halogenated compounds, such as chlorofluorocarbons (CFCs) and hydrofluorocarbons (HFCs), which are potent greenhouse gases and ozone – depleting substances. Monitoring these compounds is essential in studying the causes and effects of ozone depletion and global warming.
- **Indoor Air Quality Monitoring :** Indoor air pollution is a growing concern, particularly in urban environments where people spend a significant portion of their time indoors. GC is employed to analyze VOCs and other pollutants in indoor air, such as formaldehyde from building materials, solvents from paints and emissions from household cleaning products.

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2015 Short Questions

1. Define eutrophication?

Eutrophication is a process where water bodies, such as lakes, rivers or coastal areas, receive an excessive amount of nutrients, particularly nitrogen and phosphorus, which leads to an increase in the growth of algae and other aquatic plants. The consequences of eutrophica – tion include oxygen depletion, loss of biodiversity and the formation of harmful algal blooms (HABs).

Causes of Eutrophication : The main contributors to eutrophication include :

- Agricultural Runoff
- Wastewater Discharge
- Urban Runoff
- Detergents and Phosphates

The Process of Eutrophication : The process of eutrophication can be broken down into several key stages :

- Nutrient Enrichment
- Algal Blooms
- Oxygen Depletion
- Fish Kills and Loss of Biodiversity
- Formation of Dead Zones

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2. What do you mean by cation exchange capacity of soil?

Cation Exchange Capacity is defined as the total number of exchangeable cations that a soil can hold per unit mass or volume, typically measured in centimoles of charge per kilogram (cmol/kg). Cations such as Ca^{2+}, Mg^{2+}, K^+, Na^+, NH_4^+ and H^+ are held by soil particles (mainly clay and organic matter) through electrostatic forces. These cations can be exchanged with other cations in the soil solution, thereby providing nutrients to plants.

- In soils with a high CEC, there is a greater capacity to hold and release nutrients needed by plants, while soils with low CEC may be more prone to nutrient leaching, especially in regions with heavy rainfall or irrigation.

How CEC Works in Soil : The process of cation exchange in soil works through the interaction between soil particles (which are negatively charged) and the cations in the soil solution. The negatively charged soil particles attract positively charged ions, holding them in place. When plant roots release hydrogen ions (H^+) or other cations into the soil solution, they displace the exchangeable cations, which are then absorbed by the plant roots for nutrition.

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3. Give the role of humus in soil conservation?

Humus plays a critical role in soil conservation, both by protecting the soil from degradation and by enhancing its fertility, structure and resilience. Its contributions to the health of soil ecosystems and sustainable agriculture make humus an essential component of effective soil conservation strategies. Some roles are as :

- Improvement of soil structure
- Water retention and infiltration
- Enhancing nutrient retention and supply
- Stimulation of microbial activity
- Reduction of soil erosion
- Carbon sequestration and climate resilience
- Supports sustainable agriculture

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4. Differentiate between primary and secondary minerals?

See question no. 02 part # 04 of 2014

5. What are aflatoxin?

See question no. 02 part # 05 of 2014

6. Give the detrimental effects of pesticides in aqueous environment?

Detrimental effects of pesticides in aqueous environments are as :

i. **Toxicity to Aquatic Life :** Pesticides can be acutely or chronically toxic to a range of aquatic organisms, including fish, amphibians, crustaceans and phytoplankton.

ii. **Bioaccumulation and Biomagnification :** Certain pesticides, such as organochlorines (e.g., DDT, aldrin), are persistent and lipophilic, allowing them to accumulate in the fatty tissues of organisms and magnify along the food chain.

iii. **Disruption of Aquatic Food Webs :** Pesticides often target non – pest organisms in aquatic systems : Algae and plankton, the primary producers, are highly sensitive to herbicides (e.g., atrazine), leading to reduced photosynthesis and oxygen production. This disrupts the food supply for higher trophic levels, collapsing local food webs.

Other detrimental effects of pesticides are listed as :

- Alteration of Water Quality
- Groundwater Contamination
- Endocrine Disruption
- Alter hormone synthesis, transport and receptor binding.
- Affect sexual development in fish (e.g., feminization in male fish due to DDT, PCBs).

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7. Give the principle Of AAS?

The fundamental principle of AAS is based on Beer – Lambert Law, which states :

> **"The amount of light absorbed by a sample is directly proportional to the concentration of the absorbing species in the path of the light."**

In AAS, a sample containing metal ions is atomized (converted into free atoms) in a flame or graphite furnace. A light beam, emitted from a hollow cathode lamp (HCL) specific to the metal of interest, passes through the atomized sample. These atoms absorb light at specific wavelengths unique to each element. The decrease in light intensity is measured and used to determine the metal concentration.

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8. Why environmental monitoring is important?

See question no. 02 part # 08 of 2014

9. Give various classification of soil erosion caused by water?

Water – induced erosion can be classified into several types based on the intensity, scale and mechanism of erosion :

- Splash Erosion
- Sheet Erosion
- Rill Erosion
- Gully Erosion
- Streambank Erosion
- Tunnel or Pipe Erosion

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10. Why micronutrients are important for plant growth?

See question no. 02 part # 10 of 2014

LONG QUESTIONS (2015)

1. What is the principle of UV/VIS ? Give its role in environmental monitoring?

Ultraviolet – Visible (UV/Vis) Spectroscopy is a quantitative and qualitative analytical technique used to measure the absorption of ultraviolet (190 – 400 nm) and visible light (400 –

800 nm) by chemical substances. It is based on electronic transitions of molecules and atoms when they absorb photons of specific energies.

Basic Principle : Electronic Transitions :

The core principle of UV/Vis spectroscopy is :

When a molecule absorbs UV or visible light, electrons in a lower energy molecular orbital are excited to a higher energy molecular orbital.

This process is called an electronic transition. The energy absorbed during this transition corresponds to the energy difference between the two orbitals and matches the energy of the incoming photon, according to Planck's Equation :

$$E = h\nu = hc\lambda$$

Where :

- E = Energy of the photon
- h = Planck's constant
- ν = Frequency of light
- c = Speed of light
- λ = Wavelength of light

Only certain wavelengths are absorbed, depending on the molecule's electronic structure, resulting in a characteristic absorption spectrum.

Types of Electronic Transitions :

In organic and inorganic molecules, UV/Vis absorption involves transitions between valence electrons :

- **$\sigma \rightarrow \sigma^*$** : Very high energy ; found in saturated compounds.
- **$n \rightarrow \sigma^*$** : In molecules with lone pairs (e.g., alcohols, amines).
- **$\pi \rightarrow \pi^*$** : Found in compounds with double/triple bonds (e.g., alkenes, aromatic rings).
- **$n \rightarrow \pi^*$** : Common in carbonyls and other compounds with lone pairs and double bonds.

Note : **$\pi \rightarrow \pi^*$** and **$n \rightarrow \pi^*$** transitions typically fall in the UV-Visible range.

Chromophores and Auxochromes :

- Chromophores are parts of molecules responsible for light absorption (e.g., C=C, C=O, N=N).
- Auxochromes are groups (e.g., $-OH$, $-NH_2$) that do not absorb light themselves but modify the absorption characteristics of chromophores.

Beer – Lambert Law : The quantitative aspect is governed by the **Beer – Lambert Law** :

$$A = \varepsilon c l$$

Where :

- A = Absorbance (unitless)
- ε = Molar absorptivity ($L \cdot mol^{-1} \cdot cm^{-1}$)
- c = Concentration ($mol \cdot L^{-1}$)
- l = Path length (cm)

This law states that absorbance is directly proportional to the concentration and path length, enabling quantification of analytes.

Applications in Environmental Monitoring :

i. Water Quality Monitoring :

- **Nitrate and Nitrite Detection** : Nitrates and nitrites absorb strongly in the UV range (around 200 – 220 nm). Their concentrations in water are directly measured using UV spectrophotometers. This is essential in monitoring agricultural runoff, wastewater discharge and drinking water quality.
- **Organic Matter (COD, TOC, BOD)** :
  UV/Vis can estimate the organic load in water through parameters like Chemical Oxygen Demand (COD) or Total Organic Carbon (TOC). Absorbance at 254 nm (UV254) is often used as a surrogate parameter.
- **Heavy Metals (Indirectly via Complexes)** :
  Although metals themselves are not UV-active, they form colored complexes with chromophoric ligands (e.g., EDTA or dithizone), allowing indirect determination.

ii. Air Pollution Monitoring

- **Monitoring of Ozone (O_3)** : Ozone strongly absorbs in the UV region (around 254 nm). UV photometry is used to measure ambient ozone concentration in the troposphere and stratosphere.
- **Detection of Aromatic Hydrocarbons and NOx** : Benzene, toluene, and nitrous oxides absorb in the UV and visible regions, making UV/Vis a useful tool in real – time air monitoring for VOCs and photochemical smog components.

iii. Soil and Sediment Monitoring :

- **Pesticides and PAHs :** Many pesticides and polycyclic aromatic hydrocarbons (PAHs) are UV-active. UV/Vis is used in pre – screening or confirming the presence of such compounds in soil extracts.

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2. **Briefly describe persistent organic pollutants?**

See question no. 02 part # 02 of 2014

3. **Discuss various sources of soil pollution?**

Soil pollution refers to the presence of toxic chemicals (pollutants or contaminants) in soil in high enough concentrations to pose a risk to human health and the ecosystem. These pollutants are introduced through various human activities and natural events.
Sources of soil pollution are explain as :

1) **Industrial Activities :**
   i. **Chemical Manufacturing Plants :** Industries that manufacture or use chemicals (e.g., fertilizers, pesticides, dyes, solvents and plastics) often discharge hazardous waste into the soil either directly or through poorly managed landfills and storage tanks.
   ii. **Mining Operations :** Mining for minerals and ores often leads to the dispersion of toxic tailings and overburden containing heavy metals, which leach into surrounding soils.
2) **Agricultural Activities :**
   i. **Excessive Use of Chemical Fertilizers :** Inorganic fertilizers are applied to enhance crop yield but overuse leads to nitrate and phosphate accumulation in the soil, causing salinization and nutrient imbalance.
   ii. **Pesticides and Herbicides :** Many pesticides like DDT, aldrin and chlorpyrifos are persistent organic pollutants (POPs) that bind tightly to soil particles and resist degradation.
   iii. **Manure and Sludge Application** : While organic manure enriches the soil, untreated or improperly composted manure may contain antibiotics, hormones and pathogens that pollute soil.
3) **Urban Waste and Municipal Sources :**
   i. **Solid Waste Disposal (Landfilling) :** Unregulated dumping of domestic and commercial waste leads to leachate formation containing organic and inorganic toxins.
   ii. **Sewage Sludge and Wastewater Irrigation :** Sludge contains heavy metals, organic pollutants and pathogens. When applied untreated to soil, it causes persistent contamination.
4) **Petroleum Hydrocarbons :**
   i. **Oil Spills and Leaks :** Leakages from underground storage tanks (USTs), pipelines and accidental spills release petroleum hydrocarbons into the soil.
5) **E-Waste and Electronic Industry :**

Improper dismantling and burning of electronic components release toxic metals and flame retardants.

6) **Atmospheric Deposition :**

Pollutants released into the air from industrial and vehicular emissions settle onto soil through dry or wet deposition.

7) **Military and Nuclear Activities :**

Use of explosives, fuels, and radioactive materials pollutes the soil.

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4. How fertilizers contribute to water pollution?

How fertilizers contribute to water pollution explain as :

Surface Runoff of Nutrients :

When fertilizers are applied to soil, especially in excess or before heavy rains, the nutrients – particularly nitrates (NO_3^-) and phosphates (PO_4^{3-}) — are washed away from fields into nearby streams, rivers and lakes. This is known as surface runoff. These water bodies are then enriched with nutrients, which were never intended to reach them.

- **Example :** In the Mississippi River Basin (USA), runoff from agricultural lands has carried vast amounts of nitrogen and phosphorus into the Gulf of Mexico, leading to the formation of one of the world's largest hypoxic zones—over 15,000 km².

Leaching into Groundwater :

Nitrogen-based fertilizers, especially those containing nitrates, are highly water – soluble. When applied excessively or during rainy seasons, they move downward through the soil with percolating water and enter groundwater aquifers. This process is called leaching. Nitrate – contaminated groundwater is difficult and expensive to treat.

- **Example :** In many regions of India and Bangladesh, nitrate levels in groundwater exceed WHO's safe drinking water limit of 50 mg/L, posing a severe threat to rural populations that rely on groundwater for drinking.

Eutrophication of Water Bodies :

When excessive nutrients enter lakes and ponds, they trigger a process called eutrophication (enrichment of water that leads to rapid algal growth). These algae block sunlight and outcompete aquatic plants while their decay consumes oxygen affecting aquatic fauna.

- **Example :** Lake Erie experienced severe eutrophication in 2011, mainly from phosphorus-rich fertilizer runoff. This caused massive algal blooms, turning the lake green and making the water undrinkable for thousands.

Hypoxia and Dead Zones :

Following eutrophication, dead algae settle and decompose at the bottom of water bodies. Microbes use up large amounts of dissolved oxygen during decomposition, creating hypoxic (low oxygen) conditions. Areas with very low oxygen become **"dead zones"** where aquatic life cannot survive.

- **Example :** The Baltic Sea often suffers from hypoxic zones due to fertilizer runoff from nearby agricultural areas. These zones have led to mass die – offs of fish and invertebrates.

Contamination of Drinking Water :

Groundwater polluted with nitrates from fertilizers can become unsafe for human consumption. High levels of nitrate in drinking water interfere with oxygen transport in the blood, especially in infants, causing a condition called blue baby syndrome or methemoglobinemia.

- **Example :** In Iowa, USA, rural areas heavily reliant on well water often experience nitrate concentrations above safe limits due to intensive corn farming and fertilizer use.

Formation of Harmful Algal Blooms (HABs) :

When nutrients from fertilizers reach lakes and coastal waters, they often stimulate the growth of harmful algae, such as cyanobacteria. These blooms not only consume oxygen but also release toxins that harm fish, wildlife, pets and humans.

- **Example :** In Florida's Lake Okeechobee, HABs triggered by fertilizer runoff led to the spread of toxic algae into rivers and coastal areas, resulting in public health alerts and ecological damage.

Acidification of Aquatic Ecosystems :

Ammonium – based fertilizers release hydrogen ions (H^+) during microbial nitrification in soil. When these ions enter aquatic ecosystems, they lower the pH of the water, leading to acidification. Acidic water reduces the availability of calcium for shell-forming organisms and harms fish reproduction.

- **Example :** In parts of China, acid – sensitive streams have shown decreased pH and reduced biodiversity due to extensive ammonium fertilizer use in adjacent farmlands.

Biodiversity Loss in Aquatic Habitats :

Eutrophication and hypoxia alter the ecological balance of water bodies. Native aquatic plants die due to shading and oxygen – starved conditions cause the death or migration of fish, crustaceans and sensitive macro invertebrates, leading to a significant loss of biodiversity.

- **Example :** In the Chesapeake Bay, eutrophication from fertilizer runoff has led to declines in important fish species like striped bass and blue crabs, disrupting the entire aquatic food chain.

Bioaccumulation of Toxins :

Harmful algal blooms (HABs) can produce toxins like microcystins that accumulate in the tissues of aquatic organisms. When these organisms are consumed by predators or humans, the toxins travel up the food chain, causing long – term health effects.

- **Example :** Microcystin toxins have been found in shellfish and fish from the Great Lakes, making them unsafe for consumption and leading to temporary bans on fishing.

Economic Losses in Fisheries and Tourism :

Fertilizer-driven water pollution causes fish kills, contaminates seafood and makes water bodies unpleasant or unsafe for recreation. This negatively impacts fisheries, tourism and local economies dependent on clean water.

- **Example :** In China's Lake Taihu, eutrophication from fertilizer use led to massive fish kills, forcing authorities to shut down water supplies to millions and damaging the tourism and fisheries industries.

Reference :

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5. Briefly describe soil erosion and its types?

See question no. 03 part # 05 of 2014

6. Discuss a method to monitor air pollution?

See question no. 03 part # 06 of 2014

2016 Short Questions :

1. **Define green chemistry?**

> **"The utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of chemical products is called Green chemistry."**

- It aims to make chemistry inherently safer, cleaner and more sustainable by integrating environmental considerations from the start of chemical development.

Key objectives of green chemistry :

- ✓ Pollution Prevention at the Source
- ✓ Minimizing Toxicity and Hazard
- ✓ Use of Renewable Resources
- ✓ Energy Efficiency
- ✓ Design for Degradation

Practical Example

- **Ibuprofen Synthesis :** Traditional synthesis of ibuprofen involved six steps and generated a lot of waste. The green synthesis developed by BHC Company uses fewer steps, higher atom economy and produces less waste.

Reference :

- Anastas, P. T., & Warner, J. C. (1998). Green Chemistry: Theory and Practice. Oxford University Press.
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2. **What do you mean by cation exchange capacity of soil?**

See question no. 02 part # 02 of 2015

3. **Give various classification of soil erosion caused by water?**

See question no. 02 part # 09 of 2015

4. **Differentiate between primary and secondary minerals?**

See question no. 02 part # 04 of 2014

5. **What are aflatoxin?**

See question no. 02 part # 05 of 2014

6. **Give some detrimental effects of detergents in aqueous environment?**

   See question no. 02 part # 06 of 2014

7. **What is the principle of IR spectroscopy?**

   See question no. 02 part # 07 of 2014

8. **Why environmental monitoring is important?**

   See question no. 02 part # 08 of 2014

9. **Why PCBS were banned?**

   See question no. 02 part # 09 of 2014

10. **Why micronutrients are important for plant growth?**

    See question no. 02 part # 10 of 2014

LONG QUESTIONS (2016)

1. Write a brief note on micronutrients in soil?

Micronutrients, though required in small amounts, play huge roles in plant and soil health. Their behavior is influenced by complex chemical and biological factors and their management is crucial for sustainable agriculture, food security and human health. Proper understanding and balanced application can help optimize crop yields and restore nutrient – depleted soils.

- Micronutrients in soil are essential elements required by plants in very small quantities (typically less than 100 ppm in plant tissue), but their role in plant growth, physiology, and metabolism is crucial.
- Despite their low concentration, deficiency or excess of micronutrients can significantly affect plant health, crop yield, and soil fertility.

There are eight essential micronutrients for higher plants :

1. **Iron (Fe) :** Iron plays a pivotal role in chlorophyll synthesis, though it is not part of the chlorophyll molecule. It is a key component of cytochromes and various enzymes involved in redox reactions, especially in respiration and photosynthesis.
   - **Deficiency Symptoms :** Interveinal chlorosis in young leaves due to immobility in plants.
   - **Availability Factors :** pH is crucial ; Fe becomes less available in alkaline soils (pH > 7.5) and may form insoluble hydroxides.
2. **Manganese (Mn) :** Manganese is involved in enzyme activation, particularly those involved in photosynthesis (e.g., water – splitting in photosystem II), nitrogen metabolism and synthesis of some secondary metabolites.
   - **Deficiency Symptoms :** Interveinal chlorosis, necrotic spots and stunted growth.
   - **Availability Factors :** Low pH increases availability; high pH or excessive Fe, Zn or Cu can induce deficiency.
3. **Zinc (Zn) :** Zinc is essential for the synthesis of tryptophan (a precursor of auxin) and is involved in enzyme activation, protein synthesis and growth hormone regulation.

- **Deficiency Symptoms :** Shortened internodes (rosetting), little leaf in fruit trees and chlorosis.
- **Availability Factors :** High phosphorus levels and alkaline soils can reduce Zn availability.

4. **Copper (Cu) :** Copper plays roles in lignin synthesis, electron transport in photosynthesis (plastocyanin) and several enzymatic reactions.
    - **Deficiency Symptoms :** Leaf curling, wilting and dieback of shoots.
    - **Availability Factors :** Organic soils with high pH and sandy texture may reduce Cu availability.
5. **Boron (B) :** Boron is vital for cell wall formation, membrane integrity, pollen germination and sugar transport.
    - **Deficiency Symptoms :** Brittle or thickened leaves, poor flowering and fruit deformities.
    - **Availability Factors :** Most available at pH 5–7; leaches easily in sandy soils and is less available in dry conditions.
6. **Molybdenum (Mo) :** Molybdenum is required in very small quantities and is a key component of enzymes involved in nitrogen fixation and nitrate reduction.
    - **Deficiency Symptoms :** Chlorosis and necrosis on older leaves, often resembling nitrogen deficiency.
    - **Availability Factors :** Becomes more available in neutral to alkaline soils ; deficient in acid soils.
7. **Chlorine (Cl) :** Chlorine is involved in osmosis and ionic balance as well as in the photosynthetic reaction in PSII.
    - **Deficiency Symptoms :** Rare but include wilting, chlorosis and bronzing of leaves.
    - **Availability Factors :** Usually sufficient in soils due to atmospheric deposition and fertilizers like KCl.
      – Reference:
8. **Nickel (Ni) :** Nickel is essential for the functioning of urease, an enzyme required for nitrogen metabolism in urea – fed plants.
    - **Deficiency Symptoms :** Leaf tip necrosis, especially in legumes, due to urea toxicity.
    - **Availability Factors :** Required in minute quantities and generally available in most soils unless pH is highly alkaline.

<u>Reference</u> :

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- Gupta, U.C. (1993). “Boron and its Role in Crop Production”. CRC Press.
- Mengel, K., & Kirkby, E.A. (2001). “Principles of Plant Nutrition”, 5th Edition. Springer.
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- Brown, P.H., Welch, R.M., & Cary, E.E. (1987). "Nickel: A Micronutrient Essential for Higher Plants". Plant Physiology.

2. Briefly discuss persistent organic pollutants?

See question no. 02 part # 02 of 2014

3. Discuss various sources of soil pollution?

See question no. 03 part # 03 of 2015

4. How is AAS is employed for environmental monitoring?

Atomic Absorption Spectroscopy (AAS) is one of the most widely used techniques in environmental monitoring due to its high sensitivity, accuracy and specificity for detecting trace metals.

Atomic Absorption Spectroscopy (AAS) in Environmental Monitoring

Principle of AAS in Environmental Monitoring :

Atomic Absorption Spectroscopy is an analytical technique based on the absorption of light by free metallic ions in the gaseous state. A sample is atomized (converted into free atoms), usually by a flame or graphite furnace, and then exposed to light of a specific wavelength emitted by a hollow cathode lamp (specific to the element). The amount of light absorbed is proportional to the concentration of the element in the sample.

i. **Detection of Heavy Metals in Water Samples :** One of the most significant applications of AAS in environmental monitoring is the detection of heavy metals such as lead, zinc, cadmium, arsenic, mercury , chromium and copper in drinking water, wastewater, lakes and groundwater.
   - **Example :** AAS is used to detect Pb and Cd in drinking water samples, where even concentrations as low as 1 ppb (μg/L) can be identified using Graphite Furnace AAS (GFAAS).

ii. **Soil Contamination Analysis :** Soils near industrial zones or agricultural land treated with pesticides/fertilizers are often contaminated with trace metals. AAS is used to extract and analyze elements such as Zn, Cu, Ni and Pb from soil digests.
   - **Procedure :** Soil is digested using strong acids (HNO_3 and HCl or aqua regia), filtered and the clear solution is analyzed by AAS for metal concentration.

iii. **Airborne Particulate and Dust Analysis :** Heavy metals can also be present in airborne particulates, especially near metallurgical, mining or urban areas. Air samples collected using high – volume samplers or filters are acid – digested and analyzed by AAS.
   - **Example :** Lead and arsenic in suspended particulate matter (SPM) are determined in industrial urban zones using AAS.

iv. **Monitoring of Effluents and Industrial Discharges :** Industries such as electroplating, battery manufacturing, textile dyeing and leather tanning often discharge heavy metals. AAS is a standard method used by regulatory bodies to monitor these discharges to comply with environmental protection laws.
   - **Application :** Effluent samples from electroplating units are analyzed for Cr, Ni, Cu and Zn.

v. **Biomonitoring Using Plants and Animals :** Bioaccumulation studies in indicator organisms (e.g., fish, earthworms, algae) can help monitor environmental pollution. AAS is used to quantify metal content in tissue samples.
   - **Example :** Fish tissue from polluted rivers is digested and analyzed using AAS to determine Hg and Pb accumulation.

vi. **Monitoring Agricultural Inputs and Food Safety :** AAS is used to assess contamination in fertilizers, pesticides, and food products due to environmental exposure. Crops irrigated with contaminated water or grown in polluted soils may accumulate metals.
   - **Example :** Vegetables grown in urban areas can be analyzed for Pb and Cd using Flame AAS.

vii. **Sediment and Sludge Analysis :** River, lake and estuarine sediments often act as sinks for heavy metals. AAS is used to analyze digested sediment samples to monitor long – term pollution trends.

Reference :

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- APHA Standard Methods for the Examination of Water and Wastewater (22nd Edition), Method 3111B and 3113B for AAS.
- Bryan, G.W. (1979). "Bioaccumulation of Marine Pollutants". Philosophical Transactions of the Royal Society B.
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- Springer. USEPA Method 6010D (also applicable to AAS after suitable dilution).
- Welz, B., & Sperling, M. (1999). "Atomic Absorption Spectrometry". Wiley-VCH.

5. Briefly discuss soil erosion and its types?

See question no. 03 part # 05 of 2014

6. Discuss method to monitor air pollution?

Monitoring air pollution is a critical component of air quality management. Various methods, ranging from traditional gravimetric sampling to advanced technologies like remote sensing and laser – based techniques, are employed depending on the pollutants being monitored and the scale of monitoring required. As pollution levels rise globally, especially in urban and industrial zones, these methods provide invaluable data to inform policy, mitigate environmental impacts and protect public health.

A. Gravimetric Method for Particulate Matter (PM) Monitoring

The gravimetric method is one of the most common and traditional techniques used to monitor particulate matter (PM), such as PM10 (particles with a diameter of 10 micrometers or less) and PM2.5 (particles with a diameter of 2.5 micrometers or less).

- **Process :** Air is drawn through a filter paper, which collects particulate matter. The filter paper is then weighed before and after sampling to determine the mass of the particles collected.
- **Applications :** Used for regulatory purposes by environmental agencies like the U.S. EPA and EU to monitor the levels of PM in urban areas, industrial zones and surrounding environments.

B. Passive Sampling for Gaseous Pollutants

Passive sampling is a low – cost and simple method for monitoring gases like nitrogen dioxide, ozone, sulfur dioxide and carbon monoxide in the atmosphere.

- **Process :** The passive sampler absorbs the target gas over a period (ranging from days to weeks). After exposure, the absorbent is analyzed in the laboratory to determine the concentration of the pollutant.
- **Applications :** Widely used for monitoring outdoor air quality in areas where continuous monitoring stations are not available.

C. Continuous Emission Monitoring Systems (CEMS)

CEMS are automated systems used to continuously measure the concentration of gaseous pollutants emitted by industrial stacks, chimneys or exhaust systems.

- **Process :** Gases are drawn into the analyzer, which uses techniques such as infrared absorption, chemiluminescence or non – dispersive UV absorption to determine the concentration of pollutants like NO_x, SO_2, CO and VOCs.
- **Applications :** Used by industrial facilities, power plants and municipal incinerators to comply with environmental regulations and ensure emissions do not exceed permissible limits.

D. Gas Chromatography (GC) for VOC Monitoring

Gas Chromatography (GC) is an advanced technique for analyzing volatile organic compounds (VOCs) in the air, including benzene, toluene, xylene and other industrial solvents.

- **Process :** Air samples are captured using adsorption tubes or solid – phase microextraction (SPME) fibers. The sample is then analyzed using a GC system equipped with a flame ionization detector (FID) or mass spectrometer (MS) for precise identification and quantification of VOCs.
- **Applications :** Used in air quality monitoring programs to assess the impact of vehicular emissions, industrial pollution and solvent use in urban areas.

E. Chemiluminescence Method for Nitrogen Oxides (NOx) Monitoring :

Chemiluminescence is widely used for monitoring nitrogen oxides (NO and NO_2), which are major pollutants from automobile emissions, industrial processes and agricultural activities.

- **Process :** The sample gas reacts with ozone (O_3) in the presence of UV light, emitting light that is detected by a photomultiplier tube. The light intensity is proportional to the NOx concentration.

- **Applications :** Used in environmental monitoring stations, particularly in urban areas, to track NOx emissions and their contribution to acid rain and ozone formation.

F. Remote Sensing for Large – Scale Air Pollution Monitoring :

Remote sensing is a non-invasive method used to monitor large – scale air pollution using satellites and ground-based systems.

- **Process :** Satellites such as NASA's Terra and Aqua satellites equipped with MODIS (Moderate Resolution Imaging Spectroradiometer) or Ozone Monitoring Instrument (OMI) measure pollutants like ozone, aerosols, NO_2 and particulate matter across extensive areas.
- **Applications :** Used to track wildfires, dust storms and urban air pollution on a global scale, especially in remote areas where ground – based monitoring is unavailable.

G. Laser Induced Breakdown Spectroscopy (LIBS) :

LIBS is an emerging laser – based technology used for monitoring trace metals in air, particularly in industrial zones or areas with high emission levels.

- **Process :** A high – energy laser pulse is focused on the sample, creating a plasma that emits light. The emitted light is analyzed to determine the elemental composition of the air sample.
- **Applications :** Ideal for real – time monitoring of particulate metal pollution, such as nickel, chromium and arsenic.

H. Diffusion Tubes for NO_2 and O_3 Monitoring :

Diffusion tubes are a simple passive sampling method widely used for measuring NO_2 and O_3 in ambient air over extended periods.

- **Process :** Air diffuses through a tube containing an absorbent, where pollutants like NO_2 or O_3 are chemically trapped. After exposure, the absorbent is analyzed to determine the concentration of the pollutant.
- **Applications :** Commonly used in urban areas to monitor traffic-related pollution.

I. Optical Remote Sensing for Particulate Matter (PM) :

Lidar (Light Detection and Ranging) and SODAR (Sonic Detection and Ranging) are advanced optical methods used to detect and monitor particulate matter (PM) and other aerosols in the atmosphere.

- **Process :** Lidar uses laser pulses to detect the size, concentration and vertical distribution of aerosols and particulate matter in the atmosphere.
- **Applications :** Effective for monitoring urban air pollution, especially in smog or fog – prone areas.

Reference :

- Chow, J.C., & Watson, J.G. (1998). *Guidelines for PM10 Sampling and Chemical Analysis*. Journal of the Air & Waste Management Association, 48(2), 139–148.
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2017 Short questions :

1. **Differentiate between primary and secondary minerals?**

   See question no. 02 part # 04 of 2014

2. **Define cation exchange capacity of soil?**

   See question no. 02 part # 02 of 2015

3. **What is the principle of IR spectroscopy?**

   See question no. 02 part # 07 of 2014

4. **What are macro and micronutrients?**

Macronutrients and Micronutrients are two categories of essential nutrients required by organisms for normal growth, development, and functioning.

Macronutrients : Macronutrients are nutrients that are required by the body in large amounts. They are the building blocks of growth, energy production, and metabolism. For example :

- Carbohydrates
- Proteins
- Fats (Lipids)
- Water
- Macrominerals (Calcium, Potassium and Magnesium etc.)

Micronutrients : Micronutrients are nutrients needed in small amounts but are essential for proper body function, disease prevention, and development. For example :

- Vitamins (A, C, D, B – complex and K)
- Trace Minerals (Iron, Zinc, Iodine, Selenium, Copper, Manganese, Fluoride, Chromium, and Molybdenum)

Reference :

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- World Health Organization (WHO). (2001). *Trace Elements in Human Nutrition and Health*.

5. **Give the role of pH in nutrient availability to plants?**

   See question no. 02 part # 02 of 2014

6. **Give various classification of soil erosion caused by water?**

   See question no. 02 part # 09 of 2015

7. **What are POPS ?Give details of any two POPS?**

Persistent Organic Pollutants (POPs) are toxic chemical substances that persist in the environment for long periods due to their stability, bioaccumulation, and long – range transport. These pollutants can remain in the environment for years, decades or even longer, affecting ecosystems, wildlife and human health.

- The Stockholm Convention on Persistent Organic Pollutants, adopted in 2001, aims to eliminate or restrict the production and use of POPs worldwide.

Two Examples of POPs :

1. DDT (Dichlorodiphenyltrichloroethane) :

DDT is an organochlorine pesticide. It was widely used as an agricultural pesticide and for malaria vector control in the mid – 20th century. It became famous for its effectiveness against insects like mosquitoes. DDT is highly persistent in the environment. DDT is linked to a variety of harmful health effects, including cancer, reproductive harm and immune system disruption. It can cause thinning of eggshells in birds, especially raptors like bald eagles, leading to population declines.

2. Polychlorinated Biphenyls (PCBs) :

PCBs are a group of synthetic chemicals composed of carbon, hydrogen, and chlorine atoms. PCBs are extremely stable and resistant to environmental degradation, with some forms persisting in soil and water for decades. PCBs are highly toxic and have been associated with a range of health problems, including liver damage, developmental issues, neurotoxicity and an increased risk of cancer. They are also endocrine disruptors, affecting hormone systems.

<u>Reference</u> :

- Carson, R. (1962). *Silent Spring*. Houghton Mifflin Harcourt.
- Safe, S.H. (1994). *Polychlorinated Biphenyls (PCBs): Environmental Impact, Chemical Toxicology, and Human Exposure*. Critical Reviews in Toxicology, 24(2), 87–129.

8. **Give the role of humus in soil conservation?**

   See question no. 02 part # 03 of 2015

9. **Give impacts of chemicals on buildings and monuments?**

Chemical exposure can have significant detrimental effects on buildings, monuments and other cultural heritage structures. These effects result from both natural and anthropogenic sources of chemicals, including air pollution, industrial emissions and chemical reactions with building materials. Over time, these chemical interactions can cause degradation, discoloration and structural damage ultimately leading to the loss of historical and architectural value.

1. Acid Rain : Acid rain accelerates the weathering of limestone, marble and sandstone. The acid reacts with the calcium carbonate in the stone, forming calcium sulfate and carbon dioxide, which leads to surface pitting, roughening and the eventual breakdown of the material.

2. Sulfur Dioxide (SO_2) : SO_2 reacts with oxygen and moisture in the air to form sulfurous acid, which then interacts with building materials such as marble and limestone. This reaction leads to the deterioration of these materials through corrosion and surface loss. SO_2 reacts with moisture in the air to form sulfuric acid, which can further cause the formation of black crusts on stone monuments.

3. Nitrogen Oxides (NO_x) : Direct exposure to nitrogen oxides can cause surface cracking, pitting and discoloration of building materials, particularly those made of limestone, marble and brick.

<u>Reference</u>:

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10. What is green chemistry?

See question no. 02 part # 01 of 2016

LONG QUESTIONS (2017)

1. Explain reclamation of soil?

Soil reclamation refers to the process of restoring soil that has been degraded or damaged by various environmental factors, such as salinity, acidity, erosion or contamination, back to a condition where it can be used effectively for agricultural or other purposes.

- It involves the application of different physical, chemical, biological and mechanical methods to improve the soil's fertility, structure and overall health.
- Reclamation is critical for ensuring the sustainability of agricultural productivity, particularly in regions where soil degradation has reached critical levels.
- The goal of soil reclamation is to counteract soil degradation processes such as salinization, acidification, waterlogging and erosion. This, in turn, enhances soil health, promotes plant growth and ensures the long – term fertility of the land.

Key Methods of Soil Reclamation

1. Saline and Sodic Soil Reclamation

- **Saline Soil** : Soil that contains excess soluble salts (mainly sodium chloride) in sufficient quantities to adversely affect plant growth.
- **Sodic Soil** : Soil that contains excess sodium ions which cause dispersion of clay particles and negatively affect soil structure.

Reclamation Methods :

- **Leaching** : The process of applying large amounts of water to wash out the soluble salts from the root zone. This is effective in reclaiming saline soils.

- **Soil Amendments** : The addition of soil amendments like gypsum (calcium sulfate) is commonly used to reclaim sodic soils. Gypsum helps to replace the sodium ions with calcium, thus improving soil structure and permeability.
- **Drainage Systems** : Installing subsurface drainage systems helps to prevent waterlogging and facilitate the removal of excess salts and sodium.
- **Irrigation Management** : Proper irrigation practices, including the use of good – quality water for irrigation, help prevent further salinization.

Example : In many parts of India and Pakistan, reclaimed saline and sodic soils are now used for crops like rice and wheat after proper treatment with gypsum and leaching techniques.

2. Acidic Soil Reclamation : Acidic soils are characterized by a low pH (below 5.5), often caused by excessive rainfall, leaching, or the use of acid – forming fertilizers. These soils limit plant growth by affecting nutrient availability and causing toxicity of aluminum and manganese ions.

Reclamation Methods :

- **Liming** : The most common method for reclaiming acidic soils is the application of lime (calcium carbonate). Lime neutralizes the excess acidity, raising the soil pH to a level that is more suitable for plant growth.
- **Organic Amendments** : Organic materials, such as compost or animal manure, can also be added to improve soil structure and increase microbial activity, which helps to buffer soil pH.
- **Use of Alkaline Fertilizers** : Fertilizers containing calcium or magnesium compounds may also help in neutralizing soil acidity over time.

Example : The use of lime in the reclamation of acidic soils in regions like Southeast Asia has successfully restored the soil's productivity for crops like maize and soybean.

3. Erosion Control and Reclamation : Soil erosion, caused by wind, water or human activity, is a major form of soil degradation. Erosion removes the fertile topsoil, leading to reduced agricultural productivity.

Reclamation Methods :

- **Terracing** : Constructing terraces on slopes reduces water runoff and prevents soil erosion, allowing for better water infiltration and soil conservation.
- **Afforestation and Vegetative Cover** : Planting trees, grasses and other vegetation helps stabilize the soil by preventing wind and water erosion through root systems that bind the soil particles together.
- **Check Dams and Water Diversion** : In areas prone to water erosion, the construction of check dams and diversion channels helps control the flow of water and reduces soil loss.

Example : In the Loess Plateau in China, terracing and afforestation programs have been used extensively to combat soil erosion and reclaim degraded land.

4. Waterlogged Soil Reclamation : Waterlogging occurs when the soil remains saturated with water for extended periods, resulting in poor aeration, reduced root growth and the development of toxic conditions such as the accumulation of iron and manganese.

Reclamation Methods :

- **Drainage Systems** : Installing proper drainage systems (open or subsurface) is the primary method for reclaiming waterlogged soils. These systems help to remove excess water and improve aeration.
- **Raised Beds** : In areas where drainage is difficult, raised beds or mounds can be constructed to provide better root aeration and prevent water accumulation.
- **Water Management** : Controlled irrigation and better management of water tables through regular monitoring can prevent waterlogging.

Example : The reclamation of the Nile Delta's waterlogged soils has been accomplished through the construction of drainage systems and improved water management practices, increasing the region's agricultural productivity.

5. Contaminated Soil Reclamation : Soil contamination can occur due to the accumulation of hazardous substances like heavy metals (e.g., lead, mercury, cadmium), pesticides, industrial waste and oil spills. These contaminants harm soil fertility and pose risks to human health and the environment.

Reclamation Methods :

- **Bioremediation** : The use of microorganisms to degrade organic pollutants in the soil is a sustainable method of reclaiming contaminated soils. Plants (phytoremediation) are also used to absorb heavy metals and other toxins from the soil.
- **Soil Washing** : A technique that involves the use of water or chemical solutions to wash out pollutants, especially heavy metals, from the soil.
- **Stabilization and Solidification** : The application of chemicals to bind or immobilize contaminants and prevent them from leaching into the groundwater.

Example : In industrial zones contaminated with lead and cadmium, phytoremediation techniques have been used to grow hyperaccumulator plants like sunflowers, which absorb the metals from the soil.

6. Soil Fertility Restoration :

Soil fertility is critical for agricultural production, and its degradation can lead to reduced crop yields. Loss of fertility is often caused by excessive farming, erosion, nutrient leaching and the overuse of chemical fertilizers.

Reclamation Methods :

- **Organic Farming Practices** : The use of compost, manure, and crop rotations helps restore nutrients to the soil. These organic amendments improve soil structure and enhance microbial activity.
- **Green Manure** : Planting nitrogen – fixing crops like legumes and then plowing them under the soil helps restore nitrogen content and improve overall soil health.
- **Application of Balanced Fertilizers** : Proper application of macro – and micronutrients ensures that the soil is replenished with the nutrients necessary for plant growth.

Example : In Africa, the use of green manure and composting has been an effective way of reclaiming soils that have been depleted of essential nutrients.

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2. Write a note on impacts of household chemicals on human health?

Household chemicals are commonly used for cleaning, pest control, personal care and various other domestic tasks. While these chemicals can provide convenience and efficiency in daily life, their improper use or prolonged exposure can have adverse effects on human health. Household chemicals range from cleaning agents and disinfectants to pesticides, air fresheners and personal care products.

- The potential risks associated with these chemicals largely depend on the types of substances, the duration of exposure and the specific vulnerability of individuals, such as children, pregnant women or the elderly.

1. Respiratory Problems and Allergies : Many household chemicals release volatile organic compounds (VOCs) that can irritate the respiratory system. Prolonged exposure to VOCs and other airborne chemicals, such as those found in paints, air fresheners and cleaning products, can cause a range of respiratory problems.

- **Asthma and Allergies** : Studies have shown that exposure to indoor air pollutants from household chemicals can exacerbate asthma symptoms and trigger allergic reactions. For example, cleaning products that contain bleach or ammonia can cause coughing, wheezing, and shortness of breath in sensitive individuals.

2. Skin Irritation and Chemical Burns : Several household chemicals, such as cleaning agents, detergents and bleach, contain harsh substances that can irritate or burn the skin. Prolonged contact with these chemicals can lead to redness, rashes and severe burns in some cases.

- **Skin Sensitization** : Chemicals like bleach, ammonia, and certain acids (commonly found in toilet cleaners) can cause allergic skin reactions, leading to rashes, eczema and other dermatological conditions.

3. Endocrine Disruption : Many household chemicals contain endocrine – disrupting compounds (EDCs) that interfere with hormone systems. These chemicals can mimic, block or alter the normal function of hormones, leading to long – term health effects.

- **Phthalates and Bisphenol A (BPA)** : These chemicals are commonly found in plastic containers, personal care products and air fresheners. Exposure to EDCs has been linked to developmental and reproductive issues, including reduced fertility, early puberty and abnormal fetal development.

4. Poisoning and Accidental Ingestion : Household chemicals, particularly cleaning products, pesticides and solvents, pose a risk of poisoning, especially in children who may inadvertently ingest or inhale them.

- **Pesticides and Insecticides** : These chemicals are toxic and can cause symptoms ranging from nausea and vomiting to more severe effects like organ failure, seizures and even death if ingested or inhaled in large amounts.

5. Neurological Effects : Exposure to certain household chemicals can affect the nervous system, leading to neurological problems such as headaches, dizziness and in extreme cases, cognitive impairment.

- **Solvents** : Solvents used in paints, adhesives and cleaning products, such as toluene and formaldehyde, can cause central nervous system (CNS) depression, leading to symptoms like headaches, dizziness and drowsiness.

6. Cancer Risks : Certain household chemicals are classified as carcinogenic, meaning they can cause cancer after long – term exposure. These chemicals are often found in cleaning products, paints, pesticides and air fresheners.

- **Formaldehyde and Benzene** : Formaldehyde, a common preservative in household products and benzene, found in some cleaning agents and air fresheners, are both classified as carcinogens. Prolonged exposure to these chemicals has been linked to cancers such as leukemia and lung cancer.

7. Reproductive Health Effects : Some household chemicals, particularly those that are endocrine disruptors, can have adverse effects on reproductive health. These chemicals may alter fertility, cause birth defects or affect sexual development.

- **Lead and Mercury** : Lead, found in some old paints, and mercury, present in some thermometers and electrical devices, can negatively affect reproductive health by causing hormonal imbalances and developmental issues in fetuses.

8. Hormonal Imbalances in Children : Children are particularly vulnerable to the effects of household chemicals due to their developing bodies and higher rates of exposure. Chemicals like phthalates, BPA and pesticides can disrupt hormone production and development in children, leading to long – term health issues.

- **Early Puberty** : Exposure to household chemicals such as phthalates and BPA has been linked to early puberty in children, especially girls. This condition can increase the risk of breast cancer and other hormonal diseases later in life.

9. Environmental and Cumulative Effects : The accumulation of household chemicals in the environment can also have indirect effects on human health. Many chemicals from cleaning products, pesticides and other household items enter water systems, soil and air, where they can accumulate and affect the broader ecosystem.

- **Chemical Bioaccumulation** : Chemicals like pesticides can enter the food chain, affecting not only human health but also the health of animals and plants. This can result in chronic health conditions for those consuming contaminated water or food.

10. Effects on Mental Health : Some household chemicals, especially those that release fumes or affect hormonal balance, can also have an impact on mental health. These chemicals can cause anxiety, depression and other mood disorders.

- **Volatile Organic Compounds (VOCs)** : Exposure to VOCs, found in many household products like paints and varnishes, has been linked to symptoms of depression and anxiety in both children and adults.

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3. Write a note on HPLC(Principle and applications)?

HPLC works on the principle of partitioning of analytes between a stationary phase and a mobile phase. The stationary phase is usually a column packed with a solid adsorbent (e.g., silica), and the mobile phase is a liquid solvent or mixture of solvents that flows through the column under high pressure.

When a sample mixture is injected into the system, its components interact differently with the stationary and mobile phases due to variations in polarity, charge, size or chemical structure. Those with stronger interactions with the stationary phase will elute slower, whereas those with weaker interactions will elute faster. The retention time (time taken by a compound to elute from the column) is used for identification, while the peak area or height is used for quantification.

HPLC systems typically include :

- A **pump** to maintain high pressure,
- An **injector** for sample introduction,
- A **column** for separation,

- A **detector** (e.g., UV – Vis) to identify compounds,
- A **data system** for result interpretation.

Applications of HPLC :

HPLC is an indispensable tool across many industries due to its precision, versatility, and reliability in separating complex mixtures. Its applications in pharmaceutical analysis, food safety, environmental monitoring, and clinical diagnostics highlight its crucial role in maintaining public health, safety and quality control.

- By continually advancing the sensitivity and resolution of HPLC methods, the technology is expected to play an even larger role in the future of research and industry. Some applications are explain as :

1. Pharmaceutical Industry : Drug Analysis

Quantification of Active Pharmaceutical Ingredients (APIs) : HPLC is extensively used in pharmaceutical analysis to determine the concentration of APIs in drug formulations. For example, the determination of the content of acetaminophen in a tablet or the identification of impurities in a medication.

2. Environmental Monitoring : Water Quality Testing

Monitoring Pesticide Residues in Water : HPLC is used for environmental monitoring, specifically to detect contaminants like pesticides and herbicides in water samples. For instance, it can identify and quantify pesticides like glyphosate in drinking water.

3. Clinical Diagnostics : Blood and Urine Analysis

Measuring Hormones in Blood Samples : In clinical diagnostics, HPLC is used to measure hormones like cortisol, testosterone or insulin in blood and urine. It helps diagnose disorders related to hormone imbalances.

4. Food and Beverage Industry : Quality Control

Determining Sugar Content in Beverages : HPLC is applied in the food industry to measure the concentration of sugars, additives and preservatives in food and beverages. For example, it can quantify sucrose, glucose and fructose in fruit juices.

5. Forensic Science : Drug Testing in Biological Samples

Detection of Illicit Drugs in Blood and Urine
In forensic toxicology, HPLC is employed to detect and quantify illicit drugs, such as cocaine, methamphetamine and opiates, in biological samples like blood, urine and hair.

6. Cosmetics Industry : Quality Control and Safety Testing

In the cosmetics industry, HPLC is used to test for the presence of preservatives and other additives such as parabens and phthalates, which can affect skin health.

7. Biotechnology : Protein and Peptide Analysis

Separation and Quantification of Proteins : HPLC is essential in biotechnology for separating and quantifying proteins, peptides and other biomolecules. It is widely used for analyzing recombinant proteins used in drug development.

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- Snyder, L. R., Kirkland, J. J., & Dolan, J. W. (2010). *Introduction to Modern Liquid Chromatography* (3rd ed.). Wiley. ISBN: 978-0-470-16754-0.

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4. How fertilizers contribute the water pollution?

See question no. 03 part # 04 of 2015

5. Explain mining as a source of soil pollution?

Mining is a major industrial activity that significantly contributes to soil pollution through the release of heavy metals, toxic chemicals, and physical disturbances of the land. It alters the natural composition, structure and fertility of soils, rendering them unfit for agriculture, forestry and habitation in many cases. The pollution caused by mining is long – lasting and often irreversible without costly remediation.

How Mining Pollutes Soil :

1. Heavy Metal Contamination : Mining activities release heavy metals like lead (Pb), cadmium (Cd), mercury (Hg), arsenic (As) and chromium (Cr) into the surrounding soil. These metals are either naturally present in ores or added during ore processing. When mining waste (tailings, overburden or slurry) is dumped onto the soil or leaches into nearby land, it contaminates the soil matrix.

- **Example** : In gold mining, arsenopyrite (FeAsS) releases arsenic into the soil upon oxidation.
- **Impact** : These heavy metals are toxic, non – biodegradable and can enter the food chain via crops, causing health issues in humans and animals.

2. Acid Mine Drainage (AMD) : Acid Mine Drainage occurs when sulfide minerals (e.g., pyrite) are exposed to air and water during mining, producing sulfuric acid. This acid leaches heavy metals from surrounding rocks into the soil, dramatically lowering soil pH and enhancing metal solubility.

- **Example** : Coal mining in Appalachia (USA) has led to AMD, causing acidification and metal contamination in nearby soils and water bodies.
- **Impact** : Acidic soils disrupt microbial activity, reduce nutrient availability and harm plant life.

3. Deposition of Mining Tailings : Mining tailings are finely ground rock wastes left after ore extraction. These tailings often contain toxic metals and chemicals such as cyanide or flotation reagents. Improper disposal or tailing dam failures result in direct soil contamination.

- **Example** : The 2015 Samarco dam disaster in Brazil released 60 million cubic meters of tailings, polluting vast areas of land.
- **Impact** : Soil fertility declines and tailings may release dust particles that further spread contaminants.

4. Use of Chemicals in Ore Processing : Many mines use **chemical reagents** like cyanide (gold mining), sulfuric acid (copper) or mercury (amalgamation in gold mining) that may spill or leach into soil during processing or waste disposal.

- **Example** : Cyanide leaching in gold mining contaminates soil if containment systems fail.
- **Impact** : Toxicity to soil microbes and plants, potential for groundwater contamination.

5. Physical Soil Degradation : Mining operations, especially open – pit mining, physically disturb the soil structure by removal of topsoil, compaction and erosion.

- **Example** : Bauxite mining in India leads to loss of topsoil, making land infertile.
- **Impact** : Loss of organic matter, reduced infiltration and inability to support vegetation.

6. Loss of Soil Microbial Activity : Toxic substances and physical alterations from mining severely reduce microbial biomass and diversity, which are essential for nutrient cycling and soil health.

- **Example** : Soils near lead – zinc mines often show low microbial respiration and enzyme activity.
- **Impact** : Disruption of soil food web and nutrient transformation processes.

7. Dust Deposition : Mining and blasting operations release dust containing metal particulates that settle on surrounding soil. Windborne dust spreads pollution far beyond the mining site.

- **Example** : Lead dust from smelting operations in Kabwe, Zambia, has caused widespread soil contamination.
- **Impact** : Reduces photosynthesis in plants and increases metal load in the topsoil.

8. Radioactive Soil Contamination : Uranium and rare earth element mining can lead to radioactive contamination of the soil, posing long – term environmental and health risks.

- **Example** : Uranium mines in Niger have led to elevated levels of radon and thorium in soil.
- **Impact** : Persistent contamination affecting generations and bioaccumulation in food chains.

9. Leaching into Groundwater and Soil : Toxic leachates from mines infiltrate both the soil profile and groundwater, transporting contaminants across ecosystems.

- **Example** : Leachate from nickel mines in Canada has shown contamination of soil layers down to 1 meter.
- **Impact** : Contaminated root zones affect crops and pose risks to drinking water sources.

- **<u>Reference</u> :**

Alloway, B. J. (2013). *Heavy Metals in Soils: Trace Metals and Metalloids in Soils and*

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2018 (short questions)

1. **Define eutrophication?**

   See question no. 02 part # 01 of 2015

2. **What is the importance of soil?**

Soil is one of Earth's most vital natural resources, performing a multitude of essential ecological, agricultural, and environmental functions. Soil play following functions :

- **Supports Biodiversity :** Soil is a habitat for a vast array of organisms, including bacteria, fungi, earthworms, insects and burrowing animals.
- **Regulation of Water Cycle :** Soil acts like a sponge, absorbing and storing rainfall, then gradually releasing it into rivers, streams and groundwater.
- **Supports Agriculture and Food Security :** Soil fertility is critical to sustainable agriculture. Without fertile soil, food production would drastically decline, threatening human survival.
- **Raw Materials for Construction :** Soil provides essential materials like clay, sand, and gravel used in building infrastructure such as homes, roads, and dams.

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- Brady, N. C., & Weil, R. R. (2016). *The Nature and Properties of Soils* (15th ed.). Pearson.
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3. **How is industry a source of soil pollution?**

Industry is a **major source of soil pollution** due to the release of toxic chemicals, heavy metals and non – biodegradable waste into the environment. This contamination can severely degrade soil quality, harm plant and animal life and pose health risks to humans.

- **Disposal of Industrial Waste :** Industries such as textiles, petrochemicals and tanning discharge solid and liquid waste that often contain toxic chemicals like phenols, dyes and acids. When not properly treated, these wastes are dumped onto open land or in landfills, causing long – term soil contamination.
- **Accidental Spills and Leakages :** Oil refineries, chemical plants, and manufacturing units sometimes suffer leaks and accidental spills of hazardous materials like solvents, petroleum products and acids, which seep into the ground and pollute the soil.
- **Improper Storage of Raw Materials and Byproducts :** Many industries store chemicals, fuels and raw materials in open spaces or in poorly maintained containers. Rainwater runoff from these storage areas leaches pollutants into the soil.
- **Use of Industrial Sludge in Agriculture :** Some industrial units promote the use of sludge from effluent treatment plants as fertilizers. However, such sludge often contains toxic substances that accumulate in the soil over time.

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4. Give two impacts of chemicals on visibility?

- **Formation of Smog :** Nitrogen oxides (NOx) and volatile organic compounds (VOCs) are react in sunlight to form photochemical smog, especially in urban areas. Smog scatters and absorbs sunlight, creating a brownish haze that significantly reduces visibility over long distances.
- **Particulate Matter (PM) Pollution :** Sulfates, nitrates, organic carbon, and metals are suspended in the air scatter and absorb light, reducing clarity and sharpness in the atmosphere.
- **Sulfur Dioxide and Acid Aerosols :** SO_2 from power plants and factories reacts with moisture to form sulfate aerosols, which are highly effective at scattering light. This results in a white or bluish haze that can obscure distant objects and landscapes.

<u>References</u> :

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- U.S. EPA (2020). *Air Quality and Visibility*. https://www.epa.gov/visibility
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5. What are POPS?

See question no. 02 part # 07 of 2017

6. Give impacts of silver on human beings?

Impacts of Silver on Human Beings involve both beneficial and harmful effects depending on its form, dose and duration of exposure, some of them are as :

- **Antimicrobial Benefits (Positive Impact) :** Silver ions (Ag^+) and nanoparticles (AgNPs) have broad – spectrum antimicrobial properties. They are used in wound dressings, surgical instruments, catheters and textiles to prevent bacterial infections.
- **Argyria – Skin Discoloration (Negative Impact) :** Chronic exposure to high levels of silver, especially in colloidal or salt form, can cause argyria, a condition where silver accumulates in the skin and eyes, turning them bluish – gray.
- **Neurological Effects (Potential Risk) :** Some studies suggest that exposure to silver nanoparticles may interfere with the central nervous system, causing oxidative stress and neuronal damage in animal models. Human evidence remains limited but concerning.

Reference :

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7. **What do you mean by cation exchange capacity of soil?**

   See question no. 02 part # 02 of 2015

8. **Differentiate between primary and secondary minerals?**

   See question no. 02 part # 04 of 2014

9. **Why environmental monitoring is important?**

   See question no. 02 part # 08 of 2014

10. **Give various classification of soil erosion caused by water?**

    See question no. 02 part # 09 of 2015

LONG QUESTIONS (2018)

1. Give the principle and applications of AAS in environmental monitoring?

Atomic Absorption Spectroscopy (AAS) is a powerful analytical technique widely used in environmental monitoring to detect and quantify trace metals in air, water, soil and biological samples.

Principle of AAS (Atomic Absorption Spectroscopy) :

AAS operates on the principle that free atoms (usually metals) in the gaseous state can absorb light at specific wavelengths. Each metal atom absorbs a unique wavelength, which allows for element – specific detection.

Step – by – Step Explanation :

i. **Atomization :** The sample (solid, liquid or gas) is introduced into a flame or graphite furnace, where it is vaporized and converted into free atoms.
ii. **Radiation Source (Hollow Cathode Lamp) :** A lamp specific to the metal being analyzed emits light at a wavelength that the metal atoms can absorb.
iii. **Absorption :** As the light passes through the vaporized sample, atoms of the target metal absorb part of the radiation. The amount of light absorbed is directly proportional to the concentration of the metal.
iv. **Detection and Quantification :** A detector measures the reduction in light intensity. The system calculates the concentration of the metal using a calibration curve.

Fundamental Equation (Beer – Lambert Law) :

$$A = \varepsilon l c$$

where A is absorbance, ε is the molar absorptivity, l is the path length, and c is the concentration.

Reference :

- Skoog, D.A., Holler, F.J., & Crouch, S.R. (2017). *Principles of Instrumental Analysis* (7th ed.). Cengage Learning.
- Welz, B., & Sperling, M. (1999). *Atomic Absorption Spectrometry*. Wiley-VCH.

Applications of AAS in Environmental Monitoring :

- **Detection of Heavy Metals in Drinking Water :** AAS can detect metals like lead (Pb), cadmium (Cd), arsenic (As) and mercury (Hg) in water samples at parts per billion (ppb) levels.
- **Soil Contamination Analysis :** AAS assesses heavy metal contamination in agricultural or industrial soils including zinc, copper, nickel and chromium.
- **Monitoring Effluents from Industrial Wastewater :** AAS is used to measure toxic metals in effluents from industries such as electroplating, mining, tanneries and battery manufacturing.
- **Airborne Particulate Analysis :** Air filters are digested and analyzed using AAS to identify metal particulates like lead, nickel and manganese emitted by vehicles and industries.
- **Monitoring Mining Impacts on Water Bodies :** Surface and groundwater near mining operations are analyzed for metal leaching especially arsenic, lead and cadmium.
- **Sediment Analysis in Rivers and Lakes :** AAS helps evaluate pollutant deposition in sediments, crucial for long – term environmental impact studies.
- **Monitoring of Pesticide Residues :** Some pesticides are organometallic compounds (e.g., organotin or lead arsenate) and can be quantified via AAS.
- **Leachate Analysis from Landfills :** Leachates from landfills are tested for metal ions using AAS to prevent groundwater contamination.

Reference:

- WHO. (2017). *Guidelines for Drinking – water Quality.*
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- Förstner, U., & Wittmann, G. T. W. (2012). *Metal Pollution in the Aquatic Environment*. Springer.
- Christensen, T. H., et al. (2001). *Landfill leachate : an environmental issue*. *Waste Management & Research*, 19(6), 546–568.

2. Write a note on mining and littering?

Both mining and littering are major anthropogenic contributors to environmental degradation. While mining provides essential raw materials for development, its mismanagement leads to irreversible ecological harm. Littering, often overlooked, has a pervasive impact on ecosystems, health and aesthetics. Sustainable mining practices and responsible waste disposal are crucial to protecting environmental integrity.

i. Mining :

Mining is the process of extracting valuable minerals or other geological materials from the Earth. It includes activities such as open – pit mining, underground mining, strip mining and placer mining.

Environmental Impacts of Mining :

- **Soil Pollution and Land Degradation :** Mining disturbs the topsoil and subsoil structure, causing loss of fertile land and erosion. Dumping of overburden (waste rock) leads to contamination of soils with heavy metals like lead, arsenic, mercury, cadmium and others.
- **Water Pollution :** Mining operations generate acid mine drainage (AMD) – acidic water rich in heavy metals – which pollutes rivers and groundwater.
- **Air Pollution :** Mining emits dust particles, SO_2, NO_x and heavy metals into the air. Open – pit mining causes particulate matter (PM2.5 and PM10) dispersion, affecting human and animal respiratory health.
- **Biodiversity Loss :** Mining displaces flora and fauna, fragments habitats and disrupts ecosystems. Rare species may become endangered or extinct due to habitat destruction.
- **Visual and Landscape Impact :** Mining scars the landscape, leaving behind abandoned pits, tailings and debris. These derelict lands reduce the aesthetic and ecological value of the region.

ii. Littering :

Littering is the improper disposal of waste products (solid or liquid) in public or natural spaces rather than using proper waste management systems.

Environmental Impacts of Littering :

- **Soil and Groundwater Contamination :** Littered items like batteries, plastics and electronics contain toxic substances (e.g., lead, mercury, phthalates) which leach into the soil and contaminate groundwater.
- **Air Pollution from Decomposition and Burning :** Decomposing organic litter releases methane (CH_4) and volatile organic compounds (VOCs). Burning of litter such as plastics emits dioxins and furans, which are carcinogenic.
- **Marine and Aquatic Pollution :** Litter like plastics often ends up in rivers, lakes and oceans. Microplastics are ingested by marine life, entering the food chain and affecting human health.
- **Harm to Wildlife :** Litter such as fishing lines, plastic rings, or glass shards can cause entanglement, ingestion and injury to animals.
- **Public Health Hazards :** Littering encourages the breeding of pests (rats, mosquitoes), increasing the spread of diseases such as dengue, cholera and typhoid.

<u>References</u> :

- Robinson, B. H. (2009). *E-waste: An assessment of global production and environmental impacts*. *Science of the Total Environment*, 408(2), 183–191.

- Wiedinmyer, C., Yokelson, R. J., & Gullett, B. K. (2014). *Global emissions of trace gases, particulate matter, and hazardous air pollutants from open burning of domestic waste*. *Environmental Science & Technology*, 48(16), 9523–9530.
- Jambeck, J. R., et al. (2015). *Plastic waste inputs from land into the ocean*. *Science*, 347(6223), 768–771.

3. How GC is used for pollution monitoring?

Gas Chromatography is a versatile and powerful tool in environmental science, enabling accurate, sensitive and specific analysis of a wide range of pollutants across air, water and soil matrices. Its applications in regulatory monitoring, risk assessment, environmental forensics and compliance testing make it indispensable in modern pollution control efforts.

Principle of Gas Chromatography :

GC operates on the principle of partitioning analytes between a mobile phase (carrier gas like helium, nitrogen or hydrogen) and a stationary phase (liquid or solid coated onto a column). The sample is injected into the instrument, vaporized and carried through the column where different components elute at different times (retention times) based on their interaction with the stationary phase.

- Compounds are detected using detectors such as Flame Ionization Detector (FID), Electron Capture Detector (ECD) or Mass Spectrometry (GC – MS).
- The area under each peak in the chromatogram corresponds to the concentration of a compound.

Applications of GC in Pollution Monitoring :

i. **Monitoring Volatile Organic Compounds (VOCs) in Ambient Air :** GC is widely used to detect VOCs such as benzene, toluene, ethylbenzene and xylene (BTEX) **–** common in urban smog and emitted from vehicular and industrial sources.

ii. **Detection of Pesticide Residues in Water and Soil :** GC with Electron Capture Detector (ECD) is highly sensitive to organochlorine and organophosphorus pesticides such as DDT, aldrin and chlorpyrifos.

iii. **Identification of Polycyclic Aromatic Hydrocarbons (PAHs) :** PAHs are toxic, semi – volatile compounds formed during incomplete combustion. GC, especially GC – MS is ideal for separating and identifying multiple PAHs in environmental matrices like airborne particulates, sediment and water.

iv. **Industrial Emission Monitoring :** GC is used to monitor emissions from factories, including gases like methane, ethylene, formaldehyde and various halocarbons.

v. **Indoor Air Quality Assessment :** GC is used to evaluate VOCs emitted from paints, cleaning agents, adhesives, and furniture, which affect indoor air quality.

vi. **Forensic Environmental Investigations :** GC is used in tracking pollution sources after environmental accidents or spills.

vii. **Analysis of Greenhouse Gases :** GC is applied to detect methane (CH_4), nitrous oxide (N_2O) and carbon dioxide (CO_2) in atmospheric studies and climate research.

viii. **Wastewater and Leachate Monitoring :** GC is used to measure organic pollutants in industrial effluents and landfill leachate, including phenols, solvents and aliphatic hydrocarbons.

ix. **Determination of Microplastics Additives :** GC – MS identifies plasticizers, flame retardants and stabilizers that leach from microplastics into soil and water.

x. **Monitoring Soil Volatiles at Contaminated Sites :** GC is used to assess the volatilization of pollutants like benzene or trichloroethene from contaminated soils, which aids in risk assessment and remediation planning.

References :

- McNair, H. M., & Miller, J. M. (2011). *Basic Gas Chromatography*. John Wiley & Sons.
- U.S. Environmental Protection Agency (EPA), Compendium Method TO-17.
- Alder, L., et al. (2006). *Residue analysis of 500 high priority pesticides*. *Journal of Chromatography A*, 1135(2), 161–169.
- Ravindra, K., et al. (2008). *Atmospheric polycyclic aromatic hydrocarbons: source attribution and contribution*. *Atmospheric Environment*, 42(13), 2895–2921.
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- Weschler, C. J. (2009). *Changes in indoor pollutants since the 1950s*. *Atmospheric Environment*, 43(1), 153–169.
- Reddy, C. M., et al. (2012). *Composition and fate of gas and oil released to the water column during the Deepwater Horizon oil spill*. *PNAS*, 109(50), 20229–20234.
- IPCC Guidelines for National Greenhouse Gas Inventories (2006), Vol. 1.
- IARC (2012). *Trichloroethylene, Tetrachloroethylene and Some Other Chlorinated Agents*, IARC Monographs.
- Rochman, C. M., et al. (2013). *Ingested plastic transfers hazardous chemicals to fish and induces hepatic stress*. *Scientific Reports*, 3, 3263.
- Kitanidis, P. K., & McCarty, P. L. (2012). *Delivery and mixing in the subsurface: Processes and design principles for in situ remediation*. *Springer*.

4. Give the impacts of household chemicals on human health?

See question no. 03 part # 02 of 2017

2019 Short Questions :

1. **Define soil and give its composition?**

Soil is the uppermost layer of the Earth's crust, formed through the weathering of rocks and the decomposition of organic matter over long periods. It serves as a natural medium for plant growth, a habitat for organisms and a filter for water.

- Soil is dynamic and complex composed of mineral particles, organic matter, water, air and living organisms — all interacting to support life.

Composition of Soil :

Soil is generally composed of **five main components**, each contributing to its structure and function :

- Mineral Matter (45%)
- Organic Matter (5%)
- Soil Water (25%)
- Soil Air (25%)
- Living Organisms (Trace but vital)

2. **What is meant by eutrophication? Give its main reasons?**

See question no. 02 part # 01 of 2015

3. **Differentiate between micronutrients and macronutrients giving examples?**

Micro nutrients	**Macro nutrients**
Nutrients needed in small amounts for proper physiological functioning are called micro nutrients.	Nutrients required by the body in large amounts for energy and structural functions are called macro nutrients.
These required in milligrams or micrograms per day.	These required in grams or hundreds of grams per day.
These support metabolic processes and prevent deficiencies.	These provide energy and support growth and bodily functions.
For example ; Vitamins (e.g., Vitamin C, Vitamin D, B – complex) – Minerals (e.g., iron, zinc, iodine, calcium)	For example ; Carbohydrates (e.g., rice, bread) – Proteins (e.g., meat, beans) – Fats (e.g., oils, butter) – Water

References :

- Whitney, E. & Rolfes, S.R. (2021). *Understanding Nutrition*. 16th Edition. Cengage Learning.
- World Health Organization (WHO). (2023). Micronutrients
- National Institutes of Health (NIH) Office of Dietary Supplements. https://ods.od.nih.gov
- FAO/WHO. (2004). *Human Vitamin and Mineral Requirements*. http://www.fao.org/3/y2809e/y2809e00.htm

4. **How is environment affected by PCBS?**

Polychlorinated biphenyls (PCBs) are synthetic organic chemicals once widely used in electrical equipment, hydraulic systems, and as plasticizers. Though banned in many countries, PCBs persist in the environment due to their stability and resistance to degradation. Their environmental impact is extensive and long – lasting.

- **Environmental Persistence :** PCBs are highly persistent in the environment because of their chemical stability and lipophilicity. They do not break down easily and can remain in soil, sediments and water for decades.
- **Soil and Sediment Contamination :** PCBs bind strongly to **soil and sediment particles** due to their hydrophobic nature. They can leach slowly into groundwater or be re-released into surface water under certain conditions.
- **Air and Water Transport :** PCBs can evaporate into the atmosphere and be transported over long distances. Atmospheric PCBs can be deposited into remote ecosystems, such as the Arctic.
- **Harm to Aquatic Life :** PCBs are toxic to fish and other aquatic organisms, affecting their reproduction, growth and development. Fish exposed to PCBs often show **liver damage,** endocrine disruption and behavioral changes.

References :

- Agency for Toxic Substances and Disease Registry (ATSDR). (2000). Toxicological Profile for PCBs
- Jones, K.C., & de Voogt, P. (1999). "Persistent Organic Pollutants (POPs): State of the Science." *Environmental Pollution*, 100(1-3), 209–221. https://doi.org/10.1016/S0269-7491(99)00098-6
- Erickson, M.D., & Kaley II, R.G. (2011). "Applications of polychlorinated biphenyls." *Environmental Science and Pollution Research*, 18(2), 135–151. https://doi.org/10.1007/s11356-010-0392-2
- Wania, F., & Mackay, D. (1996). "Tracking the distribution of persistent organic pollutants." *Environmental Science & Technology*, 30(9), 390A–396A. https://doi.org/10.1021/es962399q

5. How does pH of soil affect the availability of nutrients?

The pH of soil significantly affects the availability of nutrients to plants by influencing the chemical form, solubility and mobility of nutrients. Soil pH is a measure of acidity or alkalinity, typically ranging from 4 (acidic) to 8 (alkaline) in most soils. The optimal pH range for nutrient availability is 6.0 – 7.5, although this varies depending on the crop and soil type.

- **Influence on Macronutrient Availability :**
  - **Nitrogen (N)** : Low pH (<5.5) inhibits nitrification and promotes nitrogen losses via volatilization.
  - **Potassium (K), Calcium (Ca), and Magnesium (Mg)** : More available in neutral to slightly acidic soils ; leaching increases in strongly acidic soils.

- **Influence on Micronutrient Availability :**
    - **Iron (Fe), Manganese (Mn), Zinc (Zn), Copper (Cu)** : More available at lower pH (<6.5). At higher pH, they form insoluble compounds and become deficient.
    - **Molybdenum (Mo)** : Opposite trend ; more available in alkaline soils.
- **pH and Toxic Elements :**
    - **Aluminum (Al) and Manganese (Mn) Toxicity** : In strongly acidic soils (pH < 5.5) Al^{3+} and Mn^{2+} become more soluble and toxic to plant roots, reducing growth and nutrient uptake.
- **Effects on Microbial Activity :** Soil microbes involved in organic matter decomposition and nutrient cycling (especially nitrogen fixation and mineralization) are most active between pH 6.0 and 7.5.

<u>**Reference**</u> :

- Fageria, N. K., Baligar, V. C., & Jones, C. A. (2010). *Growth and Mineral Nutrition of Field Crops*. CRC Press.
- Marschner, H. (2012). *Marschner's Mineral Nutrition of Higher Plants* (3rd ed.). Academic Press.
- Kochian, L. V., Piñeros, M. A., & Hoekenga, O. A. (2005). The physiology, genetics and molecular biology of plant aluminum resistance and toxicity. *Plant and Soil*, 274(1–2), 175–195.
  https://doi.org/10.1007/s11104-004-1158-7
- Sylvia, D. M., Fuhrmann, J. J., Hartel, P. G., & Zuberer, D. A. (2005). *Principles and Applications of Soil Microbiology*. Pearson Prentice Hall.

6. Give at least 4 volatile organic compounds that pollute the soil?

Common volatile organic compounds (VOCs) that pollute soil, particularly at contaminated industrial, agricultural and landfill sites are as :

I. **Benzene :** Benzene is a carcinogenic VOC that can persist in soil and volatilize into the air, contributing to soil and groundwater contamination.
II. **Toluene :** Toluene can degrade slowly in anaerobic soils and may leach into groundwater, impacting microbial health and plant growth.
III. **Trichloroethylene (TCE) :** TCE is highly toxic, mobile in soil and known to contaminate subsurface layers and groundwater.
IV. **Vinyl Chloride :** It is carcinogenic and highly mobile in soil, it poses a serious risk to both human health and groundwater quality.
V. **Dichloromethane (Methylene Chloride) :** It evaporates quickly but may contaminate deeper soil layers and leach into groundwater if not managed.

References :

- ATSDR (2007). *Toxicological Profile for Benzene*. https://www.atsdr.cdc.gov/toxprofiles/tp3.html
- ATSDR (2017). *Toxicological Profile for Toluene*. https://www.atsdr.cdc.gov/toxprofiles/tp56.html

- EPA (2023). *Technical Fact Sheet – Trichloroethylene (TCE)*. https://www.epa.gov/sites/default/files/2014-03/documents/ffrrofactsheet_contaminant_trichloroethylene_january2014_final.pdf
- ATSDR (2006). *Toxicological Profile for Vinyl Chloride*. https://www.atsdr.cdc.gov/toxprofiles/tp20.html
- WHO (1996). *Environmental Health Criteria 164: Methylene Chloride*. https://www.who.int/publications/i/item/9241571646

7. How is biodiversity is affected by the presence of pollutants?

Biodiversity — the variety of life across genes, species, and ecosystems — is highly sensitive to environmental pollutants. Pollutants affect biodiversity directly by causing mortality and physiological stress and indirectly by disrupting habitats and ecological interactions.

How Biodiversity Is Affected by Pollutants :

- **Toxicity – Induced Mortality in Species :** Many pollutants — such as heavy metals (e.g., lead, mercury), pesticides (e.g., DDT) and industrial effluents — are directly toxic to flora and fauna. Acute or chronic exposure can lead to population declines or species extinctions. For example Organochlorine pesticides like DDT caused egg – shell thinning in birds like eagles and falcons, leading to reproductive failure.
- **Disruption of Food Webs and Trophic Cascades :** Pollutants can disproportionately affect key species (keystone, primary producers, or top predators), destabilizing entire ecosystems and food webs. For example bioaccumulation and biomagnification of mercury in aquatic food chains affects fish-eating birds and mammals. Pesticides killing insect pollinators (e.g., bees) disrupt plant reproduction and reduce biodiversity in flowering plants.

References :

- Carson, R. (1962). Silent Spring. Houghton Mifflin.
- He, Z.L. et al. (2005). “Heavy metals in soils: Sources, interactions, and remediation.” Advances in Agronomy, 94, 113–142.
- Scheuhammer, A.M. et al. (2007). “The biological effects of mercury in fish-eating wildlife.” Environmental Toxicology and Chemistry, 26(1), 20–30.
- Potts, S.G. et al. (2010). “Global pollinator declines: trends, impacts and drivers.” Trends in Ecology & Evolution, 25(6), 345–353.

8. Define the term of cation exchange capacity of soil?

See question no. 02 part # 02 of 2015

9. How is environmental monitoring different than environmental analysis?

Environmental monitoring and environmental analysis are related but distinct processes in environmental science, each serving different roles in assessing environmental quality.

- **Environmental Monitoring :**

  It is the systematic and continuous collection of environmental data (e.g., air, water, soil, noise, radiation) over time to track changes or trends.

- **Environmental Analysis :**

  It refers to the laboratory – based examination of environmental samples to quantify and identify specific pollutants or parameters.

Key Differences

Environmental monitoring	**Environmental analysis**
It is a ongoing, long – term process.	It is a one – time or short – term testing.
It detect trends, observe changes, compliance tracking.	It identify/quantify pollutants in specific samples.
It is broader (includes physical, chemical, biological)	It focused on chemical and microbiological parameters.
It includes sensors, remote sensing, field sampling	It includes GC – MS, AAS, HPLC, spectrophotometers

Reference :

- UNEP (2002). *Environmental Impact Assessment Training Resource Manual* – https://www.unep.org
- Keith, L. H. (1991). *Environmental Sampling and Analysis: A Practical Guide*. CRC Press.
- Baird, C., & Cann, M. (2012). *Environmental Chemistry* (5th ed.). W.H. Freeman.
- U.S. EPA. (2020). *Guidance for Environmental Monitoring and Analysis*. https://www.epa.gov/quality/guidelines

10. How are micronutrients beneficial for plant growth?

See question no. 02 part # 10 of 2014

LONG QUESTIONS (2019)

1. How are littering and mining helpful in increasing the soil quality?

Natural littering refers to the deposition of organic residues from vegetation onto the soil surface. This includes fallen leaves, twigs, bark, dead roots, flowers, fruits and crop residues. It is a core ecological process in forest and agricultural ecosystems and contributes to long-term soil quality enhancement through the following pathways :

- **Input of Organic Matter and Humus Formation :** The plant litter undergoes biochemical decomposition by soil biota, including bacteria, fungi (particularly saprophytes), actinomycetes and macrofauna like earthworms. During this process, simple sugars and proteins are rapidly degraded, while lignin and cellulose take longer to break down. Over time, this decomposition results in the formation of humus, a highly stable organic fraction.

  Humus binds with mineral particles (sand, silt, clay) to form soil aggregates, enhancing :
  - Water infiltration and retention,
  - Soil porosity and aeration,
  - Cation exchange capacity (CEC) for nutrient storage.

- **Nutrient Mineralization and Availability :** Plant litter contains essential macro- and micronutrients in organically bound forms. As decomposers break down these compounds, nutrients such as nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), iron (Fe), zinc (Zn) and manganese (Mn) are mineralized into inorganic forms absorbable by plant roots.
  For example :
    - Nitrogen is converted from amino acids → ammonium (NH_4^+) → nitrate (NO_3^-).
    - Organic phosphorus is transformed into orthophosphate ($H_2PO_4^-$).

This slow – release nutrient cycling is especially important in nutrient – poor soils, where synthetic fertilizers may not be available or sustainable.

- **Enhancement of Soil Biota and Ecosystem Functioning :** The constant addition of organic matter from litter feeds heterotrophic organisms in soil. Earthworms, collembolans, termites and nematodes fragment the litter, increase surface area and mix it with mineral particles. In doing so, they :
    - Enhance microbial activity,
    - Increase enzymatic degradation of complex compounds,
    - Accelerate nutrient turnover
    - Improve carbon sequestration.
- **Buffering Soil pH and Reducing Erosion :** Decomposing organic matter produces organic acids (e.g., fulvic acid, citric acid) that can chelate metals and buffer soil pH, especially in acidic or alkaline conditions. Additionally, by forming a protective mulch layer, litter reduces :
    - Rain splash impact
    - Surface runoff
    - Topsoil loss.

This directly supports soil conservation and promotes seedling emergence and root development.

Reference :

- Brady, N.C., & Weil, R.R. (2016). *The Nature and Properties of Soils*, 15th ed. Pearson.
- Gessner, M.O., Swan, C.M., Dang, C.K., et al. (2010). "Diversity meets decomposition." *Trends in Ecology & Evolution*, 25(6), 372–380.
- Lavelle, P., & Spain, A.V. (2001). *Soil Ecology*. Springer.
- Lal, R. (2005). "Soil erosion and carbon dynamics." *Soil and Tillage Research*, 81(2), 137–142.

Human Littering (Anthropogenic Waste)

Contrary to natural litter, human-induced littering—such as the dumping of plastics, metals, cans, glass, batteries and e – waste —is a significant soil contaminant. These materials are non – biodegradable and introduce toxic substances into the pedosphere.

- **Microplastic Pollution and Soil Health :** Plastics and synthetic polymers fragment into microplastics, which persist in soil for decades. These microplastics :
    - Alter soil texture and bulk density,

- Reduce water-holding capacity
- Impede root growth
- Disrupt soil microbial communities and enzymatic activity.

- **Heavy Metal Contamination :** Improper littering of batteries, e-waste, and industrial debris introduces lead (Pb), cadmium (Cd), arsenic (As), chromium (Cr) and mercury (Hg) into soils. These metals :
  - Are non – degradable and bioaccumulate,
  - Interfere with root respiration and water uptake
  - Inhibit soil enzymes like urease and dehydrogenase
  - Can leach into groundwater.

Reference :

- Chae, Y., & An, Y.-J. (2018). "Ecotoxicological effects of microplastics on soil ecosystems." *Environmental Pollution*, 240, 387–395.
- Alloway, B.J. (2013). *Heavy Metals in Soils*, 3rd ed. Springer.

Mining and Soil Quality

- **Conventional Mining Impacts (Destructive) :** Most mining operations (open – pit, strip, mountaintop removal) result in :
  - Loss of topsoil and fertile A – horizon
  - Soil compaction and erosion
  - Exposure of pyrite – bearing rocks that cause acid mine drainage (AMD)
  - Metal toxicity from waste tailings.

This causes barren land, loss of microbial life and often renders soil unsuitable for vegetation for decades.

- **Potential Positive Contributions (Under Reclamation and Sustainable Management) :**

Despite its damaging history, responsibly managed mining can indirectly contribute to soil improvement, especially when post – mining reclamation is enforced.

1. Land Reclamation Techniques :

- Replacing topsoil
- Amending with compost, manure, or biochar
- Planting nitrogen – fixing and deep – rooted species (e.g., *Acacia*, *Vetiver*, legumes).

Over time, these steps restore microbial biomass, organic matter, CEC and water retention.

2. Mining By-products as Soil Enhancers :

- Phosphate rock mining supports fertilizer production.
- Gypsum and limestone from mines are used to reclaim sodic and acidic soils.
- Biochar production from coal fines or biomass residues improves nutrient retention.

These amendments are scientifically proven to increase soil fertility and productivity.

Reference :

- Tarras – Wahlberg, N. H. (2001). "Environmental impact of mining in the tropics." *Science of the Total Environment*, 281, 29–45.

- Sheoran, V., et al. (2010). "Revegetation of mined land." *Ecological Engineering*, 36(6), 752–758.
- Lehmann, J., & Joseph, S. (2009). *Biochar for Environmental Management: Science and Technology*. Earthscan.

2. Explain the soil erosion and some methods of its prevention?

Soil erosion is the process by which the topsoil layer – rich in organic matter, nutrients and biological activity — is detached, transported and deposited elsewhere due to natural forces (wind, water, gravity) or anthropogenic activities (deforestation, overgrazing, unsustainable agriculture and mining). It is one of the most pressing environmental threats to land productivity, biodiversity and sustainable agriculture.

Mechanisms and Types of Soil Erosion

- **Water Erosion :** This is the most widespread form and occurs in multiple stages :
    - **Splash erosion** : Raindrop impact dislodges particles.
    - **Sheet erosion** : Uniform removal of a thin soil layer.
    - **Rill erosion** : Formation of small channels.
    - **Gully erosion** : Deep and wide trenches develop, often irreversibly.
- **Wind Erosion :** Occurs in dry, bare and sandy soils—especially in arid and semi-arid zones. Fine soil particles are lifted and transported by wind in:
    - Saltation (bouncing)
    - Suspension (floating)
    - Surface creep (rolling).
- **Gravitational Erosion (Mass Movement) :** Includes landslides, mudslides and slumping. It happens on steep or unstable slopes, especially after heavy rainfall or deforestation.

Environmental and Agricultural Impacts of Soil Erosion

- **Loss of fertility** : Topsoil is nutrient – rich. Its loss directly reduces crop yields.
- **Soil structure degradation** : Eroded soils become compacted and less porous.
- **Biodiversity loss** : Soil fauna and flora are destroyed, affecting food webs.
- **Desertification** : In extreme cases, fertile land becomes barren and dry.

Soil Erosion Prevention Methods

I. Vegetative Cover and Reforestation

- **Permanent Vegetation :**
    - Roots anchor the soil, increasing cohesion.
    - Canopy reduces raindrop impact.
    - Litter layer enhances water infiltration and slows runoff.

Best Practices :

- Planting cover crops (e.g., clover, rye, vetch).
- Afforestation and reforestation in deforested regions.
- Agroforestry systems **:** Integrating trees into farms.

II. Contour Farming and Terracing :

- **Contour Farming :** It involves tilling and planting across the slope rather than downslope. This reduces runoff velocity and allows more water infiltration.
- **Terracing :** It converts steep slopes into stepped Platforms with retaining walls. Water is slowed and absorbed rather than running off.

Benefits :

- Prevents gully formation.
- Enhances water retention for crops.
- Increases cultivable land in hilly regions.

III. Strip Cropping and Crop Rotation

- **Strip Cropping :** Alternating strips of erosion – prone crops (like corn) with protective ones (like grass or legumes). The latter intercept erosion.
- **Crop Rotation :** Different crops have varying root systems and residue production. Rotating crops improves soil structure and reduces pest buildup, minimizing tillage intensity.

IV. Mulching and Organic Residue Management

Applying a layer of organic mulch (straw, compost, bark, leaf litter) covers the soil, regulating temperature, reducing evaporation and buffering rain impact.

Advantages :

- Enhances microbial activity.
- Prevents crusting and compaction.
- Gradually decomposes to add nutrients and humus.

V. Conservation Tillage and No – Till Farming

Traditional tillage breaks up soil aggregates, accelerating erosion. Conservation tillage or no – till farming leaves crop residues on the surface, maintaining soil structure.

Ecological Outcomes :

- Preserves soil carbon and nitrogen.
- Reduces fuel use and soil disturbance.
- Promotes biological activity in the rhizosphere.

VI. Windbreaks and Shelterbelts

Rows of trees or shrubs are planted around agricultural fields to block wind flow and reduce wind erosion.

- Trap airborne dust and fine particles.
- Create microclimates beneficial for crops.
- Also serve as biodiversity corridors.

VII. Check Dams and Sediment Traps

On sloped or degraded lands, small check dams made of stones, brush or geotextiles are placed across gullies to trap sediments, reduce flow speed and allow for sediment deposition and vegetation regrowth. These are essential in post – mining landscapes and overgrazed watersheds.

Reference :

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3. Explain the hazardous effects of aflatoxin on environment especially on human health?

Aflatoxins are a group of naturally occurring mycotoxins produced mainly by two species of fungi ; Aspergillus flavus and Aspergillus parasiticus. These fungi grow in warm and humid climates especially in poorly stored agricultural products like maize, peanuts, cottonseed, spices and tree nuts. Among the different types, the most toxic and commonly studied is Aflatoxin B_1 (AFB_1).

Environmental Occurrence of Aflatoxins

Aflatoxins contaminate soil, water, crops and animal feed, and persist in the environment due to the robust survival of the producing fungi. They are commonly found in :

- Stored grains under humid conditions.
- Decomposing organic matter.
- Contaminated irrigation water.
- Animal excreta and waste-fed soils.

Hazardous Effects on Human Health

Aflatoxins are classified as Group 1 carcinogens by the International Agency for Research on Cancer (IARC) due to their high potency in inducing liver cancer. The toxicological effects of aflatoxins are dose – dependent**,** chronic or acute and multisystemic.

i. **Hepatotoxicity and Liver Cancer :**

Aflatoxin B_1 binds to DNA in the liver, forming aflatoxin – DNA adducts particularly targeting the p53 tumor suppressor gene leading to hepatocellular carcinoma (HCC). Risk is significantly increased in individuals with hepatitis B virus (HBV) infection due to synergistic mutagenic effects.

ii. **Acute Aflatoxicosis :**

High exposure causes acute poisoning symptoms ; vomiting, jaundice, pulmonary edema, coma and even sudden death, especially in outbreaks linked to contaminated maize. A 2004 outbreak in Kenya resulted in 125 deaths due to consumption of moldy grains.

iii. **Immune System Suppression :**

Chronic exposure reduces the ability of immune cells (T-cells, B-cells, macrophages) to fight infections. It leads to increased susceptibility to malaria, HIV/AIDS and respiratory diseases, particularly in low – income populations.

iv. **Growth Impairment and Malnutrition in Children :**

Aflatoxins impair protein synthesis and nutrient absorption. Epidemiological studies link chronic exposure to stunted growth, underweight children and impaired cognitive development.

v. **Reproductive and Developmental Toxicity :** It interferes with hormone regulation and placental function. It linked to spontaneous abortions, congenital defects and low birth weight.

Hazardous Effects on the Environment

Aflatoxins pose a significant threat to agro – ecosystems and biodiversity, particularly when they enter soil, water bodies and animal feed chains.

- **Soil and Crop Contamination :**

  Aflatoxins persist in soil from infected crop residues. Fungal spores germinate in favorable conditions, re – infecting subsequent crops. It leads to long – term yield reduction especially in staple foods like maize, sorghum and groundnuts.

- **Water Contamination :**

  Aflatoxins leach from decomposing plant matter or run off from storage areas, entering aquatic ecosystems. They harm aquatic organisms, particularly plankton and benthic invertebrates, disrupting the aquatic food web.

- **Animal Health and Food Chain Entry :**

  Animals fed contaminated feed (e.g., cattle, poultry, fish) suffer from :

  - Liver damage.
  - Decreased milk yield.
  - Poor weight gain.

Aflatoxin M_1 residues appear in milk and dairy products, posing a secondary risk to humans.

<u>**Reference**</u> **:**

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4. Explain the concept of green chemistry giving its principles?

Green Chemistry also referred to as **sustainable chemistry**, is a scientific discipline and design philosophy aimed at developing chemical products and processes that minimize or eliminate the use and generation of hazardous substances. It represents a paradigm shift from traditional chemistry, focusing not just on efficiency and yield but also on environmental and human health impacts across a chemical's entire life cycle — from raw material sourcing to disposal. Coined and formalized in the early 1990s by Paul Anastas and John Warner, green chemistry serves as a proactive approach to pollution prevention at the molecular level, rather than managing pollutants after their creation.

Core Philosophy :

The central idea of green chemistry is the **“benign by design”** approach — designing chemical processes and products in such a way that they :

- Use safer chemicals
- Require less energy
- Generate less waste and
- Have minimal impact on human and ecological systems.

This approach aligns chemistry with the goals of sustainability and is crucial in tackling major environmental issues such as climate change, toxic pollution and resource depletion.

12 Principles of Green Chemistry :

Anastas and Warner articulated twelve guiding principles, which serve as a framework for chemists, engineers, and industries to innovate more sustainably :

- **Prevention :** Avoid waste rather than treating or cleaning it up afterward. Designing processes that prevent pollution at the source saves energy, money and environmental harm.
    - Example : Using in situ reactions to avoid isolating intermediate chemicals.

- **Atom Economy :** Maximize incorporation of all materials used into the final product. Encourages synthetic efficiency and minimizes by – products.
- **Less Hazardous Chemical Syntheses :** Design synthetic methods that are non – toxic or less harmful to humans and the environment. Focuses on risk reduction from the beginning of the process.
    - **Example :** Using water or ethanol as safer reaction solvents.
- **Designing Safer Chemicals :** Chemical products should be designed to achieve their function while minimizing toxicity. Emphasizes understanding of structure – activity relationships (SARs).
    - Example : Designing biodegradable pesticides with selective toxicity.
- **Safer Solvents and Auxiliaries :** Avoid using auxiliary substances (e.g., solvents, separation agents) or make them non **–** hazardous. Solvents often contribute significantly to the environmental footprint of processes.
- **Energy Efficiency :** Minimize energy use by conducting reactions at ambient temperature and pressure whenever possible. Encourages the use of microwave or photocatalysis instead of heating.
    - Example : Room – temperature catalysis using enzymes or organocatalysts.
- **Use of Renewable Feedstocks :** Prefer renewable raw materials over finite, depleting ones. Supports circular economy and carbon neutrality.
    - **Example :** Using plant – based ethanol instead of petroleum – based ethylene.
- **Reduce Derivatives :** Avoid unnecessary derivatization (e.g., use of protecting groups). Derivatization steps often require additional reagents and produce more waste.
- **Catalysis :** Favor catalytic reagents over stoichiometric ones. Catalysts can be reused, are often more selective and reduce side reactions.
- **Design for Degradation :** Products should break down into non – toxic components after use. Prevents persistence in the environment and bioaccumulation.
    - **Example :** Developing biodegradable polymers like polylactic acid (PLA).
- **Real – time Analysis for Pollution Prevention :** Implement real – time monitoring to control hazardous substances during synthesis. Prevents unexpected emissions and improves process safety.
- **Inherently Safer Chemistry for Accident Prevention :** Design chemicals and processes to minimize risks of explosions, fires or toxic releases. Focuses on inherent safety not just compliance with regulations.
    - **Example :** Replacing volatile organic solvents with supercritical CO_2.

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5. Explain the principle and applications of UV&VIS spectroscopy for environmental analysis?

UV – Visible (UV – Vis) spectroscopy is a quantitative and qualitative analytical technique that measures the absorbance or transmittance of ultraviolet (200 – 400 nm) and visible light (400 – 700 nm) by chemical substances. The fundamental principle is based on the interaction of electromagnetic radiation with matter, especially the excitation of electrons in atoms or molecules from a lower energy level (ground state) to a higher energy level (excited state) when they absorb specific wavelengths of light.

Key Principle :

Molecules absorb UV or visible light depending on their electronic **structure**, particularly the presence of chromophores such as π-electrons (in double bonds, conjugated systems) or lone pairs in non-bonding orbitals ($n \rightarrow \pi^*$ transitions).

The amount of light absorbed follows Beer – Lambert's Law :

$$A = \varepsilon c l$$

Where :

- A = Absorbance (no units)
- ε = Molar absorptivity ($L \cdot mol^{-1} \cdot cm^{-1}$)
- c = Concentration of the analyte ($mol \cdot L^{-1}$)
- l = Path length of the cell (cm)

Applications of UV – Vis Spectroscopy in Environmental Analysis

UV – Vis spectroscopy plays a critical role in environmental science because of its simplicity, sensitivity, low cost and ability to detect pollutants and changes in water, air and soil chemistry.

- **Water Pollution Monitoring :** UV – Vis spectroscopy is used to quantify nitrates (NO_3^-) and nitrites (NO_2^-) in surface and groundwater. Nitrates absorb UV light at ~220 nm; correction at 275 nm eliminates interference from organics.
- **Detection of Heavy Metals via Complexation :** While metals like Pb^{2+} or Cd^{2+} don't directly absorb UV – Vis light, they form colored complexes with reagents like dithizone or EDTA, enabling indirect quantification.
- **Organic Pollutants and Pesticides :** Many organic contaminants especially aromatic compounds (e.g., benzene, phenols), absorb in the UV region due to $\pi \rightarrow \pi^*$ transitions.
- **Monitoring Wastewater Treatment Efficiency :** UV – Vis spectroscopy helps track :
  - Chemical Oxygen Demand (COD)

- Total Organic Carbon (TOC)
- Aromatic hydrocarbons degradation during treatment.

- **Air Quality Monitoring :** Some gaseous pollutants like ozone (O_3), nitrogen dioxide (NO_2) and sulfur dioxide (SO_2) have strong absorption bands in the UV and visible regions.
- **Soil Contaminant Analysis :** Soil leachates can be analyzed for chromophores including humic substances, herbicides and petroleum – derived pollutants.
- **Phytochemical and Chlorophyll Analysis :** In ecological and agricultural monitoring, UV – Vis spectroscopy quantifies :
  - Chlorophyll a and b (~665 and 649 nm)
  - Carotenoids (around 470 nm)
- **Microbial and Algal Bloom Analysis** : UV – Vis absorbance helps determine algal density (via pigments) and biomass concentrations. It can be used to monitor algal **blooms** caused by nutrient overloading (eutrophication).

<u>Reference</u> :

- Lichtenthaler, H. K., & Buschmann, C. (2001). “Chlorophylls and carotenoids: Measurement and characterization by UV – Vis spectroscopy.” *Current Protocols in Food Analytical Chemistry*.

6. **What is AAS? Give its role in environmental monitoring?**

See question no. 03 part # 01 of 2014

2020 Short questions :

1. **Toxicity of Cr is attributed to Cr(6) not to Cr (3),why?**

   The toxicity of chromium is attributed to Cr(VI) because :
   - It is chemically reactive, soluble and easily absorbed.
   - It induces oxidative stress and DNA damage during its intracellular reduction.
   - It is a proven carcinogen, while Cr(III) is largely non – toxic and sometimes nutritionally essential.

- **Oxidation State and Environmental Behavior :** Cr(III) is relatively stable, poorly soluble and tends to form inert complexes with organic matter and soil particles while Cr(VI) exists primarily as chromate (CrO_4^{2-}) or dichromate ($Cr_2O_7^{2-}$) ions in water, which are highly soluble, mobile and readily absorbed by living organisms.
- **Cellular Uptake :** Cr(VI) mimics essential anions such as sulfate (SO_4^{2-}) and phosphate (PO_4^{3-}), which allows it to readily cross cell membranes via anion transporters. In contrast, Cr(III) is not easily transported across cell membranes due to its positive charge, large hydration sphere and low membrane permeability.
- **Genotoxic and Carcinogenic Potential :** Cr(VI) has been classified as a Group 1 carcinogen by the International Agency for Research on Cancer (IARC) due to its mutagenic and carcinogenic activity, especially in the lungs from inhalation. Cr(III) is not carcinogenic and is even considered essential in trace amounts for glucose and lipid metabolism in humans.

Reference :

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- IARC Monographs on the Evaluation of Carcinogenic Risks to Humans, Volume 49: Chromium, Nickel and Welding. World Health Organization (1990).

2. **What is CEC? Why soil can hold only cations?**

   See question no. 02 part # 02 of 2015

3. **What are major causes of soil erosion?**

Soil erosion is the detachment and removal of topsoil by natural forces such as wind and water or by anthropogenic (human) activities. It significantly reduces soil fertility, affects water quality and threatens sustainable agriculture. The major causes of soil erosion can be categorized into natural and human – induced factors, each explained as :

- Deforestation
- Overgrazing
- Unsustainable Agricultural Practices
- Urbanization and Construction
- Mining Activities

<u>Reference</u> :

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4. What are aflatoxin?

See question no. 02 part # 05 of 2014

5. Discuss control measure for water erosion?

Water erosion (the removal of soil by rainfall, runoff and floodwaters) is one of the most severe threats to soil fertility and land productivity. Controlling water erosion involves implementing biological, structural and agronomic practices to reduce the impact of raindrops, surface runoff and soil displacement.

- Contour Plowing
- Terracing
- Vegetative Cover and Mulching
- Strip Cropping
- Check Dams and Sediment Traps
- Agroforestry
- Cover Cropping and Crop Rotation
- Rainwater Harvesting

Reference :

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6. Differentiate between micronutrients and macronutrients in soil?

See question no. 02 part # 03 of 2019

7. Describe the principle of UV&VIS spectroscopy?

See question no. 03 part # 01 of 2015

8. Describe the main sources of lead in environment?

Lead (Pb) is a toxic heavy metal that poses serious risks to both the environment and human health. It enters the environment through both natural processes and anthropogenic (human – made) activities.

- **Industrial Emissions :** Lead is widely used in industries such as :
  - Battery manufacturing (especially lead – acid batteries)
  - Metal smelting and refining
  - Paint and pigment production
  - Ammunition manufacturing

During these processes, lead dust and fumes are emitted into the air, which eventually deposit into soil and water bodies contaminating them.

- **Lead – Based Paints :** Old buildings and infrastructure (especially before the 1970s) often contain lead – containing paints. As the paint deteriorates, it releases :
  - Lead – containing dust
  - Paint flakes, which settle into nearby soil or indoor environments

  This is a major cause of lead exposure in urban environments.
- **Mining and Ore Processing :** Lead is often released into the environment during :
  - Mining of lead and other ores (e.g., zinc, silver)
  - Crushing, washing and disposal of waste tailings

  These activities contaminate surface water**,** groundwater and soils near mines.

<u>Reference</u> :

- Jarup, L. (2003). Hazards of heavy metal contamination. *British Medical Bulletin*, 68(1), 167–182.
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  https://doi.org/10.1126/science.272.5259.223

9. Why PCBs were banned?

See question no. 02 part # 09 of 2014

10. What is atom economy?

Atom economy (also called atom utilization) refers to the proportion of reactant atoms that end up in the desired final product, rather than in by – products. It was first introduced by Barry Trost in 1991 to promote more sustainable chemical practices.

Higher atom economy = less waste = greener process

- Atom economy is one of the **12 principles of green chemistry** as proposed by Paul Anastas and John Warner. It encourages chemists to **design synthetic methods** that maximize the incorporation of all materials used in the process into the final product.
- Why Atom Economy Matters :
    - Reduces chemical waste
    - Improves efficiency and cost – effectiveness
    - Promotes sustainable manufacturing
    - Minimizes environmental and health hazards

Unlike percent yield, which focuses only on how much product is formed, atom economy considers how much of the reactants' mass actually contributes to the final product.

<u>References</u> :

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LONG QUESTIONS (2020)

1. What is soil pollution? What are major causes of soil pollution?

Soil pollution is the presence of toxic chemicals (pollutants or contaminants) in the soil, in high enough concentrations to pose risks to human health, plant growth, and ecosystems. It results in the degradation of soil quality, making it unsuitable for agriculture, construction and ecological functions. Unlike visible waste or water pollution, soil pollution is often hidden but has long – lasting and often irreversible effects.

- According to the Food and Agriculture Organization (FAO), "Soil pollution refers to the presence in soil of a chemical or substance out of place and/or present at a higher than normal concentration that has adverse effects on any non – target organism".

Types of Soil Pollutants :

- **Organic pollutants :** pesticides, herbicides, PCBs, PAHs
- **Inorganic pollutants :** heavy metals (lead, cadmium, arsenic, mercury), salts, acids
- **Biological agents :** pathogens from sewage and organic waste
- **Emerging contaminants :** pharmaceuticals, microplastics, PFAS

Major Causes of Soil Pollution :

1. **Excessive Use of Agrochemicals :**

    The extensive use of synthetic fertilizers, pesticides, herbicides and fungicides in modern agriculture introduces toxic substances to soil.

Effects :

- Accumulation of nitrates, phosphates and toxic metals
- Disruption of microbial activity and nutrient cycles
- Bioaccumulation in crops, causing health issues in humans

2. **Industrial Activities and Waste Disposal :**

Industrial waste often includes heavy metals, hydrocarbons, solvents, and other toxic residues. Improper disposal of solid and hazardous wastes leads to soil contamination.

Effects :

- Soil becomes acidic or saline
- Heavy metal poisoning of plants and groundwater
- Long – term ecological damage and reduced fertility

3. **Oil Spills and Petrochemical Leaks :**

Leakage from pipelines, underground storage tanks, and transportation accidents release petroleum hydrocarbons into soil.

Effects :

- Reduced oxygen diffusion in soil
- Soil microbial death and loss of biodiversity
- Hydrocarbon buildup makes soil hydrophobic

4. **Improper Disposal of Solid and Hazardous Waste :**

Unregulated dumping of municipal waste, hospital waste and e-waste introduces plastics, pathogens and heavy metals into soil.

Effects :

- Soil acidification or alkalization
- Spread of antibiotic resistance and disease vectors
- Leaching into groundwater sources

5. **Urbanization and Construction :**

Rapid urban expansion leads to sealing of soil surfaces and contamination from construction debris, paints, solvents and metals.

Effects :

- Loss of fertile topsoil
- Soil compaction and reduced water infiltration
- Altered pH and mineral profiles

6. **Mining Activities :**

Mining exposes soil to tailings, overburden, and runoff containing arsenic, mercury and sulfuric acid.

Effects :

- Soil acid mine drainage (AMD)
- Long – term contamination with toxic elements
- Reduced productivity and vegetation cover

References :

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2. Briefly discuss POPS?

See question no. 03 part # 02 of 2014

3. Discuss principle and applications of HPLC in environmental monitoring?

See question no. 03 part # 03 of 2017

4. Briefly describe the composition of typical soil?

Soil is a dynamic, natural body composed of minerals, organic matter, air, water and living organisms. Its composition determines its fertility, structure, drainage and biological activity. The typical composition of surface soil by volume is approximately :

- 45% Mineral Matter (inorganic particles)
- 25% Water
- 25% Air
- 5% Organic Matter (including living organisms)

This balance varies depending on climate, vegetation, topography, parent material and land use practices.

- **Mineral Matter (~45%) :**

These are the inorganic components derived from the weathering of rocks. They give soil its structure and texture and are classified by particle size :

- **Sand (0.05 – 2.00 mm) :** Coarse particles; improve drainage and aeration.
- **Silt (0.002 – 0.05 mm) :** Medium particles; smooth texture and moderate water retention.
- **Clay (<0.002 mm) :** Fine particles; high water and nutrient retention, poor drainage.

The relative proportions of sand, silt and clay define soil texture (e.g., loam, clay loam, sandy soil).

- **Water (~25%) :**

Soil water exists in several forms :

- **Gravitational Water :** Drains through soil due to gravity ; unavailable to plants.
- **Capillary Water :** Held in pores and available for plant uptake.
- **Hygroscopic Water :** Thin layer tightly bound to particles ; unavailable to plants.

Water is crucial for transporting nutrients, supporting microbial life and maintaining plant metabolism.

- **Air (~25%) :**

  Soil air fills the pore spaces not occupied by water. Its composition differs from atmospheric air :

  - Lower O_2 due to root and microbial respiration
  - Higher CO_2 levels
  - Important for aerobic microbial activity and root respiration

  Well – aerated soils promote plant growth and decomposition of organic matter. Poor aeration leads to anaerobic conditions, affecting nutrient cycling and plant health.

- **Organic Matter (~5%) :**

  Organic matter includes :

  - **Humus :** Stable, decomposed organic residues critical for soil fertility.
  - **Fresh Residues :** Leaves, roots and animal waste that are decomposing.
  - **Living Organisms :** Bacteria, fungi, earthworms, insects and other fauna.

Functions of organic matter :

- Improves soil structure and porosity
- Enhances nutrient and water – holding capacity
- Supports soil microbial life and nutrient cycling

Even though it is a small fraction by volume, it has a disproportionate effect on soil health and productivity.

- **Living Organisms (component of organic matter) :**

A teaspoon of healthy soil can contain billions of microorganisms including :

- **Bacteria :** Fix nitrogen, decompose organic matter
- **Fungi :** Form symbiotic relationships (e.g., mycorrhizae)
- **Earthworms :** Aerate soil and recycle nutrients

These biota regulate nutrient cycling, disease suppression and soil structure.

Reference :

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5. Write a note on legislation aspects of environmental pollution?

Environmental pollution legislation refers to the legal frameworks, regulations, policies and enforcement mechanisms established by governments to control, reduce and prevent

pollution of air, water, soil and ecosystems. These laws aim to ensure sustainable development, protect public health, conserve biodiversity and maintain ecological balance.
Legislation is essential because environmental degradation — if left unchecked—can cause irreversible damage to ecosystems and human health and impose long-term economic burdens.

1. Objectives of Environmental Legislation

- Regulate emission and discharge of pollutants
- Establish environmental standards and guidelines
- Promote sustainable development and cleaner technologies
- Penalize polluters and compensate victims
- Encourage public participation and environmental awareness

2. Key Legislative Instruments (International & National)

International Conventions & Agreements

- **Stockholm Declaration (1972) :** The first major international document that recognized the human right to a healthy environment and laid the foundation for global environmental law.
- **Basel Convention (1989) :** Regulates the transboundary movement and disposal of hazardous wastes.
- **Kyoto Protocol (1997) and Paris Agreement (2015) :** Legally binding international efforts to reduce greenhouse gas emissions and combat climate change.

US Example – United States Environmental Laws

- **Clean Air Act (1970) :** Controls air emissions from stationary and mobile sources and authorizes the Environmental Protection Agency (EPA) to set National Ambient Air Quality Standards.
- **Clean Water Act (1972) :** Regulates discharges into U.S. waters and enforces water quality standards.
- **Resource Conservation and Recovery Act (RCRA, 1976) :** Governs the disposal of solid and hazardous waste.

3. Implementation and Enforcement :

Effective legislation requires :

- Clear regulatory institutions (e.g., EPA, Pollution Control Boards)
- Monitoring and data collection systems
- Penalties and fines for non-compliance
- Public access to environmental justice (e.g., National Green Tribunals, Environmental Courts).
- Regular updates to environmental laws in response to emerging pollutants (e.g., microplastics, PFAS).

4. Challenges in Environmental Legislation

- Lack of enforcement in developing countries
- Corruption and political interference
- Inadequate funding and technical capacity
- Weak public participation and awareness
- Difficulty regulating diffuse sources like agricultural runoff and vehicular pollution

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2021 SHORT QUESTIONS

1. Discuss two impacts of chemicals on buildings?

See question no. 02 part # 09 of 2017

2. What are different types of aflatoxin?

Aflatoxins are a group of highly toxic and carcinogenic compounds produced by certain molds, particularly Aspergillus flavus and Aspergillus parasiticus. These molds commonly contaminate food and feed crops especially in warm and humid climates. There are several structurally related aflatoxins, classified into major and minor types based on their fluorescence properties (blue or green) under UV light and their chemical structures.

Major Types of Aflatoxins include :

- Aflatoxin B1 (AFB1)
- Aflatoxin B2 (AFB2)
- Aflatoxin G1 (AFG1)
- Aflatoxin G2 (AFG2)
- Aflatoxin M1 (AFM1)
- Aflatoxin M2 (AFM2)

Other Rare Aflatoxins :

- Aflatoxin P1
- Aflatoxin Q1

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3. What is significance of soil pH?

The significance of soil pH is profound in agriculture, environmental science and soil health management because it influences nearly every biological, chemical and physical property of the soil.

Significance of Soil pH

- **Nutrient Availability :** Soil pH directly affects the solubility of essential nutrients and their availability to plants.
    - At low pH (acidic soils), nutrients like phosphorus , calcium and magnesium become less available, while toxic metals like aluminum (Al^{3+}) and manganese (Mn^{2+}) may increase to harmful levels.
    - At high pH (alkaline soils), micronutrients like iron , manganese , zinc and copper become deficient due to precipitation or adsorption.
- **Microbial Activity :** Soil pH controls the diversity and activity of soil microorganisms involved in organic matter decomposition, nitrogen fixation and nutrient cycling. Neutral to slightly acidic pH (6.0–7.0) is optimal for most microbial processes.
- **Toxicity and Metal Solubility :** Acidic soils (< pH 5.5) increase the solubility of toxic elements like aluminum , iron and manganese which can damage root systems. Alkaline soils (> pH 8) may lead to salinity issues and reduced micronutrient availability.

References :

- Brady, N.C., & Weil, R.R. (2016). The Nature and Properties of Soils (15th ed.). Pearson.
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4. Why soil can hold only cations?

Soils can hold primarily cations (positively charged ions) because of the negatively charged nature of most soil colloids particularly clay minerals and organic matter (humus). This electrostatic attraction allows soils to adsorb and retain cations on particle surfaces through a process called cation exchange capacity (CEC).

- **Why Cations Are Attracted and Retained ?** Because like charges repel and opposite charges attract, the negatively charged soil colloids electrostatically attract and hold positively charged ions (cations) such as :
    - Calcium ($Ca2+$)
    - Magnesium ($Mg2+$)
    - Potassium ($K+$)
    - Ammonium ($NH4+$)
    - Hydrogen ($H+$)
    - Aluminum ($Al3+$)

These cations are adsorbed onto colloid surfaces but are exchangeable — meaning they can be replaced by other cations from the soil solution.

- **Anions Are Less Retained or Leached :** Negatively charged ions (anions) such as:
    - Nitrate ($NO3^-$)
    - Chloride (Cl^-)
    - Sulfate ($SO4^{2-}$)
    - Phosphate ($H2PO4^-$)

are repelled by negatively charged colloid surfaces. As a result they remain in the soil solution and are easily leached away with water percolation.

References :

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- Foth, H.D. (1990). Fundamentals of Soil Science (8th ed.). John Wiley & Sons.
- USDA Soil Survey Manual, Chapter 6 (2017): Soil Chemical Properties.

5. How soil is holding water molecules?

Soil holds water molecules through a combination of physical and physicochemical forces involving adhesion, cohesion, capillarity and surface tension within the soil's solid matrix. These interactions determine water availability to plants, soil aeration and microbial activity.

Forces Responsible for Water Retention

- **Adhesion :** Water molecules are attracted to charged surfaces of soil particles (especially clay and organic matter). This forms a thin film of tightly held water around the particles.
- **Cohesion :** Water molecules are polar and form hydrogen bonds with each other, allowing them to stick together and be retained in pore spaces.
- **Capillary Action :** Water rises in small pores due to surface tension and the adhesive-cohesive properties. This allows water to be held against gravity.
  - The smaller the pore, the greater the capillary force.
  - Finer – textured soils (e.g., clays) hold more water than sandy soils, though less of it may be plant-available.

Reference :

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6. Enlist micronutrients in soil?

Micronutrients in soil are essential elements required by plants in very small (micro) quantities, typically less than 100 ppm (parts per million). Despite their small amounts, they are crucial for various physiological and biochemical processes in plants such as enzyme activation, photosynthesis, nitrogen fixation and reproductive development.

List of Essential Micronutrients in Soil :

Here are the eight primary essential micronutrients found in soil and needed by plants :

1) Iron (Fe)
2) Manganese (Mn)
3) Zinc (Zn)
4) Copper (Cu)
5) Boron (B)
6) Molybdenum (Mo)
7) Chlorine (Cl)
8) Nickel (Ni)

References :

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- Marschner, H. (2012). Marschner's Mineral Nutrition of Higher Plants (3rd ed.). Academic Press.

7. **Describe the principle of UV/VIS spectroscopy?**

   See question no. 02 part # 07 of 2020

8. **Write down any two principles of green chemistry?**

   See question no. 02 part # 10 of 2017

9. **Write any two contributions of AGROCHRMICALS in soil pollution?**

Agrochemicals, including fertilizers, pesticides, herbicides, and fungicides, play an essential role in modern agriculture by enhancing crop production. However, their excessive or improper use can contribute significantly to soil pollution.

1. Nutrient Imbalance and Soil Acidification Due to Fertilizer Overuse :

Excessive use of chemical fertilizers, especially nitrogen, phosphorus and potassium can lead to nutrient imbalances and soil acidification. The primary impacts are :

- **Soil Acidification :** High levels of ammonium – based fertilizers (e.g., ammonium nitrate, urea) release hydrogen ions (H^+) when they are transformed in the soil, which lowers soil pH, making it more acidic.
- **Nutrient Imbalance :** When fertilizers are over – applied, specific nutrients can accumulate in the soil while others become deficient, disrupting the balance needed for optimal plant growth.

2. Pesticide and Herbicide Residues Contaminating Soil and Water :

Pesticides and herbicides, when applied in excess or improperly, can remain in the soil for extended periods, leading to the following impacts:

- **Persistence and Bioaccumulation :** Many chemical pesticides and herbicides, such as organochlorine compounds (e.g., DDT), persist in soil for long periods. Over time, these chemicals can bioaccumulate in the soil and enter the food chain through plants and animals.
- **Soil Microbial Disruption :** Pesticides can disrupt soil microbial communities, reducing biodiversity and affecting important processes like nitrogen fixation, organic matter decomposition and nutrient cycling.

3. Contamination of Soil with Heavy Metals from Pesticides and Fungicides :

Certain pesticides and fungicides contain heavy metals such as lead , arsenic and cadmium which can be toxic to both plants and microorganisms in the soil.

- **Heavy Metal Accumulation :** Pesticides such as copper – based fungicides and certain insecticides contain heavy metals that can accumulate in the soil over time. These metals bind to soil particles and are less likely to be degraded or leached away.
- **Toxicity to Soil Organisms :** Heavy metals also affect soil organisms, particularly beneficial microbes, worms and fungi that are essential for nutrient cycling and soil aeration.

Reference :

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10. What is AAS? Which type of pollutants is analyzed by AAS?

Atomic Absorption Spectroscopy (AAS) is an analytical technique used for the determination of elements in various samples by measuring the absorption of light (electromagnetic radiation) by free atoms in the ground state. It is particularly effective for detecting metals and some metalloids, offering high sensitivity and precision.

Principle of AAS : In AAS, a sample is first atomized, typically in a flame or graphite furnace. A light beam of a specific wavelength is passed through the sample and the amount of light absorbed by the atoms in the sample is measured. This absorption is proportional to the concentration of the analyte (element) in the sample.

Types of Pollutants Analyzed by AAS :

AAS is commonly used to analyze **metallic pollutants** in a variety of environmental and biological samples. These pollutants can come from industrial activities, agricultural practices, or natural processes. Here are the main types of pollutants that AAS can analyze :

- Heavy Metals in Water
- Soil Pollutants
- Airborne Pollutants (Particulate Matter)
- Food Contamination
- Industrial Effluent

References :

- Harris, D.C. (2015). *Quantitative Chemical Analysis* (9th ed.). W.H. Freeman.
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LONG QUESTIONS (2021)

1. Discuss INDUSTRY as a source of soil pollution?

Industrial activities are one of the most significant and persistent sources of soil pollution globally. They introduce a wide variety of toxic substances into the soil, either directly through disposal of industrial waste or indirectly via atmospheric deposition and wastewater discharge.

❖ **Direct Discharge of Industrial Waste into Soil :**

Many industries dispose of their solid or semi – solid wastes, including sludges, tailings and by – products, directly into landfills or open ground without adequate treatment.

Examples and Impacts :

- Heavy metal pollution (e.g., lead, mercury, cadmium, chromium) from battery manufacturing, metal plating, and mining industries can persist in soil for decades and bioaccumulate in crops and humans.
- Persistent organic pollutants (POPs) such as dioxins and polychlorinated biphenyls (PCBs) from chemical and electronic industries are non – biodegradable and toxic.
- Dye and pigment industries release aromatic amines and other synthetic compounds that are mutagenic and carcinogenic.

❖ **Industrial Effluents and Wastewater Contamination :**

Industries frequently release untreated or partially treated wastewater onto nearby soils and fields, often for irrigation or disposal, particularly in developing countries.

Examples and Impacts :

- Textile industry effluents contain heavy metals (like chromium from dyeing processes), surfactants and colorants that deteriorate soil structure and harm microbial communities.
- Tannery effluents rich in hexavalent chromium (Cr^{6+}) disrupt nitrogen cycling, inhibit plant growth and are carcinogenic to humans.
- Pharmaceutical industry waste introduces antibiotic residues into soil, promoting antimicrobial resistance in microbes.

❖ **Atmospheric Deposition from Industrial Emissions :**

Industrial smokestacks and chimneys emit airborne particulates, volatile chemicals, and acid – forming gases which settle onto land through dry or wet deposition.

Examples and Impacts :

- Acid rain from sulfur dioxide (SO_2) and nitrogen oxides (NO_x) emissions reduces soil pH, leading to leaching of essential nutrients like calcium and magnesium.
- Fly ash from coal – fired power plants settles on topsoil, increasing concentrations of heavy metals and polycyclic aromatic hydrocarbons (PAHs).
- Soot and particulate matter may blanket agricultural soils, reducing photosynthesis and altering microbial dynamics.

❖ **Accidental Spills and Leakage :**

Leaks from industrial pipelines, storage tanks, and accidents (e.g., oil spills, chemical spills) contaminate surrounding soils.

Examples and Impacts :

- Oil spills from petroleum refineries or pipelines lead to hydrocarbon contamination, reducing soil porosity and oxygen availability.
- Chemical plant accidents may release acids, bases, solvents and nitrates that drastically alter soil chemistry and kill beneficial soil biota.
- Land near industrial zones is often contaminated with complex mixtures of pollutants, making remediation difficult.

❖ **Use of Industrial By – products in Agriculture :**

Certain industries supply by-products like fly ash, phosphogypsum or slag as soil conditioners or fertilizers. Improper use or overuse leads to soil contamination.

Examples and Impacts :

- Fly ash may contain arsenic, boron and selenium, which can accumulate in soil and crops.
- Phosphogypsum contains radioactive elements like radium – 226 and fluorides that can enter the food chain.
- Steel industry slag can raise soil pH excessively and introduce metals like vanadium and chromium.

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2. Briefly describe the composition of a typical soil?

See question no. 03 part # 04 of 2020

3. **Discuss principle and applications of HPLC in environmental monitoring?**

See question no. 03 part # 03 of 2017

4. **Explain the impacts of household chemicals on environment?**

Household chemicals – such as cleaning agents, detergents, disinfectants, air fresheners and pesticides – are commonly used to maintain hygiene, comfort and aesthetics in homes. However, improper use, storage or disposal of these chemicals can lead to serious environmental consequences.

1. Water Pollution from Domestic Drainage :

Many household chemicals end up in wastewater systems after being rinsed down sinks, bathtubs or toilets. These compounds are not always fully removed by conventional sewage treatment processes, leading to contamination of rivers, lakes and groundwater.

Examples and Effects :

- Phosphates from detergents promote eutrophication—a process in which water bodies become overly enriched with nutrients, leading to excessive algae growth and oxygen depletion.
- Triclosan, a common antibacterial agent in soaps, disrupts aquatic ecosystems by affecting algae and invertebrate reproduction.
- Surfactants from dishwashing liquids and shampoos can reduce surface tension in water, harming aquatic life like fish and amphibians.

2. Air Pollution from Volatile Organic Compounds (VOCs) :

Many household products contain volatile organic compounds (VOCs) such as formaldehyde, benzene, toluene and limonene. These chemicals are emitted as gases from certain solids or liquids and significantly contribute to indoor and outdoor air pollution.

Examples and Effects :

- VOCs can react with nitrogen oxides (NO_x) in the presence of sunlight to form ground – level ozone, a major component of smog.
- Indoor use of air fresheners, aerosol sprays and certain cleaners contributes to poor indoor air quality and respiratory irritation.
- Long-term exposure to VOCs has been associated with neurological effects, hormonal disruption and increased risk of cancer.

3. Soil Contamination through Improper Disposal :

When household chemicals are disposed of by pouring them down the drain, into backyard soils or dumping into municipal garbage, they can leach into the soil and degrade its quality.

Examples and Effects :

- Bleach and ammonia – based products disrupt soil microbial life, reducing the breakdown of organic matter and nutrient cycling.
- Batteries and old electronics contain heavy metals like lead, cadmium and mercury that leach into soil, affecting plant health and potentially entering the food chain.

- Pesticides used in home gardens or stored inappropriately can accumulate in soil and persist for long periods, affecting soil biodiversity.

4. Harm to Aquatic and Terrestrial Life :

Many chemical compounds found in household products are toxic to living organisms even at low concentrations.

Examples and Effects :

- Pesticides used indoors or around homes (e.g., ant sprays, mosquito repellents) can kill beneficial insects, contaminate bird food sources and wash into waterways.
- Chlorine – based bleaches and disinfectants can form chlorinated organics, which are toxic to fish and amphibians.
- Nonylphenol ethoxylates (from laundry detergents) mimic estrogen and disrupt hormonal systems in aquatic species, leading to reproductive issues.

5. Bioaccumulation and Persistent Organic Pollutants (POPs) :

Some compounds used in household products, like polychlorinated biphenyls (PCBs), dioxins and certain flame retardants, are classified as persistent organic pollutants.

Examples and Effects :

- These compounds resist degradation and accumulate in the fatty tissues of living organisms.
- POPs biomagnify up the food chain, meaning top predators (including humans) face the highest concentrations.
- They are linked to cancer, immune suppression and developmental issues in both wildlife and humans.

<u>References</u> :

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- Fatta, D., Papadopoulos, A., & Loizidou, M. (1999). "A study on the landfill leachate impact on soil properties." Environmental Geochemistry and Health, 21(2), 175–190.
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5. How POPS affect the environment?

Persistent Organic Pollutants (POPs) are a class of toxic chemical substances that are highly resistant to environmental degradation through chemical, biological and photolytic processes. Due to their persistence, they accumulate in the environment and pose serious threats to ecosystems and human health.

- POPs include substances such as :
  - Polychlorinated biphenyls (PCBs)
  - Dioxins and furans
  - Organochlorine pesticides (e.g., DDT, aldrin, dieldrin, heptachlor)
  - Polycyclic aromatic hydrocarbons (PAHs)
  - Perfluorinated compounds (e.g., PFOS, PFOA)
- They are characterized by :
  - High chemical stability
  - Low water solubility but high lipid solubility
  - Long – range atmospheric transport
  - High bioaccumulation and biomagnification potential

How POPs Affect the Environment :

1. **Contamination of Soil :** POPs enter soil through pesticide application, industrial discharge and atmospheric deposition.

Impacts :

- Persist for decades in soil due to their low degradation rates.
- Alter soil microbial communities and inhibit enzymatic activity.
- Reduce nitrogen fixation and nutrient cycling.

2. **Pollution of Water Bodies :** POPs enter aquatic ecosystems via runoff, wastewater and atmospheric deposition.

Impacts :

- Accumulate in sediments and aquatic organisms (e.g., fish, plankton).
- Disrupt aquatic food chains and reproductive cycles of aquatic fauna.
- Lead to hypoxic conditions by altering microbial populations and biochemical oxygen demand.

3. **Atmospheric Transport and Global Dispersion :** POPs are semi – volatile, allowing them to evaporate in warm regions and condense in cooler climates, a process known as the “grasshopper effect.”

Impacts :

- POPs released in industrialized regions are transported to the Arctic and Antarctic, contaminating pristine ecosystems.

- Accumulate in snow, ice and permafrost, affecting polar wildlife.

4. **Bioaccumulation and Biomagnification :** Because POPs are lipophilic (fat – soluble), they accumulate in the fatty tissues of living organisms and magnify up the food chain.

Impacts :

- Top predators (e.g., birds of prey, marine mammals, humans) are most affected.
- Adverse effects include hormonal disruption, immune suppression, reproductive issues and cancer.

5. **Adverse Effects on Wildlife :** POPs interfere with endocrine systems, reproduction and development in various animal species.

Examples :

- DDT causes eggshell thinning in birds like eagles and falcons, leading to population decline.
- PCBs impair thyroid function in fish and amphibians, affecting growth and metamorphosis.
- Dioxins affect immune system function in marine mammals and birds.

6. **Long – Term Ecosystem Disruption :** Even after use is banned, POPs continue to circulate in the environment due to their persistence.

Impacts :

- Chronic exposure causes subtle but long – lasting changes in species diversity, soil health and aquatic ecosystems.
- Delayed recovery of contaminated ecosystems due to the long half – life of POPs.

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2022 Short Questions :

1. What is HUMUS?

Humus is the stable, dark – colored, organic material that remains in soil after the majority of plant and animal residues have decomposed. It is composed of complex organic molecules, primarily formed from lignin**,** cellulose**,** proteins and other biopolymers that are transformed and stabilized through microbial action.

- ➢ It acts as a nutrient reservoir, especially for nitrogen, phosphorus and sulfur .
- ➢ It increases the water – holding capacity of soils.

Composition of Humus : Humus is not a single chemical compound but a mixture of many large, complex organic molecules, including :

- Humic substances, mainly :
  - Humic acids
  - Fulvic acids
  - Humin
- Non – humic substances like carbohydrates, fats, proteins, waxes and resins.

Formation of Humus (Humification Process) : Humus is formed via humification, which occurs in two main stages :

- Decomposition of plant and animal residues by bacteria, fungi, and other decomposers into simpler organic compounds (e.g., amino acids, sugars).
- Synthesis and polymerization of these compounds into stable humic substances through microbial activity and chemical reactions (e.g., Maillard reactions).

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2. How soil air is important for fertility?

Soil air is essential for maintaining soil fertility, as it directly affects the respiration of plant roots, microbial activity, nutrient availability and overall biological productivity of soil. Without adequate soil aeration, even nutrient – rich soils can become infertile.

- ➢ Soil air refers to the gaseous phase present in the soil pore spaces not occupied by water. It contains a mix of gases, including :
- Oxygen (O_2)
- Carbon dioxide (CO_2)
- Nitrogen (N_2), water vapor and trace gases.

Importance of Soil Air for Soil Fertility

- Root Respiration and Growth
- Microbial Activity and Organic Matter Decomposition
- Nitrification Process
- Redox Reactions and Nutrient Availability
- Prevention of Toxic Gases

References :

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3. **Suggest some control measures for WATER erosion.**

   See question no. 02 part # 05 of 2020

4. **What type of pollutants can be analyzed by AAS.**

   See question no. 02 part # 10 of 2021

5. **What's meant by AFLATOXINS?**

   See question no. 02 part # 05 of 2014

6. **How availability of Phosphorous is affected by soil pH?**

The availability of phosphorus (P) in soil is highly dependent on soil pH because pH controls the chemical form of phosphorus and its solubility. Improper pH levels can lead to phosphorus being fixed into unavailable forms, reducing its uptake by plants – even if phosphorus is present in sufficient total amounts.

Optimal pH Range for Phosphorus Availability : 6.0 – 7.0

In this range, phosphorus exists mostly as $H_2PO_4^-$ and HPO_4^{2-}, which are soluble and readily available for plant uptake.

Low pH (Acidic Soils: < 5.5) :

Phosphorus reacts with aluminum (Al^{3+}) and iron (Fe^{3+}) ions and forms insoluble aluminum and iron phosphates such as :

- $AlPO_4$ (Aluminum phosphate)
- $FePO_4$ (Iron phosphate)

These compounds are not available to plants.

High pH (Alkaline Soils: > 7.5) :

Phosphorus reacts with calcium (Ca^{2+}) to form calcium phosphates including :

- Dicalcium phosphate
- Hydroxyapatite ($Ca_5(PO_4)_3OH$)

These compounds have low solubility especially above pH 8.0.

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- Havlin, J. L., et al. (2014). *Soil Fertility and Fertilizers* (8th ed.). Pearson Education.

7. What is chemical speciation?

Chemical speciation refers to the identification and quantification of different chemical forms (species) of an element present in a sample, including its oxidation states, complexed forms, bound forms and free ions. It is essential because the toxicity, bioavailability, mobility and reactivity of an element often depend more on its chemical form than on its total concentration.

Techniques for Speciation Analysis

- Chromatography – based techniques (HPLC – ICP – MS and GC – MS)
- Electrochemical methods (Voltammetry for redox species (e.g., Fe^{2+}/Fe^{3+})
- Spectroscopic techniques (X-ray Absorption Spectroscopy (XAS) for oxidation state analysis

Applications of Chemical Speciation

1. **Environmental Chemistry** : Arsenic speciation in drinking water and Mercury speciation in fish and sediments
2. **Soil and Agriculture** : Determining bioavailable nutrients like Zn^{2+}, Cu^{2+}, PO_4^{3-} and assessing heavy metal mobility

References :

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8. Propose the reclamation of Basic soil.

Reclamation of basic (alkaline or sodic) soils is a multidimensional process involving chemical, biological and hydrological interventions. Among these, gypsum amendment and leaching form the backbone of effective treatment, while organic inputs and salt – tolerant crops enhance the long – term fertility and sustainability of the reclaimed soil.

Objectives of Reclamation :

- Replace excess Na^+ ions with Ca^{2+}
- Improve soil structure, infiltration and drainage
- Lower pH and electrical conductivity
- Enhance microbial activity and nutrient availability

Reclamation Strategies :

- Chemical Amendments (Gypsum Application)
- Organic Matter Addition
- Leaching and Drainage
- Use of Sulfur or Sulfuric Acid (where gypsum is not effective)
- Crop Selection and Agronomic Practices

References :

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- Qadir, M., & Oster, J. D. (2004). *Crop and irrigation management strategies for saline-sodic soils and waters*. Soil Use and Management, 20(1), 4–14.
- Maas, E. V., & Hoffman, G. J. (1977). *Crop salt tolerance — current assessment*. Journal of the Irrigation and Drainage Division.

9. What in principle of UV spectroscopy?

See question no. 02 part # 07 of 2020

10. Mention the salient feature of HPLC.

High – Performance Liquid Chromatography (HPLC) is a sophisticated and powerful analytical technique used for the separation, identification and quantification of components in complex mixtures. One of its most notable features is high resolution which allows for the effective separation of compounds that are structurally similar, due to the efficiency of the stationary and mobile phases involved. This results in sharp and narrow peaks, facilitating precise analysis.

- Another major strength of HPLC is its high sensitivity, which enables detection of substances in the nanogram to picogram range, especially when using sensitive detectors like UV – Vis, fluorescence or mass spectrometry (MS).
- HPLC is versatile, accommodating a wide range of chemical types through various modes, such as reverse – phase, normal – phase, ion – exchange and size – exclusion chromatography.
- Importantly, HPLC can analyze thermally labile compounds that would degrade under the high temperatures required for gas chromatography, making it ideal for biological or pharmaceutical substances.
- Furthermore, HPLC is fully automatable, supporting automatic sample injection, data acquisition and processing.

References :

- Snyder, L. R., Kirkland, J. J., & Dolan, J. W. (2010). *Introduction to Modern Liquid Chromatography* (3rd ed.). Wiley.

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- Meyer, V. R. (2010). *Practical High-Performance Liquid Chromatography* (5th ed.). Wiley.
- Kazakevich, Y., & Lobrutto, R. (2007). *HPLC for Pharmaceutical Scientists*. Wiley.

11. What are primary & secondary minerals of soil?

Primary and secondary minerals in soil refer to two distinct categories of minerals based on their origin and transformation within the soil environment. These minerals play critical roles in determining soil properties such as fertility, texture, water retention, and nutrient availability.

Primary Minerals : Primary minerals are those that are formed directly from the solidification (crystallization) of molten magma and are found in igneous rocks. They have not been chemically altered since their formation and are typically inherited by soils from the parent rock material through physical weathering.

Key Characteristics :

- Generally resistant to weathering.
- Coarse – grained and crystalline in structure.

Common Examples of Primary Minerals :

i. Quartz (SiO_2)
ii. Feldspars (orthoclase, plagioclase) etc.

Secondary Minerals : Secondary minerals are formed from the chemical weathering, hydrolysis, or alteration of primary minerals. They develop in the soil environment and are typically finer, more reactive, and more important for soil fertility.

Key Characteristics :

- Usually fine – grained (clay – size particles).
- More reactive and less crystalline than primary minerals.

Common Types of Secondary Minerals :

i. Clay Minerals (Phyllosilicates)
ii. Oxides and Hydroxides etc.

<u>Reference</u> :

- Brady, N.C. & Weil, R.R. (2016). “The Nature and Properties of Soils”, 15th Ed., Pearson.
- Foth, H.D. (1990). “Fundamentals of Soil Science”, 8th Ed., Wiley.
- Bohn, H.L., McNeal, B.L., & O’Connor, G.A. (2001). “Soil Chemistry”, 3rd Ed., Wiley.

12. What are limitations of GC?

Gas Chromatography (GC) is a highly efficient and widely used technique for separating and analyzing volatile compounds. However, like all analytical methods, it has limitations that can restrict its use in certain applications.

- Analyte Must Be Volatile
- Analyte must be Thermally Stable
- Limited to Low Molecular Weight Compounds
- Derivatization May Be Required
- Limited Applicability for Aqueous Samples
- Sample Preparation Can Be Intensive
- Cannot Analyze Ionic or Inorganic Compounds

References :

- Poole, C. F. (2012). *Gas Chromatography* (2nd ed.). Elsevier.
- McNair, H. M., & Miller, J. M. (2009). *Basic Gas Chromatography* (2nd ed.). Wiley.
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- Dettmer – Wilde, K., & Engewald, W. (2014). *Practical Gas Chromatography: A Comprehensive Reference*. Springer.

13. What is DIRTY DOZEN?

The **"Dirty Dozen"** is a term commonly used in environmental science and public health to describe two distinct but related lists :

- **Dirty Dozen of Persistent Organic Pollutants (POPs) – Stockholm Convention :**

  The original Dirty Dozen refers to twelve highly toxic chemicals identified by the Stockholm Convention on Persistent Organic Pollutants (2001).

The Original Dirty Dozen POPs :

- **Pesticides :**
  1) Aldrin
  2) Chlordane
  3) DDT
  4) Dieldrin
  5) Endrin
  6) Heptachlor
  7) Hexachlorobenzene (also an industrial chemical)
  8) Mirex
  9) Toxaphene
- **Industrial Chemicals :**
  10) Polychlorinated biphenyls (PCBs)
- **Industrial By – products :**
  11) Dioxins
  12) Furans

These chemicals are banned or strictly restricted under the **Stockholm Convention**, a global treaty aimed at protecting human health and the environment.

- **Dirty Dozen List of Produce – Environmental Working Group (EWG)**

In a food safety and public health context, the term "Dirty Dozen" is also used annually by the Environmental Working Group (EWG) to describe 12 fruits and vegetables most contaminated with pesticide residues. This is based on data from the U.S. Department of Agriculture (USDA).

Example (2024 list – may vary by year) :

1) Strawberries
2) Spinach
3) Kale, Collard & Mustard Greens
4) Grapes
5) Peaches
6) Pears
7) Nectarines
8) Apples
9) Bell & Hot Peppers
10) Cherries
11) Blueberries
12) Green Beans

References :

- Stockholm Convention on Persistent Organic Pollutants. (2001). https://www.pops.int
- UNEP. (2005). *Ridding the World of POPs: A Guide to the Stockholm Convention on Persistent Organic Pollutants*. United Nations Environment Programme.
- WHO. (2010). *Health risks of persistent organic pollutants*. https://www.who.int
- Environmental Working Group (EWG). (2024). *Dirty Dozen List*. https://www.ewg.org/foodnews
- USDA Pesticide Data Program (PDP). https://www.ams.usda.gov/pdp

14. Where is the word SOIL from?

The **word "soil"** originates from the Latin word **solum**, which means **"ground" or "bottom"**. Over time, the term evolved through various languages and meanings.

Etymological Origin of "Soil" :

- **Latin :** *Solum* — meaning *bottom*, *foundation*, or *ground*.
- **Old French (12th century) :** Soil *or* sol — *referring to* ground *or* territory.
- **Middle English (13th–14th century) :** The word soile referred to the upper layer of earth, including meanings related to land and dirt.

The term eventually took on its modern meaning in English, referring to the top layer of the earth where plants grow, composed of organic matter, minerals, gases, liquids and organisms.

References :

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15. What is importance of clay in nutrient holding capacity of soil?

Clay plays a crucial role in the nutrient – holding capacity of soil due to its high surface area, small particle size and negative charge, which enable it to retain and exchange essential nutrients for plant growth.

- Clay is essential for holding and exchanging nutrients, especially cations.
- It enhances soil fertility by reducing nutrient loss, storing micronutrients and buffering pH.
- Soils rich in clay generally require less frequent fertilization than sandy soils due to better nutrient retention.

References :

- Brady, N. C., & Weil, R. R. (2016). *The Nature and Properties of Soils* (15th ed.). Pearson.
- Foth, H. D. (1990). *Fundamentals of Soil Science* (8th ed.). Wiley.
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- Tisdale, S. L., Nelson, W. L., Beaton, J. D., & Havlin, J. L. (1993). *Soil Fertility and Fertilizers* (5th ed.). Macmillan.

Long questions (2022)

1. Describe the LIQUID and GASEOUS components of soil.

Soil is a complex mixture of solid, liquid and gaseous components, each playing a vital role in supporting plant growth, regulating water and nutrient cycles and maintaining soil structure.

❖ Liquid Component of Soil :

The liquid component of soil primarily consists of soil water, which is crucial for plant growth, microbial activity, and nutrient transport. Soil water is found in the pore spaces between soil particles, and it can exist in various forms depending on its movement and availability.

a. Soil Water :

Soil water is the moisture present in the soil and can be classified into different types based on its availability to plants and its movement through the soil. These types are :

1. **Gravitational Water :** This is the water that drains out of the soil due to gravity and is not available to plants. It tends to flow through the larger soil pores and is usually found just after heavy rainfall or irrigation.
2. **Capillary Water :** This water is held within the microprobes of the soil, and it is the most accessible form of water for plants. It is held by capillary forces and can be taken up by plant roots for hydration and nutrient absorption.
3. **Hygroscopic Water :** Hygroscopic water is the water that adheres to soil particles tightly, making it unavailable to plants. It is found in very dry soils and is held by strong molecular attraction.
4. **Interstitial Water (Soil Solution) :** This is the water in the soil that contains dissolved nutrients, minerals and gases forming the soil solution. It is essential for nutrient transport and its composition directly affects the cation exchange capacity (CEC) and soil fertility.

b. Role of Soil Water :

- **Nutrient Transport** : Water dissolves nutrients from the soil minerals, allowing them to be absorbed by plant roots.
- **Microbial Activity** : Soil organisms, including bacteria and fungi, rely on water for their metabolic functions and reproduction.
- **Plant Growth** : Water is crucial for photosynthesis and the transport of sugars and other nutrients in plants.

❖ Gaseous Component of Soil :

The gaseous component of soil mainly consists of air, which occupies the pore spaces in the soil along with water. The soil atmosphere plays a key role in supporting root respiration, microbial activity and maintaining soil health. The composition of soil air differs from atmospheric air due to biological processes occurring within the soil.

a. Soil Air :

Soil air is composed of gases that exist in the pore spaces of soil. The main components of soil air are :

1. **Oxygen (O_2) :** Oxygen is essential for root respiration and microbial activity in the soil. Soil oxygen levels are highest in well – aerated soils and are crucial for the growth of plant roots and aerobic soil organisms. Low oxygen levels in compacted soils can lead to anaerobic conditions, which are harmful to plant roots.
2. **Carbon Dioxide (CO_2) :** Carbon dioxide is produced during the respiration of plant roots and soil microorganisms. High concentrations of CO_2 can lead to soil acidification and are a key factor in regulating soil pH and nutrient availability.
3. **Nitrogen (N_2) :** Nitrogen gas (N_2) is the most abundant gas in soil air, accounting for approximately 78% of the soil atmosphere. While nitrogen is abundant, it must be converted into usable forms, such as nitrate (NO_3^-) or ammonium (NH_4^+), through microbial processes like nitrogen fixation.
4. **Methane (CH_4) :** Methane is an important trace gas produced in soils under anaerobic conditions, particularly in waterlogged soils. It is generated by methanogenic archaea during the decomposition of organic matter in the absence of oxygen.
5. **Other Trace Gases :** Other gases in the soil atmosphere include hydrogen sulfide (H_2S) and ammonia (NH_3), which are typically found in areas with anaerobic conditions. These gases are products of microbial processes, particularly in wetlands or waterlogged soils.

b. Role of Soil Air

- **Respiration** : Soil air is crucial for the **respiration of plant roots** and soil microorganisms, which consume oxygen and release carbon dioxide.
- **Microbial Processes** : The composition of soil gases, such as oxygen and carbon dioxide, affects microbial processes such as nitrification, denitrification and methanogenesis.
- **Nutrient Cycling** : The gaseous exchange in the soil supports nutrient cycles like the nitrogen cycle and carbon cycle, influencing soil fertility.

References :

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- Foth, H. D. (1990). *Fundamentals of Soil Science* (8th ed.). Wiley.
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2. How MINING is contributing in soil pollution?

See question no. 03 part # 05 of 2017

3. Write down the salient features of POPs?

See question no. 03 part # 02 of 2014

4. Describe any six principles of GREEN chemistry.

See question no. 03 part # 04 of 2019

5. Describe the importance of SPECTROSCOPIC techniques in environmental monitoring.

Spectroscopic techniques play a crucial role in environmental monitoring by providing accurate, reliable, and non-destructive methods for analyzing a wide range of environmental samples, including air, water, soil and sediments. These techniques are invaluable in assessing the quality of the environment, detecting pollutants and monitoring changes in ecosystems. Their sensitivity, selectivity and versatility make them ideal for a variety of environmental applications.

- Spectroscopy refers to the study of the interaction between electromagnetic radiation and matter, where the absorption, emission, or scattering of light is measured. Spectroscopic techniques can be categorized based on the type of electromagnetic radiation used, such as :
  - Ultraviolet – Visible (UV – Vis) Spectroscopy
  - Infrared (IR) Spectroscopy
  - Atomic Absorption Spectroscopy (AAS)
  - Fluorescence Spectroscopy
  - Nuclear Magnetic Resonance (NMR) Spectroscopy
  - Raman Spectroscopy
  - Mass Spectrometry (MS)

These methods can be applied to identify and quantify pollutants and trace elements in various environmental samples, providing critical data for regulatory compliance, ecological studies, and pollution control.

Applications of Spectroscopic Techniques in Environmental Monitoring

a. Water Quality Monitoring : Water quality monitoring is one of the most important applications of spectroscopic techniques, as clean water is essential for human health, agriculture, and aquatic ecosystems.

- **UV – Vis Spectroscopy** is frequently used to monitor chemical oxygen demand (COD), biochemical oxygen demand (BOD), and pollutants like pesticides, heavy metals and organic compounds. The absorption spectra of these compounds in the UV and visible regions enable quick detection.
- **Fluorescence Spectroscopy** is effective in detecting organic pollutants, including polycyclic aromatic hydrocarbons (PAHs) and aromatic hydrocarbons in water, as they exhibit characteristic fluorescence when exposed to UV light.
- **Atomic Absorption Spectroscopy (AAS)** is widely used for measuring trace metals such as lead, mercury, cadmium and arsenic in water samples. AAS provides high sensitivity and is able to detect metals at parts per billion (ppb) levels.

b. Air Quality Monitoring : Air pollution is a global issue, and spectroscopic techniques are pivotal for measuring pollutants in the atmosphere, such as particulate matter, nitrogen oxides, sulfur dioxide, ozone and volatile organic compounds.

- **UV-Vis Spectroscopy** is widely used to determine ozone levels in the atmosphere. The absorption peak of ozone at 254 nm allows for precise quantification in real – time air monitoring.
- **Fourier Transform Infrared (FTIR) Spectroscopy** allows the identification and quantification of a wide range of gaseous pollutants by analyzing their vibrational spectra. FTIR is highly effective for monitoring VOCs, CO_2, NO_2, and CH_4 in ambient air.
- **Raman Spectroscopy** is used to monitor particulate matter in the air, such as black carbon and dust. It is particularly beneficial for detecting nano – scale particles, which can have detrimental health effects.

c. Soil and Sediment Monitoring : Soil health is vital for agriculture and ecosystem sustainability. Spectroscopic techniques are used to detect soil contaminants and assess soil properties.

- **X-ray Fluorescence (XRF) Spectroscopy** is widely used for analyzing heavy metals and trace elements in soils, such as lead, arsenic, cadmium and zinc . XRF provides a non – destructive, rapid analysis of elemental concentrations without the need for sample digestion.
- **Near Infrared (NIR) Spectroscopy** is used for analyzing soil properties, including moisture content, organic matter, and nutrient availability. It works by analyzing the absorption of near-infrared light by soil samples, providing information on their chemical composition.

d. Ecosystem Monitoring : Spectroscopy is also employed to monitor the health of entire ecosystems, including forests, wetlands and agricultural fields, by evaluating the levels of various environmental parameters.

- **Chlorophyll Fluorescence Spectroscopy** is used to assess plant health and stress in ecosystems. By measuring the fluorescence emitted by chlorophyll under specific light conditions, it is possible to monitor photosynthetic activity and detect the effects of environmental stressors such as drought, pollution or temperature changes.
- **Multispectral and Hyperspectral Imaging** (related to remote sensing) are spectroscopic techniques that provide high-resolution spectral data across many wavelengths. These are used for land use monitoring, vegetation health and pollution detection from satellites or drones.

Advantages of Spectroscopic Techniques in Environmental Monitoring :

1. **Non-destructive** : Spectroscopic methods do not alter the sample, allowing for repeated analysis of the same sample.
2. **High Sensitivity** : Many spectroscopic techniques, like AAS and fluorescence spectroscopy, can detect pollutants at very low concentrations (ppb or even ppt levels).
3. **Real-time Monitoring** : Spectroscopic techniques can be employed in situ for real – time environmental monitoring, which is particularly valuable for air quality assessments.
4. **Wide Range of Analytes** : Spectroscopic techniques can be used to analyze a broad spectrum of pollutants, including metals, organic compounds, gases and particulates.
5. **High Throughput** : These techniques can process a large number of samples quickly, which is essential for large-scale environmental monitoring.

References :

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- Shifrina, T. A., & Goryacheva, I. Y. (2015). *Application of Atomic Absorption Spectrometry in Water Analysis*. *Journal of Analytical Chemistry*, 70(9), 847-860.
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- Esposito, G., & Amici, M. (2006). *Raman Spectroscopy for Airborne Particulate Matter Detection*. *Analytical Chemistry*, 78(1), 232-237.
- Shepherd, K. D., & Walsh, M. G. (2002). *Development of Reflectance Spectroscopic Methods for Soil Analysis*. *Soil Science Society of America Journal*, 66(3), 488-496.

2023 Short Questions :

1. **Mention the percentage of organic and inorganic components of a typical fertile soil.**

- **Inorganic Components (Mineral Matter) :**
    - **Percentage** : 45%
    - This includes the mineral particles derived from the weathering of rocks, including sand, silt and clay.
- **Organic Components (Organic Matter) :**
    - **Percentage** : 5%
    - This consists of decomposed plant and animal material, including humus, microorganisms, and decaying organic residues.

References :

- Brady, N. C., & Weil, R. R. (2016). *The Nature and Properties of Soils* (15th ed.). Pearson.
- Sposito, G. (2008). *The Chemistry of Soils*. Oxford University Press.

2. **What is HUMUS?**

See question no. 01 part # 01 of 2022

3. **Which analytical technique is suitable for analysis of heavy metals in the environment?**

Heavy metals are toxic at elevated concentrations and pose significant environmental and health risks. Detecting and quantifying these metals in environmental samples such as soil, water and air is critical for monitoring and controlling pollution. Various analytical techniques are employed for the accurate and sensitive analysis of heavy metals.

- ICP – MS is the most comprehensive technique, offering excellent sensitivity, multi – element capability and precision making it ideal for a wide variety of environmental samples.
- AAS and GFAAS are cost – effective, suitable for routine analysis with high sensitivity for specific heavy metals.
- XRF is perfect for non – destructive, rapid field – based analysis, especially for large – scale environmental monitoring.
- Electrochemical methods are useful for in situ analysis and portable field measurements of certain metals.

References :

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4. What are main types of soil erosion?

Soil erosion is the process of the removal of the topsoil layer due to various natural and anthropogenic forces. It is a significant environmental problem as it depletes the fertile layer of the soil, leading to reduced agricultural productivity, loss of nutrients and desertification. The main types of soil erosion are :

i. Water Erosion (Sheet Erosion, Rill Erosion and Gully Erosion)
ii. Wind Erosion
iii. Mass Erosion (or Gravity Erosion) (Landslides, Mudslides and Soil Creep)
iv. Coastal Erosion
v. Ice Erosion (Glacial Erosion)

References :

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- Bennett, M. R., & Glasser, N. F. (2009). "Glacial Geology: Ice Sheets and Landforms." *Wiley-Blackwell*.

5. Mention the names of fungi which are producing 'AFLATOXINS?

Aflatoxins are a group of highly toxic and carcinogenic secondary metabolites produced by certain species of fungi, particularly *Aspergillus*, which contaminate agricultural crops such as peanuts, maize, and tree nuts. The primary species known for producing aflatoxins include :

- The primary fungi responsible for aflatoxin production are members of the *Aspergillus* genus, with *Aspergillus flavus* and *Aspergillus parasiticus* being the most well – known and studied species.
- Other species such as Aspergillus nomius and Aspergillus sojae also produce aflatoxins, though less frequently.

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- Yabe, K., & Toyosato, T. (2007). "Molecular Biology of Aflatoxin Biosynthesis." *International Journal of Food Microbiology*, 119(1-2), 12-16.

6. **Enlist micronutrients of plants.**

See question no. 02 part # 06 of 2021

7. **What is littering?**

Littering refers to the act of discarding waste or trash in public or private areas without proper disposal, such as throwing items like paper, plastic, bottles or food wrappers onto streets, parks, or other places where waste should not be present. It is an environmentally harmful activity that contributes to pollution, especially in urban and natural environments. Littering not only impacts the aesthetic value of the environment but also has serious consequences for wildlife, soil, water quality, and human health.

Types of Litter :

Litter can be broadly categorized into the following types :

- **Solid Waste** : Includes items like plastic bottles, cans and cigarette butts etc.
- **Biodegradable Waste** : Includes food scraps, paper products and organic materials etc.
- **Hazardous Waste** : Includes items like chemicals, batteries and medical waste etc.
- **E-waste** : Electronic items like batteries, phones and computers

Impacts of Littering

1. **Environmental Impact** : Littering contributes significantly to pollution in land, water and air. Plastics, for example, take hundreds of years to decompose and often end up in oceans, harming marine life.
2. **Economic Impact** : Areas that are littered often see a decline in property values as they become less desirable for residents and visitors.
3. **Health Impact** : Litter, particularly food scraps and other organic matter, can attract pests like rats and mosquitoes, which can spread diseases.

References :

- Hernandez, M. A., & Salazar, R. (2015). "The Impact of Littering on the Environment and Public Health." *Environmental Science and Technology*, 49(12), 7234-7241.
- United Nations Environment Programme (UNEP). (2017). "Single-Use Plastics: A Roadmap for Sustainability." UNEP, Nairobi.
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8. **Enlist any two sources of soil pollution.**

- **Industrial Activities :** Factories release pollutants such as heavy metals (lead, mercury, arsenic), petroleum hydrocarbons and chemical solvents, which contaminate soil through improper waste disposal, spills and leaks.
- **Agricultural Chemicals :** Excessive use of pesticides, insecticides, herbicides and fertilizers leads to chemical buildup in soil, which can leach into groundwater and disrupt soil microbial balance.
- **Municipal Solid Waste and Sewage Sludge :** Improper disposal of household garbage and sewage sludge can lead to the accumulation of plastics, pharmaceuticals and pathogens in soil.

References :

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- Ghosh, P., et al. (2015). "Contamination of Soil Due to Solid Waste Disposal—A Review." *Environmental Science and Pollution Research*, 22(10), 7415–7424. https://doi.org/10.1007/s11356-014-3945-z

9. Where does the term soil come from?

See question no. 01 part # 14 of 2022

10. What is soil texture?

Soil texture refers to the relative proportion of different – sized mineral particles in the soil – sand, silt and clay – which collectively determine the soil's physical characteristics. It is a fundamental property of soil that affects water retention, drainage, aeration, nutrient availability and root penetration.

Components of Soil Texture :

1. Sand (Particle size: 0.05 – 2.00 mm)
2. Silt (Particle size: 0.002 – 0.05 mm)
3. Clay (Particle size: < 0.002 mm)

Soil Textural Classes : Using these three components, soil scientists classify soil texture into **12** textural classes (e.g., sandy loam, silty clay loam, clay) using the soil textural triangle, a tool that helps visualize and determine the classification based on the percentage of sand, silt and clay.

References :

- **Brady, N.C., & Weil, R.R. (2016).** *The Nature and Properties of Soils* (15th ed.). Pearson Education.
  ISBN: 978-0133254488
- **USDA NRCS (2020).** "Soil Texture."
  https://www.nrcs.usda.gov
- **FAO (Food and Agriculture Organization).** "Soil Texture and Structure."
  http://www.fao.org/3/a0100e/a0100e07.htm

11. What is CEC of the soil?

See question no. 02 part # 02 of 2015

12. What is dirty dozen?

See question no. 01 part # 13 of 2022

13. What are PCBs?

Polychlorinated Biphenyls (PCBs) are a group of synthetic organic chemicals composed of carbon, hydrogen and chlorine atoms. These compounds belong to a class of chemicals known as chlorinated hydrocarbons and were widely used in various industrial and commercial applications due to their chemical stability, non – flammability and insulating properties.

- PCBs were banned or restricted in many countries starting in the late 1970s.

- Included in the **Stockholm Convention on Persistent Organic Pollutants** (2001), which seeks to eliminate or restrict their production and use globally.

Chemical Structure and Types : PCBs consist of two linked benzene rings (a biphenyl structure) with varying numbers (1–10) of chlorine atoms attached, resulting in 209 possible congeners (structural variants).

General Formula : $C_{12}H_{10-x}Cl_x$, Where *x* can range from 1 to 10.

Properties of PCBs :

- Highly lipophilic (fat-soluble)
- Chemically stable and resistant to degradation
- Low water solubility
- Persistent in the environment and bioaccumulative

Major Uses of PCBs (Before Ban) :

- Electrical transformers and capacitors (as insulating oils)
- Plasticizers in paints, sealants and caulking

References :

- **U.S. Environmental Protection Agency (EPA)**
  "Learn about Polychlorinated Biphenyls (PCBs)."
  https://www.epa.gov/pcbs
- **World Health Organization (WHO)**
  "Polychlorinated biphenyls: human health aspects." Environmental Health Criteria 140, 1993.
  https://www.who.int/ipcs/publications/ehc/ehc_140/en/
- **Agency for Toxic Substances and Disease Registry (ATSDR)**
  "Toxicological Profile for Polychlorinated Biphenyls (PCBs)."
  https://www.atsdr.cdc.gov/toxprofiles/tp17.html

14. Mention any two applications of IR spectroscopy in environmental monitoring.

- **Detection of Air Pollutants (e.g., CO_2, NO_x, SO_2, CH_4) :** IR spectroscopy, especially Fourier – Transform Infrared (FTIR) spectroscopy, is widely used for continuous, real-time monitoring of gaseous pollutants in the atmosphere.
- **Analysis of Organic Contaminants in Water and Soil :** IR spectroscopy can detect and quantify organic pollutants such as pesticides, herbicides and petroleum hydrocarbons in soil and water samples.
- Microplastics Identification in Environmental Samples : Attenuated Total Reflectance (ATR) – IR spectroscopy is commonly used to identify and characterize microplastics in water, sediment and soil.

References :

- Griffith, D. W. T. (1996). “Synthetic Calibration and Quantitative Analysis of Gas-Phase FT-IR Spectra.” *Applied Spectroscopy*, 50(1), 59–70.
  https://doi.org/10.1366/0003702963906129

Banat, F., & Al-Asheh, S. (2001). “Assessment of the Use of Infrared Spectroscopy for Monitoring the Biodegradation of Petroleum Hydrocarbons in Soils.” *International Biodeterioration & Biodegradation*, 47(2), 179–185. https://doi.org/10.1016/S0964-8305(01)00058-0

- Mintenig, S. M., et al. (2017). “Identification of Microplastic in Effluents of Waste Water Treatment Plants Using FTIR Spectroscopy.” *Water Research*, 108, 365–372. https://doi.org/10.1016/j.watres.2016.11.015

15. What is ATOM ECONOMY in Green Chemistry?

See question no. 02 part # 10 of 2020

Long Questions (2023)

1. Describe the SOLID components of soil.

Soil solids constitute the non – fluid portion of soil and represent approximately 50% of soil volume in well – structured soils. They include :

1. **Inorganic (mineral) materials** (~45% by volume)
2. **Organic matter** (~3–5% by weight in fertile mineral soils)

These two fractions determine the chemical reactivity, nutrient availability, water retention and structural integrity of the soil.

1. INORGANIC (MINERAL) MATTER

Origin : It is derived from the parent rock (bedrock or transported material) through :

- **Physical weathering** : mechanical breakdown due to temperature fluctuations, freeze – thaw cycles, abrasion and root penetration.
- **Chemical weathering** : alteration of minerals via hydrolysis (reaction with water), oxidation (reaction with oxygen), dissolution (reaction with acids or bases) and carbonation (reaction with CO_2).
- **Biological weathering** : lichen activity, microbial metabolism and organic acid release from roots.

Classification of Soil Minerals

A. **Primary Minerals :** Minerals inherited from the parent material, relatively unweathered. These are predominantly present in sand and silt fractions.

- **Examples** :
  - **Quartz (SiO_2)** : Chemically inert, dominates sandy soils. Provides bulk but low nutrient retention.
  - **Feldspars ($KAlSi_3O_8$, $NaAlSi_3O_8$)** : Weather into clays, releasing essential ions like K^+ and Na^+.
  - **Micas (biotite, muscovite)** : Flaky silicate minerals, rich in **K^+, Mg^{2+}, and Fe^{2+}**, which become bioavailable upon breakdown.

B. Secondary Minerals : New minerals formed through the alteration of primary minerals. These are dominant in clay – sized particles (<2 µm).

- **Types** :
  - **Phyllosilicates (clay minerals)** :
    - ✓ **Kaolinite (1:1)** : Low surface area, low cation exchange capacity (CEC).

    - ✓ **Smectite (2:1)** : High surface area, high swelling capacity, very high CEC.
    - ✓ **Illite** : Intermediate characteristics.
  - **Iron and aluminum oxides** (e.g., goethite FeOOH, gibbsite $Al(OH)_3$): Highly reactive with phosphorus and organic matter.

Key Functions of Mineral Matter :

- **Textural classification** : Determined by sand (2.0 – 0.05 mm), silt (0.05 – 0.002 mm) and clay (<0.002 mm). Texture influences :
  - Water – holding capacity
  - Aeration
  - Root penetration
  - Microbial habitat
- **Chemical reactivity** :
  - CEC from negatively charged clay surfaces allows retention of nutrients like K^+, NH_4^+, Ca^{2+}.
  - pH buffering via exchange and surface reactions with H^+ and OH^- ions.
- **Nutrient supply** :
  - Weathering releases micronutrients (Fe, Zn, Mn, Cu) and macronutrients (K, Mg, Ca, Si).
  - Clay minerals can adsorb phosphate and ammonium, controlling their mobility and availability.

2. ORGANIC MATTER

Organic matter is the most biologically active and chemically complex component of the soil. It plays a disproportionate role in nutrient cycling**,** soil structure and biological health despite its small quantity.

Origin : Organic matter derived from :

- Plant residues (roots, leaves, stems)
- Animal residues (manure, carcasses, secretions)
- Microbial biomass (bacteria, fungi, protozoa)
- Root exudates (sugars, organic acids)

Classification of Organic Matter

A. Litter and Detritus :

- Fresh or partially decomposed organic remains.
- Subject to rapid microbial decomposition.
- Source of labile (readily decomposable) carbon and nutrients.

B. Humus :

- Highly decomposed, dark – colored, stable fraction of organic matter.
- Composed of humic substances :
  - Fulvic acids (lightest, most soluble)
  - Humic acids (intermediate)
  - Humin (dark, insoluble)
- High in carboxylic and phenolic functional groups → very high **CEC**.

C. Microbial Biomass :

- Living portion of organic matter : bacteria, fungi, actinomycetes, protozoa.
- Acts as a biological engine of soil — decomposes litter, mineralizes nutrients.

Key Functions of Organic Matter

- **Nutrient cycling** : Organic matter is a reservoir of N, P and S. Microbial decomposition releases nutrients in plant-available forms (e.g., NH_4^+, PO_4^{3-}).
- **Water retention** : Organic matter can hold up to 90% of its weight in water. It enhances water infiltration and reduces erosion.
- **Soil structure** : Polysaccharides and humic substances bind particles into aggregates → improves porosity and tilth.
- **Biological activity** : Provides food and energy for microbes, earthworms and arthropods.

References :

- Brady, N.C. & Weil, R.R. (2016). *The Nature and Properties of Soils*, 15th ed. Pearson Education.
- FAO (2005). *The importance of soil organic matter*. http://www.fao.org/3/a0100e/a0100e00.htm
- Stevenson, F.J. & Cole, M.A. (1999). *Cycles of Soil: Carbon, Nitrogen, Phosphorus, Sulfur, Micronutrients*, 2nd ed. Wiley.
- Jenny, H. (1941). *Factors of Soil Formation*. McGraw-Hill.
- USDA NRCS (2020). *Soil Health Technical Note: Soil Organic Matter* https://www.nrcs.usda.gov

2. How INDUSTRIES are contributing in soil pollution?

See question no. 0 part # 01 of 2021

3. What are POPs? How these are affecting the environment?

Persistent Organic Pollutants (POPs) are organic compounds that are resistant to environmental degradation through chemical, biological and photolytic processes. Due to their persistence, POPs bioaccumulate through the food web and pose a significant threat to human health and the environment.

Characteristics of POPs :

I. **Persistence** : POPs resist degradation and remain in the environment for years or even decades. For example DDT has a half – life of up to 15 years in soil.
II. **Bioaccumulation** : POPs accumulate in the fatty tissues of living organisms and increase in concentration up the food chain.
III. **Long – range environmental transport** : POPs can travel great distances from their source through atmospheric and oceanic transport.
IV. **Toxicity** : POPs are highly toxic and can cause cancer, reproductive disorders, immune system damage and endocrine disruption.

Examples of POPs :

- **Pesticides** : DDT, Aldrin, Chlordane
- **Industrial Chemicals** : PCBs (Polychlorinated Biphenyls), HCB (Hexachlorobenzene)
- **By-products** : Dioxins, Furans

How POPs Affect the Environment :

i. **Soil Contamination :** POPs persist in soils, reducing microbial diversity and affecting plant growth due to their toxicity. For example DDT residues affect nitrogen – fixing bacteria in agricultural soil.

ii. **Water Pollution :** POPs leach into groundwater and surface water systems, threatening aquatic life and contaminating drinking water. POPs in water can bioaccumulate in fish, affecting entire aquatic food chains.

iii. **Air Pollution and Long – Range Transport :** POPs evaporate from soil or water surfaces and travel long distances via the atmosphere. Polar regions are significantly contaminated despite no local sources.

iv. **Biodiversity Loss :** POPs interfere with reproductive success, immunity and development in wildlife, leading to population declines. For example PCBs affect reproduction in seals and dolphins.

v. **Food Chain Contamination :** Because POPs accumulate in fat tissues, they biomagnify up the food chain, posing risks to apex predators and humans. For example Inuit populations consuming marine mammals show high POP levels.

vi. **Human Health Risks :** Humans are exposed through food, air and water. POPs are linked to :

- Endocrine disruption
- Neurodevelopmental disorders
- Cancer
- Immune suppression

Reference :

- UNEP, "Stockholm Convention on Persistent Organic Pollutants (POPs)," 2001. https://www.unep.org
- ATSDR. "Toxicological Profile for DDT, DDE, and DDD," 2002. https://www.atsdr.cdc.gov
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- Wania, F., & Mackay, D. (1996). "Tracking the distribution of persistent organic pollutants." *Environmental Science & Technology*, 30(9), 390A–396A. https://doi.org/10.1021/es962399q
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- Ross, P. S., et al. (1996). "High PCB concentrations in free-ranging Pacific killer whales." *Marine Pollution Bulletin*, 32(8–9), 534–541. https://doi.org/10.1016/0025-326X(96)00006-7
- Dewailly, E., et al. (1993). "High levels of PCBs in Inuit mothers and their infants in Arctic Quebec." *Environmental Health Perspectives*, 101(3), 216–220. https://doi.org/10.1289/ehp.93101216
- WHO, "Health risks of persistent organic pollutants," Fact Sheet, 2018. https://www.who.int

4. Elaborate any six importance of GREEN CHEMISTRY.

See question no. 02 part # 04 of 2019

5. Write a brief note on GC.

Gas Chromatography (GC) is a powerful analytical separation technique used for analyzing **volatile** and **semi-volatile compounds** in complex mixtures. It is widely employed in environmental monitoring, pharmaceuticals, food testing, forensic science and petrochemical industries due to its high resolution, sensitivity, and speed.

Basic Principle :

Gas Chromatography operates on the principle of partitioning a sample between a mobile phase (carrier gas) and a stationary phase (column coating or packed material). The sample, after injection, is vaporized and carried by an inert gas (like helium or nitrogen). As it travels through the column, the compounds interact differently with the stationary phase based on their volatility and polarity. Compounds that interact weakly elute (exit) faster, while strongly retained compounds elute later. The retention time (t_r) – the time a compound takes to pass through the column – is used to identify and quantify components.

Main Components of a GC System

1. **Carrier Gas Supply**
   - **Common gases :** Helium, Nitrogen and Hydrogen.
   - **Function :** Acts as the mobile phase ; must be inert and pure.
   - Flow controlled via pressure regulators.
2. **Sample Injection System :**
   - Introduces the liquid or gaseous sample into the GC.
   - Heated to instantaneously vaporize the sample.
   - **Injection types :** Split, splitless, on-column, and PTV (programmable temperature vaporizer).
3. **Column :**
   - Heart of the GC; where separation occurs.
   - **Types :**
     - Packed columns : Shorter, filled with solid support.

        - Capillary (open tubular) columns : Long, narrow, coated with stationary phase (e.g., polyethylene glycol, dimethylpolysiloxane).
    - Temperature is programmed (isothermal or temperature gradient) to enhance separation.

4. **Oven :**
    - Thermally controls the column temperature.
    - Isothermal for constant separation or programmed ramping for complex mixtures.
5. **Detector :**
    - Identifies and quantifies eluting components.
    - **Types :**
        - Flame Ionization Detector (FID) : detects hydrocarbons.
        - Electron Capture Detector (ECD) **:** for halogenated compounds, pesticides.
        - Thermal Conductivity Detector (TCD): universal, less sensitive.
        - Mass Spectrometer (GC-MS) : for structural analysis and high sensitivity.
6. **Data System :**
    - Computer software records and analyzes chromatograms.
    - Calculates **retention times**, **peak areas**, and **concentrations**.

Types of Stationary Phases in GC :

- Nonpolar columns (e.g., dimethylpolysiloxane) : for nonpolar analytes.
- **Polar** columns (e.g., polyethylene glycol) : for alcohols, amines and acids.
- Selection depends on the **"like dissolves like"** rule to enhance selectivity.

Applications of Gas Chromatography :

i. **Environmental Chemistry**
    - **Air Quality Monitoring** : Identification and quantification of volatile organic compounds (VOCs), ozone precursors, and industrial pollutants.
    - **Water and Wastewater Testing** : GC is widely used to detect halogenated solvents, pesticides, and phenolic compounds in surface water and groundwater.
    - **Soil and Sediment Analysis** : Detects residual hydrocarbons, persistent organic pollutants (POPs) and pesticide residues.
    - **Indoor Air and Workplace Exposure Assessment** : Detects benzene, toluene, xylene (BTX) compounds and formaldehyde.
    - **Hazardous Waste Analysis** : Used under EPA Methods (e.g., 8015, 8021, 8270) for characterization of toxic substances in industrial and municipal waste.

ii. **Pharmaceuticals** : Impurity profiling, residual solvent analysis (ICH Q3C compliance).

iii. **Food Industry** : Analysis of flavors**,** preservatives, fatty acid methyl esters (FAMEs) and contaminants.

iv. **Petrochemicals** : Hydrocarbon composition, gasoline analysis, sulfur compounds.

v. **Forensic Science** : Detection of drugs**,** accelerants, explosives and poisons.

vi. **Clinical Chemistry** : Blood alcohol analysis, screening for toxic volatiles.

Limitations of GC :

Despite its strengths, GC has certain limitations :

i. **Only for volatile and thermally stable compounds** : Non-volatile compounds require derivatization or alternative techniques (like HPLC).
ii. **Sample Preparation** : Complex matrices require **extensive cleanup**, which is time – consuming.
iii. **Destructive Detection** : Detectors like FID and ECD destroy the sample.
iv. **Limited polarity range** : Polar, ionic, or high molecular weight compounds are difficult to analyze directly.
v. **Instrument Cost and Maintenance** : High-performance systems (GC-MS) are expensive and require regular calibration.

References :

- McNair, H. M., & Miller, J. M. (2009). *Basic Gas Chromatography* (2nd ed.). Wiley-Interscience.
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- Sparkman, O. D., Penton, Z. E., & Kitson, F. G. (2011). *Gas Chromatography and Mass Spectrometry: A Practical Guide* (2nd ed.). Academic Press.
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- Pawliszyn, J. (2012). *Comprehensive Sampling and Sample Preparation: Analytical Techniques for Scientists*. Academic Press.

SHORT QUESTIONS (2024)

1. What is DIRTY DOZEN?

See question no. 01 part # 13 of 2022

2. What is composition of soil air?

Soil air refers to the mixture of gases found in the pore spaces of soil that are not filled with water. It is a critical component of soil health because it supplies oxygen to plant roots and soil organisms, supports microbial activity and influences chemical reactions in the soil.

- **Nitrogen (N_2) :** Soil air, comprising about 78% nitrogen.
- **Oxygen (O_2) :** Soil air typically contains 10% to 15% oxygen.
- **Carbon Dioxide (CO_2) :** It can range from 1% to 10% in soil air.
- **Water Vapor :** Soil air contains more water vapor than atmospheric air, especially in moist and warm soils.
- **Trace Gases :** Soil air also contains small amounts of other gases :
    - Methane (CH_4)
    - Ammonia (NH_3) and Hydrogen sulfide (H_2S)
    - Nitrous oxide (N_2O)

Reference :

- Brady, N.C., & Weil, R.R. (2016). *The Nature and Properties of Soils*. Pearson.
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- Hillel, D. (1998). *Environmental Soil Physics*. Academic Press.
- Conrad, R. (1996). "Soil microorganisms as controllers of atmospheric trace gases." *Microbiological Reviews*, 60(4), 609–640.

3. What is WATER erosion?

Water erosion is the process by which soil particles are detached and transported by the action of water, particularly rainfall, runoff, rivers or melting snow. It is one of the most significant forms of soil erosion and poses a major threat to agriculture, infrastructure and natural ecosystems.

Causes of Water Erosion :

- Rainfall Intensity and Frequency
- Slope Gradient
- Lack of Vegetation
- Soil Properties

Types of Water Erosion :

- Splash Erosion
- Sheet Erosion
- Rill Erosion
- Gully Erosion
- Streambank and Riverbank Erosion

References :

- Morgan, R.P.C. (2005). *Soil Erosion and Conservation*. Blackwell Publishing.
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4. **What type of pollutants can be analyzed by AAS?**

See question no. 02 part # 10 of 2021

5. **Which type of fungi produce AFLATOXINS?**

See question no. 01 part # 05 of 2023

6. **How availability of Phosphorous is affected by soil pH?**

See question no. 01 part # 06 of 2022

7. **Which type of bacteria are involved in the production of Nitrogen in the soil?**

The production and transformation of nitrogen in the soil is mainly carried out by nitrogen – fixing bacteria and other nitrifying and denitrifying microorganisms that participate in the nitrogen cycle. These bacteria help convert atmospheric nitrogen (N_2) and organic nitrogen into plant – available forms such as ammonium (NH_4^+) and nitrate (NO_3^-).

- **i.** Nitrogen – Fixing Bacteria (Biological Nitrogen Fixation)
- **ii.** Ammonifying Bacteria (Ammonification)
- **iii.** Nitrifying Bacteria (Nitrification)
- iv. Denitrifying Bacteria (Denitrification)

Reference :

- DzSprent, J.I. (2001). *Nodulation in Legumes*. Royal Botanic Gardens, Kew
- Alexander, M. (1977). *Introduction to Soil Microbiology*. Wiley-Interscience.
- Prosser, J.I. (1986). *Nitrification*. IRL Press.

- Zumft, W.G. (1997). *Cell biology and molecular basis of denitrification*. *Microbiology and Molecular Biology Reviews*, 61(4), 533–616.

8. **Propose the reclamation of Basic soil.**

   See question no. 01 part # 08 of 2022

9. **Enlist any two applications of UV – Vis spectroscopy for environmental monitoring?**

I. **Detection of Nitrate and Nitrite in Water :** Nitrate and nitrite absorb UV light around 200 – 230 nm, allowing for quantitative analysis in rivers, lakes and drinking water.

II. **Monitoring of Organic Pollutants (e.g., Aromatic Hydrocarbons) :** UV – Vis detects **aromatic compounds** like benzene, toluene and phenols due to their $\pi \rightarrow \pi^*$ transitions around 250 – 280 nm.

III. **Measurement of Chemical Oxygen Demand (COD) :** UV – Vis is used after oxidation reactions to measure absorbance of chromic ions, correlating with COD levels.

<u>Reference</u> :

- Clesceri, L.S., Greenberg, A.E., & Eaton, A.D. (1998). *Standard Methods for the Examination of Water and Wastewater*, APHA.
- Richardson, S.D. (2009). *Environmental mass spectrometry: emerging contaminants and current issues*. *Analytical Chemistry*, 81(12), 4645–4677.
- Radojević, M., & Bashkin, V.N. (2006). *Practical Environmental Analysis*, Royal Society of Chemistry.

10. **Mention any two special features of HPLC.**

    See question no. 01 part # 06 of 2022

11. **Enlist primary macronutrients of the soil?**

The primary macronutrients are the essential elements that plants require in large amounts for growth, development, and reproduction. These elements are integral to plant metabolism, enzyme activation and overall plant health. The primary macronutrients in the soil include nitrogen (N), phosphorus (P)**,** and potassium (K) each nutrient plays a distinct role in plant metabolism and productivity :

i. **Nitrogen** : Supports vegetative growth, photosynthesis and protein formation.
ii. **Phosphorus** : Key for root development, energy transfer and reproduction.
iii. **Potassium** : Regulates water balance, enzyme activation and stress resistance.

Sufficient availability of these nutrients in soil is essential for optimum crop yield and **soil fertility** management.

<u>References</u> :

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12. **Is UV – Vis spectroscopy appropriate technique for analysis of heavy metal?**

Yes, UV – Vis spectroscopy can be an appropriate technique for the analysis of heavy metals under certain conditions, especially when complexed with specific ligands or in the form of chromophore – containing metal complexes. UV – Vis spectroscopy is often used for quantifying the concentration of metals in environmental samples, such as water, soil or air, due to its simplicity, cost-effectiveness and ability to provide rapid analysis.

i. **Formation of Metal – Ligand Complexes :** Copper (Cu^{2+}) forms a complex with diethyldithiocarbamate (DDTC), which absorbs UV light at a wavelength of around 440 nm. This allows for the determination of copper concentrations.
ii. **Specificity for Metal Ions :** Lead (Pb^{2+}) can form a complex with pararosaniline that absorbs at 530 nm and this absorbance is directly related to the concentration of lead in the sample.
iii. **Sensitivity and Detection Limits :** The complexation of mercury (Hg^{2+}) with 5 – Br – PADAP results in an intense absorption peak, enabling detection of very low mercury concentrations in water samples.

Reference :

- Snell, F.D., & Snell, C.T. (2000). *Colorimetric Methods of Analysis*. CRC Press.
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- Zygmunt, B., & Walentowicz, M. (2001). *Applications of UV-Vis Spectroscopy in Environmental Monitoring*. *Analytical Methods*, 9(1), 104–108.

13. What do you know about "HUMUS"?

See question no. 01 part # 01 of 2022

14. How clay is involved in nutrient holding capacity of soil?

See question no. 01 part # 15 of 2022

15. How acidic waste is produced by MINING?

Mining activities, particularly metal ore extraction and coal mining, can generate acidic waste that poses a significant environmental threat. This acidic waste is commonly referred to as acid mine drainage (AMD**)**, which occurs when sulfide minerals, primarily pyrite (FeS_2), are exposed to oxygen and water during mining operations. The chemical reactions that occur during mining processes can lead to the formation of acidic, toxic leachates that contaminate surrounding soil and water sources. Acidic waste produced by mining done by following ways :

I. Exposure of Sulfide Minerals to Oxygen and Water
II. Weathering of Sulfide Minerals
III. Ore Processing (Flotation and Leaching)
IV. Coal Mining and Acidic Waste
V. Leaching of Acidic Metals into Surrounding Environments

References :

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LONG QUESTIONS (2024)

1. **Elaborate the SOLID components of soil.**

   See question no. 02 part # 01 of 2023

2. **How MINING is contributing in soil pollution?**

   See question no. 03 part # 05 of 2017

3. **Describe the principle and application of AAS for environmental monitoring.**

   See question no. 03 part # 01 of 2014

4. **What are POPs? How they are polluting environment?**

   See question no. 02 part # 03 of 2023

5. **Write a note on micronutrients of soil.**

   See question no. 03 part # 01 of 2016